

School of Computing and Information Technology
CSIT321 — Project

uiabo.

Preliminary Technical Documentation

Project topic: CSIT-26-S3-30 — AI-Based Misinformation Detection for Short-Form Content

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The report retains the agreed product requirements and design, and records the implemented text prototype separately. Current and proposed functions are distinguished throughout.

Document Control

Field	Record
Document owner	FYP-26-S3-30 project team
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Record of Revision

Date	Version	Recorded change
4 September 2026	Initial section draft	Date carried by the existing UIABO PTD section files.
7 September 2026	0.2	Compiled UIABO sections; added coverage absent from the new sample comparison; refreshed implementation, testing, scope and remaining-work records.

Evidence categories

- Implemented: observed in the current workspace and/or a dated test record.
- Planned requirement: retained from PRD/URS/TDM; its presence in a design does not establish completion.
- Proposed addition: a draft analysis or management process prepared for team review, with no invented agreement or sign-off.
- Pending evidence: information or validation not yet available. See Section 18 and PTD_MISSING_INFORMATION.md.

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1. Product Functionality Overview

uiabo is an elderly-friendly Android application for assessing potentially misleading English short-form content. The agreed product scope covers text, public webpage links and static images, with free and premium access. The September text prototype implements a narrower working slice; the scope and status table below distinguishes it from the full planned product.

Core Functions

The following functional descriptions retain the planned product scope from the PRD. They are not a claim that every function has been implemented. Audio and video analysis are excluded; required deepfake detection concerns supported static images.

Content Submission

- Free users and premium users may paste short-form text or captions.
- Premium users may upload images, screenshots and memes together with an accompanying caption.
- Premium users may add or edit the caption associated with an uploaded image.
- Premium users may review and correct OCR-extracted text before analysis.
- Free users and premium users may submit a supported public webpage or article link. The system will also inspect the submitted URL for malicious-link indicators.

Text and Image Processing

- Extract visible text from premium image submissions using Optical Character Recognition (OCR).
- Identify objectively checkable factual claims.
- Distinguish factual claims from opinions, jokes and other non-verifiable statements.
- Analyse the relationship between a premium user's uploaded image and its caption.
- Identify possible contextual mismatches involving date, location, event or description, and generate an image deepfake or manipulation likelihood score where supported.

Caption-image analysis will focus on contextual inconsistency. The premium deepfake detector may identify indicators associated with AI generation or digital manipulation, but its score will be presented as probabilistic supporting evidence rather than conclusive proof. Its media-forensics explanation will translate the detector's available technical output into plain language by describing the signals that influenced the score, their limitations and, when supported by the detector, any suspicious visual regions. It will not treat a heatmap or individual artefact as proof of manipulation.

Fact-check and Evidence Retrieval

- Search for semantically similar claims that have already been reviewed by fact-checking organisations.
- Retrieve supporting or contradicting evidence from available sources.
- Prioritise authoritative and credible sources.
- Assess the credibility of the original source when sufficient information is available.
- Display links to the evidence used in the assessment.

Result Generation

- Display a clear result page with a simple concern label.
- Display a misinformation or fake-news risk score.
- Display an AI-generation or manipulation likelihood score where applicable and communicate the uncertainty of both scores.
- Show the factual claim identified by the system.
- Provide a plain-language explanation.

- Present supporting and contradicting evidence.
- Provide source links or citations.
- Allow the user to report a result that appears inaccurate and leave a review of the service.
- Provide a shareable result link and recommend an appropriate next action, such as advising the user not to forward the content yet.

The proposed concern labels are Low Concern, Needs Caution, High Concern and Not Enough Information. Numerical values will be presented as risk or likelihood indicators rather than mathematical proof that a claim is false or that an image was AI-generated.

Account and Usage Functions

- Register an account, log in, log out, reset a password, update a profile and delete an account.
- Free users may submit text or a public webpage link once per day and cannot access image upload or OCR.
- Premium users may perform up to 60 successful submissions per month and may access all supported input and analysis functions.
- Display the user's remaining daily or monthly submission allowance.
- Allow users to view and delete result history and share a result through a link.
- Allow users to report incorrect results, leave a review and provide feedback for investigation.
- Allow free users to upgrade to premium access.

Administrative and Data-management Functions

- System administrators may log in, log out, query and view user accounts, and suspend or unsuspend accounts.
- System administrators may review user feedback and reported problems.
- Data engineers may log in, log out and update their profiles.
- Data engineers will use a dashboard to review reported incorrect results and data-quality issues.
- Data engineers will monitor data-ingestion pipelines so that failed or incomplete updates can be identified.
- Data engineers will maintain the data, source provenance and evaluation records required by the verification workflow.

1.1 Current implementation status

Area	Status	Evidence / limit
Account access	Available	Free registration, verification-link gating, sign-in, password reset and sign-out; profile name editing.
Input preparation	Available	English detection/validation, meaning-preserving normalization, length checks and warnings.
Claim analysis	Available with limits	Ollama Cloud classification ensemble and source-span extraction; satire/disagreement limits remain.
Evidence retrieval	Available with limits	Google Fact Check discovery, Tavily extraction and scoped search, citations and controlled failures.
Assessment	Available baseline	Lexical evidence stance and heuristic scores; semantic mistakes observed, accuracy not established.
Android text flow	Available	TDM-based Home, text submission, progress, result detail, source links and personal history.
Firestore/allowance	Available	Owner-bound saved results; atomic completion/charge; idempotent retries; free daily and premium monthly counters.
Webpage submission	Planned	Direct fetching and malicious-link screening are not implemented; UI offers copied-text checking.
Image/OCR/context/deepfake	Required/planned	TDM designs exist; image pipelines remain unimplemented. Deepfake remains required.
Payments	Planned	Premium may be granted for demonstration; no checkout, paid renewal or cancellation service.
Feedback/deletion/public sharing	Partial/planned	Native text/source sharing exists. Public result links, reviews, reports and account/history deletion remain planned.
Operational portal	Planned	System-administrator and data-engineer designs exist; working portals are not implemented.

Evidence / source: PRD 1.2; URS 2.2–2.3; current source files and evaluation/reports/app-integration/README.md, dated 7 September 2026.

2. Project Overview

2.1 Project Description

uiabo will provide a guided misinformation-checking Android application designed for Singapore citizens aged 65 and above who may have limited experience evaluating online information. The initial release will support English and will provide separate free-user, premium-user, system-administrator and data-engineer access.

After registering or logging in, a free user may paste text or submit a public webpage or article link. Free users receive one successful submission per day and cannot upload images or use OCR. The submitted URL will be checked for malicious-link indicators before the article content is analysed.

The system will identify the principal factual claim or claims in the submitted text or linked article. Statements consisting mainly of opinion, humour, personal experience or unverifiable predictions will not automatically be treated as misinformation. If no checkable claim is found, the result page will state that there is Not Enough Information.

For a checkable claim, the system will search for similar claims previously investigated by recognised fact-checking organisations and retrieve current supporting or contradicting evidence. Source credibility may contribute to the assessment, but it will not replace claim-specific evidence.

Premium users will receive up to 60 successful submissions per month and may use all supported functions. They may upload images with captions, review and correct OCR-extracted text, analyse caption-image context and run an image deepfake detector. The deepfake detector will provide an AI-generation or manipulation likelihood score rather than claim conclusive proof. The accompanying media-forensics explanation will describe, in plain language, the available detection signals that influenced the score, the limitations of those signals and any suspicious visual regions supplied by the detector. It will not imply that an individual signal or highlighted region conclusively proves manipulation.

After analysis, the result page will display a concern label, a fake-news or misinformation risk score, uncertainty information, a plain-language explanation and evidence links. When relevant to the input, it will also display an AI-generation or manipulation likelihood score. The two scores will remain separate because media authenticity and factual accuracy are different questions.

Free and premium users will be able to view and delete result history, share a result through a link, report a result that appears wrong and leave a review. The dashboard will display the user's remaining daily or monthly submission allowance, and free users will be able to upgrade to premium access.

The interface will prioritise readable text, high contrast, large interaction targets, clear labels and a small number of steps. System administrators will query and manage accounts, suspend or unsuspend users and review feedback. Data engineers will monitor ingestion pipelines through a dashboard and investigate reported incorrect results and data-quality failures. World Wide Web Consortium guidance notes that declining vision, dexterity, concentration and short-term memory can affect some older users, and that established accessibility standards address many of these needs (World Wide Web Consortium, 2025).

2.2 Learning Objectives

Proposed learning objectives for team review. These are goals aligned with the work, not retrospective claims that every member has demonstrated each outcome.

- Design and integrate an Android client, authenticated Python API and cloud persistence using explicit component contracts.

- Compare claim-analysis, retrieval and evidence-assessment methods using labelled examples and documented error analysis.
- Communicate uncertainty and evidence in an interface suitable for older Android users, then validate it through usability testing.
- Apply credential separation, user ownership, failure handling and reproducible testing in a multi-service application.
- Coordinate a five-person project using version control, documented handoffs, scope decisions and reviewable progress evidence.

2.3 Target Audience

The primary target users are Singapore citizens aged 65 and above who can use Android smartphones but may have limited experience with formal fact-checking. They may regularly encounter forwarded text, webpage links, screenshots, images or captions and prefer clear explanations over professional investigation tools. Family members may recommend the application or help an older user interpret a result.

2.4 Business Model

uiabo will adopt a social-impact freemium subscription model. Free access will provide one text or webpage-link submission per day, while premium access will provide up to 60 successful submissions per month together with image upload, OCR, caption-image analysis and image deepfake detection. A future API-licensing model may allow organisations to integrate selected verification functions into their own systems.

Key Partners

- Government and public agencies involved in digital inclusion, media literacy, public communication and online safety.
- Trusted news and fact-checking organisations that publish verified claims and evidence.
- Community centres, senior activity centres, libraries and elderly-support organisations.
- Universities and research organisations that can support evaluation and dataset development.
- Technology providers supplying AI, OCR, search, authentication, cloud hosting, database and payment services.
- Social-media and messaging platforms as possible future integration partners.

These organisations are potential partners only. The project must not imply formal endorsement or access to private platform data unless an agreement is established.

Key Activities

- Designing and testing an elderly-friendly mobile interface.
- Analysing text and webpage links, screening malicious URLs, extracting OCR text and assessing caption-image context.
- Matching claims against previous fact-checks and retrieving current evidence.
- Ranking source relevance and credibility.
- Generating misinformation risk, AI-generation or manipulation likelihood, uncertainty and claim-level explanations.
- Validating citations and maintaining trusted-source collections.
- Evaluating OCR, retrieval and misinformation-assessment performance.
- Managing free daily allowances, premium monthly allowances, accounts, subscriptions, user reports, privacy and security.
- Conducting usability testing and improving the system through feedback.

Key Resources

- A five-person development team.

- Mobile application development frameworks.
- AI and natural-language processing models.
- OCR, malicious-link screening, image deepfake detection, search and fact-check retrieval services.
- Trusted government, academic, news and fact-check sources.
- Backend, authentication, database and cloud infrastructure.
- Labelled evaluation datasets and data-engineering pipelines.
- Older adult participants for usability testing.

Value Proposition

The principal value proposition is to provide Singapore citizens aged 65 and above with a simple and transparent method of assessing suspicious short-form content before believing or forwarding it.

- A guided workflow that does not require professional fact-checking knowledge.
- Free text and webpage-link checking, with premium image, caption, OCR and deepfake-detection functions.
- Plain-language explanations rather than an unexplained verdict.
- Links to relevant evidence so users can verify the result independently.
- Visible uncertainty and a Not Enough Information outcome when evidence is weak.
- Premium caption-image context analysis and a separate AI-generation or manipulation likelihood score.
- Shareable results that support check-before-sharing behaviour.
- A no-cost daily free tier and a predictable premium monthly allowance of 60 successful submissions.

Individual Users

Free Users

Free users are registered users who receive one successful submission per day. They may submit pasted text or a public webpage/article link and use the malicious-link detector. They can view the result page, misinformation assessments, explanations, evidence links and remaining allowance. They may also delete history, report incorrect results, leave a review, share a result through a link and upgrade to premium access. Free users cannot upload images, use OCR or access the image deepfake detector and its AI-generation or manipulation likelihood score.

Premium Users

Premium users pay for monthly access and receive up to 60 successful submissions per month. They retain all free-user functions and may additionally upload images with captions, review and correct OCR-extracted text, analyse caption-image context and use the image deepfake detector. Their dashboard will display the remaining monthly submission allowance.

Operational Users

System administrators will query and view accounts, suspend or unsuspend users and review user feedback. Data engineers will access a monitoring dashboard, review reported incorrect results, maintain relevant account details and monitor data-ingestion pipelines for failed or incomplete updates.

Channels

- Google Play Store for supported Android smartphones.
- Community centres and senior activity centres.
- Public libraries and digital-literacy programmes.
- Government and community awareness campaigns.

- Family-member recommendations.
- News and social-media campaigns.
- Workshops and demonstrations for older users.
- Future developer and organisational outreach for API licensing.

Revenue Streams

Free Introductory Access

Free users will receive one successful text or webpage-link submission per day. This free access allows users to experience the core verification, explanation, sharing and feedback functions before upgrading.

Premium User Access

Premium users will pay for monthly access that includes up to 60 successful submissions per month. Premium access unlocks image uploads with captions, OCR correction, caption-image analysis and image deepfake detection. The subscription price should be validated through user research.

Future API Licensing

A future version may license selected claim-analysis, evidence-retrieval, malicious-link and image-analysis functions through an API for approved organisational or platform integrations.

Cost Structure

Table 3. Proposed Cost Structure

Cost category	Examples
Development	Mobile application development; backend and database development; UI/UX and accessibility design; AI, OCR, malicious-link, deepfake and retrieval integration; testing tools.
Operations	AI-model usage; OCR, malicious-link, image deepfake and web-search charges; cloud hosting; database and image storage; app-store accounts; monitoring and subscription payment-processing fees.
Data and evaluation	Dataset collection, cleaning and labelling; source validation; retrieval and model evaluation; accessibility and elderly-user usability testing; possible participant incentives.
Marketing and support	Community demonstrations; onboarding material; digital-literacy workshops; customer support; public-awareness campaigns.
Contingency	Unexpected API usage; changes to third-party services; additional cloud capacity; replacement technology and security improvements.

Customer Segments

- Free-user segment: Singapore citizens aged 65 and above who use Android smartphones, want to check text or webpage links and can work within one successful submission per day.
- Premium-user segment: Older Android users who require more frequent checks or image analysis and are willing to pay for up to 60 successful submissions per month.
- Future API-customer segment: Approved organisations or platforms that may license selected verification functions. Family members, community groups and libraries remain important recommendation and adoption channels rather than separate first-version account types.

2.5 Problem statement, purpose and vision

Short forwarded messages, captions and screenshots can circulate without their original context. Checking them requires users to identify the claim, find relevant sources and interpret competing evidence. UIABO focuses on older Singapore Android users who may benefit from a simpler guided workflow and readable explanations.

The project purpose is to support evidence review before sharing. Its proposed vision is a clear, accessible checking assistant that communicates what is known, what is uncertain and where the evidence can be inspected. It does not promise an automatic guarantee of truth.

2.6 Project objectives and measurable evidence

Objective	How completion is assessed	Current position
Complete the basic text journey	Verified account → English text → cited result → owned history in Android.	Demonstrated in the local emulator and live Firestore checks.
Preserve evidence and ownership	Save user_id, timestamps, passages and URLs; deny other users access.	Implemented; automated and live checks recorded.
Improve assessment quality	Labelled held-out claims, evidence relevance and stance/verdict metrics.	Dataset and acceptance thresholds pending; no accuracy percentage claimed.
Deliver required image features	OCR, caption/context and separate deepfake likelihood with limits.	Designs exist; detector selection and implementation pending.
Validate usability and performance	Older-user task study, response-time distributions and load checks against URS targets.	Not yet established.

2.7 Conceptualization and product rationale

The concept combines a guided Android input flow with claim-level evidence and plain-language uncertainty. The PRD compares existing monitoring, source-rating, fact-check retrieval and multimedia tools. UIABO's proposed differentiator is their selected combination for older users, rather than a claim that it outperforms those tools. The staged pipeline enables individual components to be evaluated and replaced without redesigning the entire app.

2.8 SWOT analysis

Draft qualitative analysis derived from the agreed design and current implementation; review with the team.

Area	Analysis
Strengths	Focused user group; readable TDM-based flow; source citations and uncertainty; explicit contracts; working authenticated text-to-history integration.
Weaknesses	Lexical stance errors; small synthetic evaluation sets; restricted English source catalogue; unfinished image, payment and portal work.
Opportunities	Evaluate with older users/community helpers; improve Singapore-context examples; replace weak assessment rules while retaining the app/API contracts.
Threats	Provider outages/quotas; missing or outdated evidence; prompt injection; user over-trust; limited time before submission; competing verification tools.

2.9 Proposed unique selling points

- Guided English Android checking for Singapore users aged 65 and above.
- Readable concern labels, uncertainty, claim-level explanations and inspectable evidence in one workflow.
- Free daily access with a defined premium allowance; image-context and authenticity functions remain part of the planned product.
- A distinction between factual misinformation risk and image-manipulation likelihood.

These are design propositions, not validated market advantages, formal partnerships or demonstrated accuracy superiority.

2.10 Stakeholders

Stakeholder	Need / responsibility
Older free/premium users	Understand submissions, evidence, uncertainty and allowance through accessible interaction.
Family/community helpers	Assist onboarding and interpretation; not a separate account role.
Supervisor: Mr Liaw Chun Huei	Academic and technical guidance; scope and progress review.
Assessor: Mr Terrence Chew	Review the project deliverables and demonstrated outcomes.
Five-person project team	Implement, evaluate, integrate and document the system; roles in Section 11.
System administrator / data engineer	Planned operational access, feedback investigation and data-quality monitoring.
Service and evidence providers	External dependencies with access, availability and usage constraints; no partnership asserted.

Evidence / source: PRD 1–2; URS 1.2/2.3; existing PTD 2.1–2.4; project discussion. SWOT, explicit learning objectives and objective evidence criteria are proposed additions.

3. Similar Existing Products

3.1 AI Misinformation Products on the Market

The research focused on five products that directly support misinformation verification: Full Fact AI, Factiveverse, NewsGuard, Google Fact Check Explorer and InVID-WeVerify. These products were selected because they support one or more stages of the verification process, including claim monitoring, factual assessment, retrieval of previous fact-checks, source evaluation, evidence search or multimedia verification. Supporting technologies such as general-purpose AI APIs and standalone deepfake-detection services are outside the scope of this direct market comparison.

The findings were verified against official product information on 31 July 2026. A feature is classified as documented only when the vendor identifies it as a current and substantive capability. A partial classification means that the feature is indirect, optional, human-operated or dependent on another component.

3.2 Research Findings and Analysis

Full Fact AI supports large-scale monitoring and human fact-checking. It monitors text, audio, video, social media, radio and broadcast content; generates transcripts; extracts and ranks check-worthy statements; and matches new statements with claims that were previously fact-checked. However, the product is intended to help professional fact-checkers identify what requires investigation rather than automatically deciding whether every new claim is true or false (Corney, 2025; Full Fact AI, n.d.; Johnston, 2025).

Factiveverse is the closest direct comparison to an automated evidence-backed workflow. It analyses text, audio and video; transcribes speech; detects claims; searches multiple engines and databases; prioritises credible sources; and provides Supported or Disputed assessments with linked evidence. Current documentation does not establish image or meme forensics, and Factiveverse Live describes user correction and interaction as a future feature rather than a current workflow (Factiveverse, n.d.-a, n.d.-b, n.d.-c).

NewsGuard specialises in source reliability and human-vetted false claims. Its News Reliability Ratings assess publishers using journalistic criteria, while False Claim Fingerprints document known false narratives, evidence, provenance, examples and spread. The source rating and the claim record serve different purposes. A publisher's 0–100 rating must not be presented as the probability that an individual post is false (NewsGuard, n.d.-a, n.d.-b, n.d.-c).

Google Fact Check Explorer is a retrieval service rather than a new-claim verifier. Through the Fact Check Tools API, users and systems can search published ClaimReview records using text or an image. This is valuable when a rumour has already been checked, but it does not extract claims, evaluate a new statement, assess the original publisher or conduct media forensics independently (Google, 2024).

InVID-WeVerify is a human-guided multimedia verification toolkit with automated components. It provides OCR, video keyframe extraction, reverse and cross-network search, fact-check and known-fake search, context aggregation, source-credibility lookup, deepfake detection and image-forensics outputs such as probability scores and heatmaps. A journalist or analyst chooses the relevant tool and interprets the evidence; the plugin does not normally produce a single automatic truth verdict for an entire post (WeVerify, n.d.-a, n.d.-b, n.d.-c, n.d.-d).

Overall market gap. None of the official documentation reviewed for the five direct-verification products describes the exact proposed combination of tiered consumer access, elderly-friendly text and link checking, malicious-link screening, evidence-backed misinformation assessment, premium image and OCR analysis, and image deepfake-risk scoring within one mobile workflow. This conclusion is limited to the documented capabilities reviewed and does not establish that every listed product lacks all of these characteristics. Factiveverse is strongest for claim assessment and

evidence search, NewsGuard is strongest for source reliability and curated false-claim data, Google is strongest for retrieving existing ClaimReview records, Full Fact AI is strongest for professional monitoring and repeat-claim detection, and InVID-WeVerify provides the broadest human-guided multimedia verification toolkit.

4. Product Comparison

4.1 Direct Product Comparison

To preserve readability in portrait format, the direct comparison is divided into two tables. “Yes” means that a capability is documented, “Partial” indicates an indirect or component-dependent capability, and “No” means that no current official documentation was identified. In the uiabo column, “Planned” represents a selected FYP capability, “Future” indicates a possible later enhancement, and a blank cell means that the feature is not included in the currently selected scope.

Table 1A. Direct Verification Comparison: Full Fact AI, Factiveverse and NewsGuard

Feature group	Feature	Full Fact AI	Factiveverse	NewsGuard
Claim analysis	Automated factual-claim extraction	Yes	Yes	Partial
Claim analysis	AI-assisted misinformation risk assessment	Partial	Yes	Partial
Multimodal input	Image and meme analysis	No	No	No
Multimodal input	Short-form video analysis	Yes	Yes	No
Multimodal input	OCR: extract text from images	No	No	No
Multimodal input	Speech-to-text / transcription	Yes	Yes	No
Context analysis	Caption-image consistency or context analysis	No	No	No
Media authenticity	AI-generated or manipulated-media detection	No	No	No
Evidence	Match claims with previous fact-checks	Yes	Yes	Yes
Evidence	Retrieve supporting or contradicting evidence	Partial	Yes	Yes
Evidence	Assess credibility of the original source	No	Yes	Yes
Explainability	Claim-level factual explanation	Partial	Partial	Yes
Explainability	Media-forensics explanation	No	No	No
Explainability	Links or citations to supporting sources	Partial	Yes	Yes
Explainability	Confidence or uncertainty score	No	Partial	Partial
Explainability	Highlight suspicious claims or visual regions	Yes	Yes	Partial
Workflow	Human-review or user-feedback workflow	Yes	No	Yes
Integration	API or external-system integration	Partial	Yes	Yes

Table 1B. Direct Verification Comparison: Google Fact Check Explorer, InVID-WeVerify and uiabo

Feature group	Feature	Google Fact Check Explorer	InVID-WeVerify	uiabo (FYP)
Claim analysis	Automated factual-claim extraction	No	No	Planned
Claim analysis	AI-assisted misinformation risk assessment	No	Partial	Planned
Multimodal	Image and meme analysis	Partial	Yes	Planned

Feature group	Feature	Google Fact Check Explorer	InVID-WeVerify	uiabo (FYP)
input				(Premium)
Multimodal input	Short-form video analysis	No	Yes	
Multimodal input	OCR: extract text from images	No	Yes	Planned (Premium)
Multimodal input	Speech-to-text / transcription	No	No	
Context analysis	Caption-image consistency or context analysis	No	Yes	Planned (Premium)
Media authenticity	AI-generated or manipulated-media detection	No	Yes	Planned (Premium)
Evidence	Match claims with previous fact-checks	Yes	Yes	Planned
Evidence	Retrieve supporting or contradicting evidence	Partial	Yes	Planned
Evidence	Assess credibility of the original source	No	Partial	Planned
Explainability	Claim-level factual explanation	Partial	Partial	Planned
Explainability	Media-forensics explanation	No	Yes	Planned (Premium)
Explainability	Links or citations to supporting sources	Yes	Yes	Planned
Explainability	Confidence or uncertainty score	No	Yes	Planned
Explainability	Highlight suspicious claims or visual regions	No	Yes	
Workflow	Human-review or user-feedback workflow	No	Partial	Planned
Integration	API or external-system integration	Yes	Partial	Future

4.2 Selected Scope for uiabo

Table 2. Selected Features and Access Tiers for uiabo

Feature group	Selected feature	Access tier
Claim analysis	Automated factual-claim extraction	Free and Premium
Claim analysis	AI-assisted misinformation risk assessment	Free and Premium
Link security	Public webpage/article analysis and malicious-link screening	Free and Premium
Multimodal input	Image and meme analysis	Premium
Multimodal input	OCR: extract and correct text from images	Premium
Context analysis	Caption-image consistency or context analysis	Premium
Media authenticity	AI-generated or manipulated-image likelihood detection	Premium
Evidence	Match claims with previous fact-checks	Free and Premium
Evidence	Retrieve supporting or contradicting evidence	Free and Premium
Evidence	Assess credibility of the original source	Free and Premium
Explainability	Claim-level factual explanation	Free and Premium
Explainability	Plain-language media-forensics explanation of the available detection signals, limitations and any supported suspicious-region indicators	Premium
Explainability	Links or citations to supporting sources	Free and Premium
Explainability	Confidence or uncertainty communication	Free and Premium

Feature group	Selected feature	Access tier
Workflow	Incorrect-result reporting, reviews and user feedback	Free and Premium

The planned application will provide one free text or webpage-link submission per day and up to 60 premium submissions per month. Both tiers will receive evidence-backed misinformation assessments, explanations, citations, uncertainty information, result sharing and incorrect-result reporting. Premium access will additionally support images with captions, OCR correction, caption-image context analysis, an image deepfake or manipulation likelihood score and a plain-language explanation of the detector's available forensic signals and limitations. Video and speech analysis remain outside the prototype scope.

5. Project Design

This chapter retains the submitted URS/TDM design, including facsimile use cases and proposed diagrams. Requirements, example screen values and future collections shown in those figures are not evidence of implementation. Section 1.1 and the additions below give the current state; Section 15 maps requirements to implementation and verification.

5.1 Functional Requirements

uiabo will cater to 2 types of users: Individual users and Operational users.

- Individual users: Free Users and Premium Users
- Operational users: System Administrators and Data Engineers

We have grouped our functional requirements for these users into:

- Base Requirements (Minimum requirements for a viable product)
- Additional Requirements (Nice-To-Have features)
- Stretch Goals (Ambitious/future features)

Base Requirements

For all 4 users:

- Log in
- Log out
- View and update profile
- Reset password

Login is a basic functionality that is required to differentiate/personalise what each user sees on the various pages of the app, based on the data stored in our database. E.g. different homepage UI for different user types, or individual users' unique results page containing content submitted for misinformation detection.

The login and logout system also ensures the security of the app for the users, ensuring only the specific user has access to their account and its functions. This is especially necessary for system admin accounts, which have the ability to suspend other accounts.

View profile is needed before updating profile is possible. The update profile function is needed so important user information can be kept up-to-date. Resetting passwords is an essential function in the case of forgotten passwords, or as a preventative measure when suspicious activity is observed.

For Free & Premium users:

- Register for an account
- Delete account
- Upload short-form text into the app for misinformation detection

- Upload a link to an article/webpage into the app for misinformation detection
- View their personal results page and individual results details
- View their remaining free credits
- Report results that are incorrect

In order to login and logout, the user must first register for an account, so this is a base requirement. Users will have the option to register as a free user or as a premium user on the spot.

Deleting an account is a common function for account-based apps, as users may decide they no longer want to use the app or receive notifications from it, so it is another base requirement.

In order to achieve a viable product, the app must be able to:

- Accept at least one type of short-form content
- Analyse the content for misinformation and generate an output result

As a base requirement, we plan to implement text-based detection for short-form text as well as links to articles/webpages. The results of users' submitted content will be saved so the user can view their results at any time on the results page. Free and premium users will be granted limited uses to the functions of the app.

Finally, users are able to report a result for providing incorrect information (e.g. misclassification of AI/deepfake images or identifying misinformation in a factually accurate article). This is a base requirement as it allows the operational team (specifically data engineers) to gather information on issues with the AI models easily.

For Free users:

- Upgrade to a premium account

For Premium users:

- Renew their premium subscription
- Cancel their premium subscription
- Upload images into the app into various functions
 - Images with captions
 - Review and correct OCR extracted text
 - Analyse caption-image context
 - Deepfake/AI generated image detector

As uiabo uses a freemium business model, an option for free users to upgrade to a premium user is a base requirement of the app. As uiabo's premium subscription is on a monthly basis, an option for premium users to renew or cancel their subscription is also necessary.

In order to differentiate between free and premium users, premium users will be given more uses of the functions within the app. They will also be able access the full range of functions, namely these four image-related functions:

Image with captions: Refers to analysing both the image and its caption to explain the content of the image and its caption.

Review and correct OCR extracted text: Refers to extracting the text present in an image and analysing it specifically for misinformation, disregarding the actual image.

Analyse caption-image context: Refers to analysing both the image and its caption but also the using the context to detect misinformation.

Deepfake/AI generated image detector: Refers to analysing an image to determine if it is deepfaked, modified using AI or generated by AI.

For System administrators:

- Query for an account
- View account details
- Suspend and un-suspend accounts

The base requirement of the system administrators is to be able to manage the accounts of users of the app, which requires the ability to view the account details as well as suspend/unsuspend an account when necessary. In a large user base, a query function is also necessary to allow system administrators to locate a specific account quickly.

For Data engineers:

- View data dashboard of the results of the AI models in the app
- View reported incorrect results

When it comes to AI based systems, there is always room for error when it comes to the results it generates. Therefore, it is a base requirement for the data engineers to be able to access an overview of the incorrect results in the form of a dashboard, as well as view the details of the individual reports.

Additional Requirements (nice-to-haves)

For Free & Premium users:

- Delete their results history
- Leave a review
- Share a result via link

For System administrators:

- View user reviews

The above features are not strictly necessary for the app to function, however they will provide benefits to the operational team as well as the individual users.

For individual users, the ability to delete their results history grants them even more control over their account, allowing them to de-clutter their results page by removing unwanted results. This also helps them to find specific results in their collection faster. Sharing results via link is the most efficient way to allow users to share information with others, and it also serves as free advertising for the app.

The individual users' ability to leave a review are common features that can be found in many apps. They are especially useful as they allow system administrators to detect flaws/issues with the functions of the app.

Stretch Goals

A function that could possibly be implemented in future versions is an API for the app, which allows other organisations' to integrate their platforms with uiabo, leveraging on the various AI models within it to perform misinformation detection.

Not only does that expand the reach of the app outside of our target group (elderly Singaporeans), but it also opens up partnership/business opportunities, where we would be given access to organisation-specific data, resulting in new AI models that are extremely robust in detecting misinformation in specific areas.

5.1.1 Use Case Diagrams and Descriptions

The following use-case diagrams, user stories and descriptions are reused from Section 3 of the approved User Requirement Specification.

3. System Features

This section describes the user-visible and operational behaviour of uiabo. Use-case identifiers UC-01 to UC-43 follow the agreed user-story list. Shared individual-user use cases retain a single identifier and apply to both free and premium users; premium-specific use cases are listed separately.

3.1 Use Case Diagram – Free User

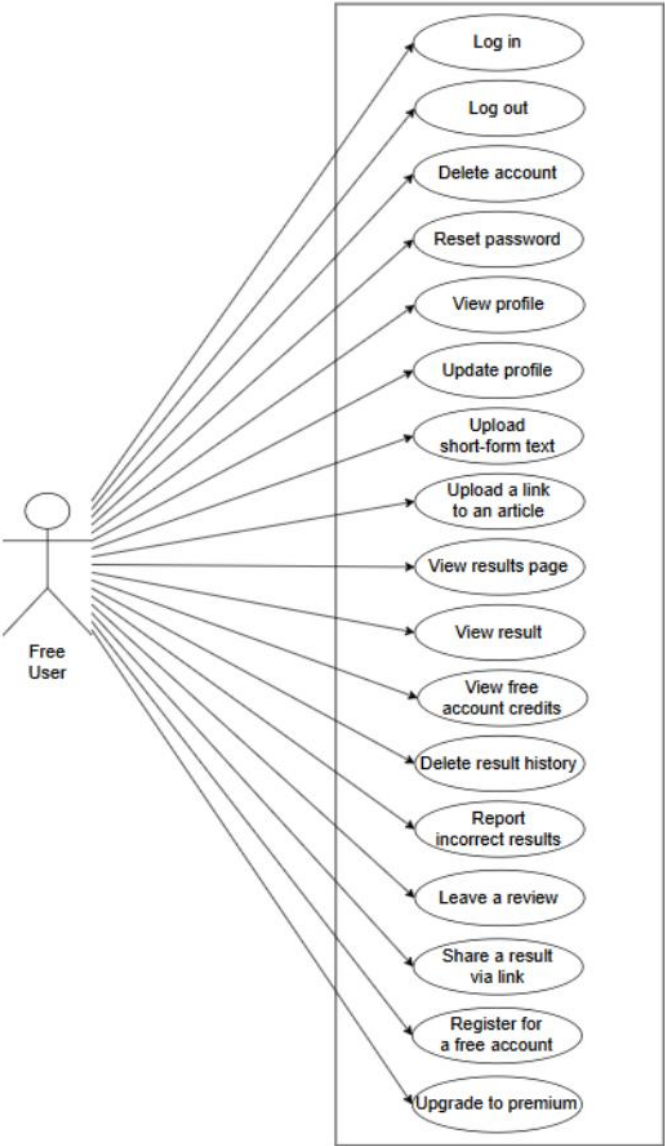


Figure 3.1. Free User Use Case Diagram

3.1.1 User Stories – Free User

- UC-01: As a free user, I want to log in to my account so that I can access the functions of the app.
- UC-02: As a free user, I want to log out of my account so that I can maintain the security of my account.
- UC-03: As a free user, I want to delete my account so that I can remove my personal data from the system.
- UC-04: As a free user, I want to reset my password so that I can improve account security or regain access when the password is forgotten.
- UC-05: As a free user, I want to view my profile so that I can check whether my account information is up to date.
- UC-06: As a free user, I want to update my profile so that I can keep my account information up to date.
- UC-07: As a free user, I want to upload short-form text so that I can find out whether the submitted text contains misinformation.
- UC-08: As a free user, I want to upload a link to an article so that I can find out whether a linked article or webpage contains misinformation.
- UC-09: As a free user, I want to view my personal results page so that I can access the results of my submitted content.
- UC-10: As a free user, I want to view the result of my submitted content so that I can view the details of a specific analysis result.
- UC-11: As a free user, I want to view my remaining account credits so that I can find out how many uses of the app I have left.
- UC-12: As a free user, I want to delete my result history so that I can remove results I no longer want to see from my results page.
- UC-13: As a free user, I want to report results that are incorrect so that I can provide data and feedback so the system can be improved in future.
- UC-14: As a free user, I want to leave a review so that I can express my experience using the app so it can be improved in the future.
- UC-15: As a free user, I want to share a result via link so that I can inform others of content containing misinformation.
- UC-16: As a free user, I want to register for a free account so that I can gain access to the free functions of the app.
- UC-17: As a free user, I want to upgrade to premium so that I can access all the functions of the app and gain more uses of the functions.

3.1.2 Use Case Description – Free User

Use Cases 01–15 apply to both free and premium users. Use Cases 16–17 apply only to free users.

Use Case 01 – User logs in to their account

Name: User logs in to their account	ID: 01
Stakeholders & Goals: Free user, Premium user – Access the functions of the app.	
Description: A registered free or premium user authenticates with their email address and password to access the personalised functions and data for their account.	
Actor(s): Free user, Premium user	
Trigger: The user selects the Log In button on the login screen.	
Pre-condition: The user has a registered, active account, is not already logged in, and has an internet connection.	
Main Flow: 1. The user opens uiabo and selects Log In. 2. The system displays the login form. 3. The user enters their registered email address and password. 4. The user selects Log In. 5. The system validates the submitted credentials using Firebase Authentication. 6. The system retrieves the user's role, access tier and account status. 7. The system creates the authenticated session and displays the user's personalised home screen.	
Sub-flow: S1. If the user has forgotten their password, they select Forgot Password and continue with Use Case 04.	
Alternative flow: 5a. If the email address or password is incorrect, the system does not create a session and displays a clear error message. 5b. If required information is missing, the system highlights the incomplete field and asks the user to correct it. 6a. If the account is suspended, the system denies access and informs the user that the account is unavailable. 7a. If authentication or account data cannot be retrieved because of a network or service error, the system keeps the user logged out and offers a retry option.	

Use Case 02 – User logs out of their account

Name: User logs out of their account	ID: 02
Stakeholders & Goals: Free user, Premium user – Maintain the security of their account.	
Description: An authenticated user ends the current uiabo session so that account functions and personal data are no longer accessible from the device without logging in again.	
Actor(s): Free user, Premium user	
Trigger: The user selects Log Out from the account or profile menu.	
Pre-condition: The user is logged in to an active uiabo session.	
Main Flow: 1. The user opens the account or profile menu. 2. The user selects Log Out. 3. The system asks the user to confirm the action. 4. The user confirms that they want to log out. 5. The system ends the authenticated session and removes locally stored session credentials. 6. The system returns the user to the login screen and protected account data is no longer displayed.	
Sub-flow: S1. After 30 minutes of inactivity, the system automatically expires the session, clears access to protected screens and returns the user to the login screen.	
Alternative flow: 4a. If the user cancels the confirmation, the system closes the prompt and keeps the session active. 5a. If the remote sign-out request fails, the system clears the local session, returns to the login screen and informs the user that the server could not be reached.	

Use Case 03 – User deletes their account

Name: User deletes their account	ID: 03
Stakeholders & Goals: Free user, Premium user – Remove their personal data from the system.	
Description: A free or premium user permanently deletes their uiabo account, associated profile information and saved result history. The action is irreversible.	
Actor(s): Free user, Premium user	
Trigger: The user selects Delete Account from the account or profile settings.	
Pre-condition: The user is logged in, can re-authenticate when requested, and has an internet connection.	
Main Flow: 1. The user opens the account or profile settings. 2. The user selects Delete Account. 3. The system explains that account data and saved result history will be permanently deleted and cannot be recovered. 4. The system asks the user to confirm the deletion and re-authenticate. 5. The user provides the required authentication and confirms Delete Permanently. 6. The system validates the user's identity. 7. The system deletes the user's account record, personal data and associated result history, then removes the authentication account. 8. The system ends the session, confirms deletion and returns to the welcome or login screen.	
Sub-flow: None.	
Alternative flow: 5a. If the user cancels, the system closes the confirmation and does not delete any data. 6a. If re-authentication fails, the system refuses the deletion and asks the user to try again or reset their password. 7a. If any required deletion operation fails, the system does not report the account as deleted, records the failure safely and offers a retry or support option. 8a. If the network connection is lost, the system informs the user that deletion could not be completed and keeps the account accessible until deletion succeeds.	

Use Case 04 – User resets their password

Name: User resets their password	ID: 04
Stakeholders & Goals: Free user, Premium user – Improve account security or regain access when the password is forgotten.	
Description: A free or premium user requests a secure password-reset link and replaces the existing password with a new valid password.	
Actor(s): Free user, Premium user	
Trigger: The user selects Forgot Password on the login screen or Reset Password in account settings.	
Pre-condition: The user has a registered email-and-password account, can access the registered email inbox, and has an internet connection.	
Main Flow: 1. The user selects Forgot Password or Reset Password. 2. The system displays the password-reset form. 3. The user enters their registered email address and submits the request. 4. The system validates the request and sends a time-limited reset link to the registered email address. 5. The system displays a neutral confirmation instructing the user to check their email. 6. The user opens the reset link and enters the new password twice. 7. The system checks that the reset link is valid and that the new password meets the password rules. 8. The system updates the password through Firebase Authentication and invalidates the reset link. 9. The system confirms the reset and directs the user to log in with the new password.	
Sub-flow: S1. A logged-in user may start the same reset process from account settings as a preventative security measure.	
Alternative flow: 3a. If the email field is empty or invalid, the system highlights the field and does not submit the request. 4a. To avoid revealing whether an account exists, the system displays the same neutral confirmation even when no matching account can be disclosed. 7a. If the reset link is invalid, expired or already used, the system rejects it and offers to send a new link. 7b. If the passwords do not match or the new password fails the password rules, the system explains the problem and asks the user to correct it. 8a. If the service or network is unavailable, the system does not change the password and offers a retry option.	

Use Case 05 – User views their profile

Name: User views their profile	ID: 05
Stakeholders & Goals: Free user, Premium user – Check whether their account information is up to date.	
Description: A logged-in free or premium user views the profile information stored for their own account.	
Actor(s): Free user, Premium user	
Trigger: The user selects Profile from the navigation or account menu.	
Pre-condition: The user is logged in to an active account.	
Main Flow: 1. The user opens the navigation or account menu. 2. The user selects Profile. 3. The system verifies the current session and retrieves only the authenticated user's profile information. 4. The system displays the profile information and the user's account tier in a readable format. 5. The user reviews the displayed information.	
Sub-flow: S1. The user may select Edit Profile and continue with Use Case 06.	
Alternative flow: 3a. If the session has expired, the system returns the user to the login screen and asks them to log in again. 3b. If the profile cannot be retrieved, the system displays an error without exposing another user's data and offers a retry option. 4a. If an optional profile field has no value, the system shows it as not provided rather than displaying incorrect information.	

Use Case 06 – User updates their profile

Name: User updates their profile	ID: 06
Stakeholders & Goals: Free user, Premium user – Keep their account information up to date.	
Description: A logged-in free or premium user changes editable personal information in their own profile.	
Actor(s): Free user, Premium user	
Trigger: The user selects Edit Profile while viewing their profile.	
Pre-condition: The user is logged in, has opened their profile, and the existing profile information is available.	
Main Flow: 1. The user views their profile and selects Edit Profile. 2. The system displays the editable profile fields with their current values. 3. The user changes one or more fields. 4. The user selects Save Changes. 5. The system validates the updated information. 6. The system saves the valid changes to the authenticated user's profile. 7. The system displays the updated profile and confirms that the changes were saved.	
Sub-flow: None.	
Alternative flow: 4a. If the user selects Cancel, the system discards unsaved changes and returns to the profile view. 5a. If required information is missing or has an invalid format, the system highlights the relevant field and does not save the changes. 5b. If a changed unique value is already used by another account, the system asks the user to provide a different value. 6a. If the update fails because of a network, authentication or database error, the system retains the previous stored profile and offers a retry option.	

Use Case 07 – User uploads short-form text

Name: User uploads short-form text	ID: 07
Stakeholders & Goals: Free user, Premium user – Find out whether the submitted text contains misinformation.	
Description: A free or premium user pastes English-language short-form text for automated claim identification, evidence retrieval and misinformation-risk analysis.	
Actor(s): Free user, Premium user	
Trigger: The user selects the short-form text analysis option and submits text.	
Pre-condition: The user is logged in, has an internet connection and has at least one successful submission remaining in the applicable allowance (one per day for free users or up to 60 per month for premium users).	
Main Flow: 1. The user selects the short-form text analysis option. 2. The system displays a text-entry screen and the user's remaining submission allowance. 3. The user pastes or enters the short-form text. 4. The user selects Analyse. 5. The system validates that the text is present and supported and confirms that an allowance remains. 6. The system creates the analysis request and shows clear progress information. 7. The system identifies the principal objectively checkable factual claim or claims. 8. The system searches recognised fact-checking and credible sources for supporting or contradicting evidence. 9. The system generates the concern label, misinformation-risk score, uncertainty information, plain-language explanation and evidence links. 10. The system saves the completed result to the user's result history and deducts one successful submission from the applicable allowance. 11. The system displays the completed result as described in Use Case 10.	
Sub-flow: S1. If no objectively checkable claim is found, the system creates a Not Enough Information result with a plain-language explanation rather than labelling opinion, humour or an unverifiable statement as misinformation.	
Alternative flow: 5a. If the text is empty, the system asks the user to enter text and does not start or deduct an analysis. 5b. If the text is unsupported or cannot be processed, the system explains the issue and allows the user to edit the submission. 5c. If no allowance remains, the system does not start the analysis, displays when access becomes available again and may offer the free user an upgrade option. 8a. If evidence services are temporarily unavailable, the system reports that the analysis could not be completed and offers a retry option. 9a. If processing exceeds the 20-minute timeout, the system cancels the request, informs the user and does not deduct a successful submission.	

Use Case 08 – User uploads a link to an article

Name: User uploads a link to an article	ID: 08
Stakeholders & Goals: Free user, Premium user – Find out whether a linked article or webpage contains misinformation.	
Description: A free or premium user submits a supported public webpage or article URL. uiabo screens the URL for malicious-link indicators, extracts available article content and analyses checkable claims.	
Actor(s): Free user, Premium user	
Trigger: The user selects the link analysis option and submits a webpage or article URL.	
Pre-condition: The user is logged in, has an internet connection and has at least one successful submission remaining in the applicable daily or monthly allowance. The target URL is intended to be public.	
Main Flow: 1. The user selects the link analysis option. 2. The system displays a URL-entry screen and the user's remaining submission allowance. 3. The user enters or pastes a public webpage or article URL. 4. The user selects Analyse. 5. The system validates the URL format, supported protocol and remaining allowance. 6. The system screens the submitted URL for malicious-link indicators before retrieving article content. 7. The system safely retrieves and extracts the available text and relevant source information from the public page. 8. The system identifies the principal objectively checkable factual claim or claims. 9. The system searches recognised fact-checking and credible sources for supporting or contradicting evidence. 10. The system generates the concern label, misinformation-risk score, uncertainty information, plain-language explanation, evidence links and available source context. 11. The system saves the completed result and deducts one successful submission from the applicable allowance. 12. The system displays the completed result as described in Use Case 10.	
Sub-flow: S1. If malicious-link indicators are detected, the system displays a prominent warning and does not open the submitted link automatically. Analysis continues only when content can be retrieved safely. S2. If no objectively checkable claim is found, the system creates a Not Enough Information result with an explanation.	
Alternative flow: 5a. If the URL is empty, malformed or uses an unsupported protocol, the system explains the required format and does not start or deduct an analysis. 5b. If no allowance remains, the system does not start the analysis, displays when access becomes available again and may offer the free user an upgrade option. 7a. If the page is private, inaccessible, unsupported, blocked or contains no extractable content, the system explains that it cannot analyse the page and does not deduct a successful submission. 9a. If evidence services are temporarily unavailable, the system reports that the analysis could not be completed and offers a retry option. 10a. If processing exceeds the 20-minute timeout, the system cancels the request, informs the user and does not deduct a successful submission.	

Use Case 09 – User views their personal results page

Name: User views their personal results page	ID: 09
Stakeholders & Goals: Free user, Premium user – Access the results of their submitted content.	
Description: A logged-in free or premium user views the saved analysis results associated with their own account.	
Actor(s): Free user, Premium user	
Trigger: The user selects Results or Result History from the app navigation.	
Pre-condition: The user is logged in to an active account.	
Main Flow: 1. The user opens the app navigation. 2. The user selects Results or Result History. 3. The system verifies the current session. 4. The system retrieves only the result records associated with the authenticated account. 5. The system displays the user's saved results in a readable list, including a useful summary such as submission type, date, concern label and analysis status. 6. The user reviews the list of results. 7. The user may select a completed result to continue with Use Case 10.	
Sub-flow: S1. If the account has no saved results, the system displays an empty-state message and a clear option to submit content for analysis. S2. If an analysis is still processing, the system displays its current status without presenting an unfinished assessment as final.	
Alternative flow: 3a. If the session has expired, the system returns the user to the login screen and asks them to log in again. 4a. If the results cannot be retrieved, the system displays an error, does not expose another user's results and offers a retry option. 5a. If an individual result was deleted, it is no longer displayed in the list.	

Use Case 10 – User views the result of their submitted content

Name: User views the result of their submitted content	ID: 10
Stakeholders & Goals: Free user, Premium user – View the details of a specific analysis result.	
Description: A free or premium user opens a completed result to review the system's assessment, reasoning, uncertainty and evidence for the submitted content.	
Actor(s): Free user, Premium user	
Trigger: The user selects a completed result from the personal results page, or the system completes a newly submitted analysis.	
Pre-condition: The user is logged in, the result belongs to the authenticated user, and the analysis has completed successfully.	
Main Flow: 1. The user selects a completed result, or a new analysis finishes. 2. The system verifies that the authenticated user is authorised to view the result. 3. The system retrieves the saved submission, assessment and evidence records. 4. The system displays the concern label: Low Concern, Needs Caution, High Concern or Not Enough Information. 5. The system displays the misinformation-risk score, uncertainty information, identified factual claim and a plain-language explanation. 6. The system displays supporting and contradicting evidence with source links or citations. 7. Where applicable to a premium image result, the system separately displays the AI-generation or manipulation likelihood score and its limitations. 8. The system displays an appropriate next-action recommendation and keeps the result available in the user's history.	
Sub-flow: S1. The user may select an evidence link to inspect the source independently; the external source opens without replacing the saved result. S2. If the result is Not Enough Information, the system explains that insufficient checkable evidence was found and does not present a definitive misinformation judgement. S3. The user may use the separate report, share, review or delete-result functions when those actions are available.	
Alternative flow: 2a. If the result belongs to another account, the system denies access and does not reveal the result data. 3a. If the result was deleted or cannot be found, the system informs the user and returns to the personal results page. 3b. If the result cannot be loaded because of a network or database error, the system offers a retry option. 6a. If an external evidence page is no longer available, the system keeps the saved assessment visible and marks the affected link as unavailable when detected.	

Use Case 11 – User views their free account credits

Name: User views their free account credits	ID: 11
Stakeholders & Goals: Free user, Premium user – Find out how many uses of the app they have left.	
Description: A logged-in free or premium user views the number of analysis submissions remaining in their applicable allowance and when that allowance next refreshes.	
Actor(s): Free user, Premium user	
Trigger: The user selects Credits from the account or profile menu, or views the allowance indicator shown on a submission screen.	
Pre-condition: The user is logged in to an active account.	
Main Flow: 1. The user opens the account or profile menu. 2. The user selects Credits. 3. The system verifies the current session. 4. The system retrieves the authenticated user's account tier and submission usage. 5. The system displays the remaining allowance, the applicable limit (one successful submission per day for free users or up to 60 successful submissions per month for premium users) and the date on which the allowance next refreshes. 6. The user reviews the displayed allowance.	
Sub-flow: S1. The system also displays the remaining allowance on the text and link submission screens.	
Alternative flow: 3a. If the session has expired, the system returns the user to the login screen and asks them to log in again. 4a. If the usage record cannot be retrieved, the system displays an error rather than an inaccurate balance and offers a retry option. 5a. If a submission is still processing, the system shows it as pending and does not deduct it from the allowance until the analysis completes successfully. 5b. If an analysis was cancelled or timed out, the system does not deduct it from the allowance and the displayed balance is unchanged.	

Use Case 12 – User deletes their result history

Name: User deletes their result history	ID: 12
Stakeholders & Goals: Free user, Premium user – Remove results they no longer want to see from their results page.	
Description: A logged-in free or premium user permanently deletes one or more saved analysis results from their own result history. The action is irreversible.	
Actor(s): Free user, Premium user	
Trigger: The user selects Delete on a saved result, or selects Delete Selected on the personal results page.	
Pre-condition: The user is logged in and has at least one saved result associated with the authenticated account.	
Main Flow: 1. The user opens their personal results page. 2. The user selects one or more saved results. 3. The user selects Delete. 4. The system explains that the selected results will be permanently deleted and cannot be recovered. 5. The user confirms the deletion. 6. The system verifies that the selected results belong to the authenticated account. 7. The system deletes the selected result records, including the stored submission, assessment and evidence records. 8. The system confirms the deletion and displays the updated results list.	
Sub-flow: S1. The user may delete a single result while viewing it, after which the system returns to the personal results page. S2. If the account has no saved results, the system displays an empty-state message and does not offer a delete option.	
Alternative flow: 5a. If the user cancels, the system closes the confirmation and does not delete any data. 6a. If a selected result does not belong to the authenticated account, the system denies the deletion and does not reveal the result data. 7a. If the deletion fails because of a network or database error, the system does not report the results as deleted and offers a retry option. 7b. Deleting a result does not restore a used submission to the user's allowance.	

Use Case 13 – User reports results that are incorrect

Name: User reports results that are incorrect	ID: 13
Stakeholders & Goals: Free user, Premium user – Provide data and feedback so the system can be improved in future.	
Description: A free or premium user flags a completed result they believe is wrong and submits a reason so that the result can be reviewed by the data engineer.	
Actor(s): Free user, Premium user	
Trigger: The user selects Report Incorrect Result while viewing a completed result.	
Pre-condition: The user is logged in, is viewing a completed result that belongs to the authenticated account, and has not already reported that result.	
Main Flow: 1. The user opens a completed result. 2. The user selects Report Incorrect Result. 3. The system displays the report form with a list of reasons and an optional comment field. 4. The user selects a reason and may enter additional comments. 5. The user submits the report. 6. The system validates the submitted report. 7. The system saves the report together with a reference to the result, the identified factual claim, the concern label and the misinformation-risk score that were produced. 8. The system confirms that the report has been submitted for review and marks the result as reported.	
Sub-flow: S1. The submitted report is made available to the data engineer for review. S2. Reporting a result does not change the displayed assessment, and the result remains visible in the user's history.	
Alternative flow: 4a. If no reason is selected, the system highlights the field and does not submit the report. 6a. If the user has already reported the result, the system informs the user and does not create a duplicate report. 7a. If the report cannot be saved because of a network or database error, the system does not confirm submission and offers a retry option. 8a. If the user deletes the result after reporting it, the system retains only the information needed for review and does not retain the deleted personal result record.	

Use Case 14 – User leaves a review

Name: User leaves a review	ID: 14
Stakeholders & Goals: Free user, Premium user – Express their experience using the app so it can be improved in the future.	
Description: A logged-in free or premium user submits a rating and written feedback about their overall experience with uiabo.	
Actor(s): Free user, Premium user	
Trigger: The user selects Leave a Review from the account menu or the feedback page.	
Pre-condition: The user is logged in to an active account.	
Main Flow: 1. The user opens the account menu or the feedback page. 2. The user selects Leave a Review. 3. The system displays the review form with a rating scale and a comment field. 4. The user selects a rating and may enter comments. 5. The user submits the review. 6. The system validates the submitted review. 7. The system saves the review against the authenticated account. 8. The system confirms that the review has been submitted.	
Sub-flow: None.	
Alternative flow: 4a. If no rating is selected, the system highlights the rating field and does not submit the review. 6a. If the comment exceeds the supported length, the system explains the limit and asks the user to shorten it. 7a. If the review cannot be saved because of a network or database error, the system does not confirm submission and offers a retry option.	

Use Case 15 – User shares a result via link

Name: User shares a result via link	ID: 15
Stakeholders & Goals: Free user, Premium user – Inform others of content containing misinformation.	
Description: A free or premium user generates a shareable link to a completed result so that other people can view the assessment without logging in to uiabo.	
Actor(s): Free user, Premium user	
Trigger: The user selects Share while viewing a completed result.	
Pre-condition: The user is logged in, and the result is completed and belongs to the authenticated account.	
Main Flow: 1. The user opens a completed result. 2. The user selects Share. 3. The system explains what the shared page will display and asks the user to confirm. 4. The user confirms. 5. The system generates a shareable link for the selected result. 6. The system displays the link together with a copy option. 7. The user copies the link and shares it outside the app.	
Sub-flow: S1. The shared page displays the assessment, explanation and evidence only, and does not display the user's profile information or result history.	
Alternative flow: 4a. If the user cancels, the system does not generate a link. 5a. If a share link already exists for the result, the system displays the existing link instead of generating a new one. 5b. If the link cannot be generated because of a network or database error, the system offers a retry option. 7a. If the result is later deleted, the shared link no longer resolves.	

Use Case 16 – User registers for a free account

Name: User registers for a free account	ID: 16
Stakeholders & Goals: Free user – Gain access to the free functions of the app.	
Description: A new user creates a uiabo free account using an email address and password, and verifies the email address before the account becomes active.	
Actor(s): Free user	
Trigger: The user selects Register on the welcome or login screen.	
Pre-condition: The user is not logged in, has an internet connection, and does not already have an account registered with the same email address.	
Main Flow: 1. The user opens uiabo and selects Register. 2. The system displays the registration form and the available account types. 3. The user selects the free account type. 4. The user enters their email address and password and confirms the password. 5. The system displays the terms of use and privacy notice, and the user gives consent. 6. The user submits the registration. 7. The system validates the submitted information and checks that the email address is not already registered. 8. The system creates the account through Firebase Authentication with the free access tier and an allowance of one successful submission per day, and sends a verification email. 9. The user opens the verification link in the registered email inbox. 10. The system activates the account, confirms registration and directs the user to log in.	
Sub-flow: None.	
Alternative flow: 7a. If required information is missing or the email format is invalid, the system highlights the relevant field and does not create an account. 7b. If the password does not meet the password rules or the two entries do not match, the system explains the requirement and asks the user to correct it. 7c. If the email address is already registered, the system informs the user and offers to log in or to reset the password. 8a. If the account cannot be created because of a network or service error, the system does not create a partial account and offers a retry option. 9a. If the verification link expires before it is used, the system rejects it and offers to send a new verification email.	

Use Case 17 – Free user upgrades to premium

Name: Free user upgrades to premium	ID: 17
Stakeholders & Goals: Free user – Access all the functions of the app and gain more uses of the functions.	
Description: A logged-in free user subscribes to the premium plan, after which the account tier and submission allowance are updated and the premium functions become available.	
Actor(s): Free user	
Trigger: The user selects Upgrade to Premium from the account menu, the subscription page or a prompt shown when the free allowance is exhausted.	
Pre-condition: The user is logged in as a free user, does not have an active premium subscription, and has an internet connection.	
Main Flow: 1. The free user selects Upgrade to Premium. 2. The system displays the premium plan, the price, the billing period and the premium allowance and functions. 3. The free user confirms the upgrade. 4. The system directs the free user to the payment provider. 5. The free user enters their payment details on the payment provider's page and authorises the payment. 6. The payment provider confirms that the payment was successful. 7. The system updates the account tier to premium, applies the premium allowance of up to 60 successful submissions per month and records the subscription expiry date. 8. The system confirms the upgrade and makes the premium functions available.	
Sub-flow: S1. Submissions already used on the free allowance are not carried over to the premium allowance.	
Alternative flow: 3a. If the free user cancels, the system returns to the previous screen and makes no changes. 6a. If the payment is declined or cancelled, the system does not upgrade the account and allows the free user to try again. 7a. If the payment succeeds but the account cannot be updated, the system does not report the upgrade as complete, records the failure for administrator review and informs the user that support has been notified. 8a. If the connection is lost after payment, the system applies the upgrade once the payment confirmation is received.	

3.2 Use Case Diagram – Premium User

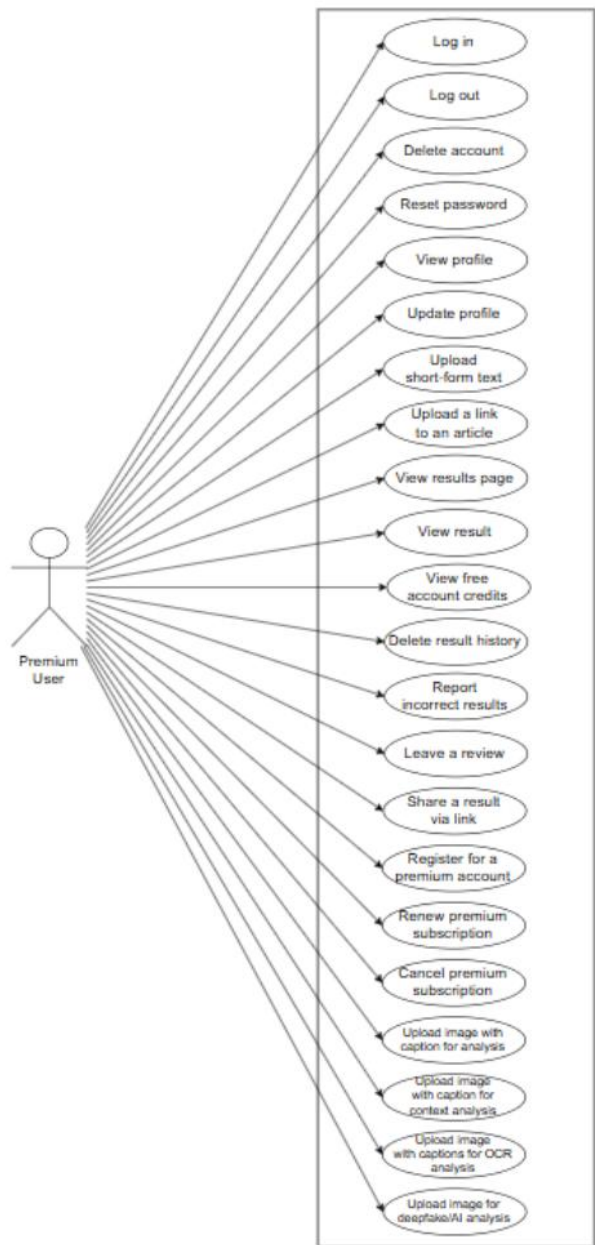


Figure 3.2. Premium User Use Case Diagram

3.2.1 User Stories – Premium User

- UC-01: As a premium user, I want to log in to my account so that I can access the functions of the app.
- UC-02: As a premium user, I want to log out of my account so that I can maintain the security of my account.
- UC-03: As a premium user, I want to delete my account so that I can remove my personal data from the system.
- UC-04: As a premium user, I want to reset my password so that I can improve account security or regain access when the password is forgotten.
- UC-05: As a premium user, I want to view my profile so that I can check whether my account information is up to date.
- UC-06: As a premium user, I want to update my profile so that I can keep my account information up to date.
- UC-07: As a premium user, I want to upload short-form text so that I can find out whether the submitted text contains misinformation.
- UC-08: As a premium user, I want to upload a link to an article so that I can find out whether a linked article or webpage contains misinformation.
- UC-09: As a premium user, I want to view my personal results page so that I can access the results of my submitted content.
- UC-10: As a premium user, I want to view the result of my submitted content so that I can view the details of a specific analysis result.
- UC-11: As a premium user, I want to view my remaining account credits so that I can find out how many uses of the app I have left.
- UC-12: As a premium user, I want to delete my result history so that I can remove results I no longer want to see from my results page.
- UC-13: As a premium user, I want to report results that are incorrect so that I can provide data and feedback so the system can be improved in future.
- UC-14: As a premium user, I want to leave a review so that I can express my experience using the app so it can be improved in the future.
- UC-15: As a premium user, I want to share a result via link so that I can inform others of content containing misinformation.
- UC-18: As a premium user, I want to register for a premium account so that I can gain access to all the functions of the app.
- UC-19: As a premium user, I want to renew my premium subscription so that I can prevent loss of access to the premium functions of the app.
- UC-20: As a premium user, I want to cancel my premium subscription so that I can stop paying for the premium subscription.
- UC-21: As a premium user, I want to upload an image with a caption for analysis so that I can understand the submitted image and caption and assess potentially misleading content.
- UC-22: As a premium user, I want to upload an image with a caption for context analysis so that I can find out whether the image and caption are contextually consistent.
- UC-23: As a premium user, I want to upload an image for OCR analysis so that I can review visible text extracted from an image and assess it for misinformation.
- UC-24: As a premium user, I want to upload an image for deepfake/AI analysis so that I can assess whether an image may have been generated or manipulated using AI.

3.2.2 Use Case Description – Premium User

Shared Use Cases 01–15 are defined in Section 3.1.2 and apply equally to premium users. The following descriptions define premium registration, subscription and image-analysis behaviour (Use Cases 18–24).

Use Case 18 – User registers for a premium account

Name: User registers for a premium account	ID: 18
Stakeholders & Goals: Premium user – Gain access to all the functions of the app.	
Description: A new user creates a uiabo account and subscribes to the premium plan within a single registration process.	
Actor(s): Premium user	
Trigger: The user selects Register on the welcome or login screen and selects the premium account type.	
Pre-condition: The user is not logged in, has an internet connection, and does not already have an account registered with the same email address.	
Main Flow: 1. The user opens uiabo and selects Register. 2. The system displays the registration form and the available account types. 3. The user selects the premium account type. 4. The user enters their email address and password and confirms the password. 5. The system displays the terms of use and privacy notice, and the user gives consent. 6. The user submits the registration. 7. The system validates the submitted information and checks that the email address is not already registered. 8. The system directs the user to the payment provider. 9. The user enters their payment details on the payment provider's page and authorises the payment. 10. The payment provider confirms that the payment was successful. 11. The system creates the account through Firebase Authentication with the premium access tier, applies the premium allowance of up to 60 successful submissions per month and sends a verification email. 12. The user opens the verification link in the registered email inbox. 13. The system activates the account, confirms registration and directs the user to log in.	
Sub-flow: None.	
Alternative flow: 7a. If required information is missing or invalid, the system highlights the relevant field and does not proceed to payment. 7b. If the email address is already registered, the system informs the user and offers to log in or to reset the password. 10a. If the payment is declined or cancelled, the system does not create an account and offers the user the option to register for a free account instead. 11a. If the payment succeeds but the account cannot be created, the system records the failure for administrator review and informs the user that support has been notified. 12a. If the verification link expires before it is used, the system rejects it and offers to send a new verification email.	

Use Case 19 – Premium user renews their premium subscription

Name: Premium user renews their premium subscription	ID: 19
Stakeholders & Goals: Premium user – Prevent loss of access to the premium functions of the app.	
Description: A premium user extends an expiring or expired subscription for a further billing period so that premium access and the premium allowance continue.	
Actor(s): Premium user	
Trigger: The premium user selects Renew Subscription on the subscription page, or acts on the renewal reminder sent by the system.	
Pre-condition: The user is logged in as a premium user, has a subscription that is due to expire or has expired, and has an internet connection.	
Main Flow: 1. The system notifies the premium user before the subscription expiry date. 2. The premium user opens the subscription page. 3. The premium user selects Renew Subscription. 4. The system displays the renewal price, the billing period and the new expiry date. 5. The premium user confirms the renewal. 6. The system directs the premium user to the payment provider. 7. The premium user enters their payment details on the payment provider's page and authorises the payment. 8. The payment provider confirms that the payment was successful. 9. The system extends the subscription, updates the expiry date and refreshes the premium allowance for the new period. 10. The system confirms the renewal and displays the new expiry date.	
Sub-flow: S1. If automatic renewal is enabled, the system attempts the renewal payment on the expiry date without further action from the premium user.	
Alternative flow: 5a. If the premium user cancels, the system makes no changes and the existing expiry date remains. 8a. If the payment is declined or cancelled, the system does not extend the subscription and informs the premium user when premium access will end. 9a. If the subscription has already been renewed, the system informs the premium user and does not take a further payment. 10a. If the subscription expired before it was renewed, the system restores premium access and applies the premium allowance from the renewal date.	

Use Case 20 – Premium user cancels their premium subscription

Name: Premium user cancels their premium subscription	ID: 20
Stakeholders & Goals: Premium user – Stop paying for the premium subscription.	
Description: A premium user cancels the automatic renewal of the premium subscription. Premium access continues until the current expiry date, after which the account reverts to the free tier.	
Actor(s): Premium user	
Trigger: The premium user selects Cancel Subscription on the subscription page.	
Pre-condition: The user is logged in as a premium user and has an active premium subscription.	
Main Flow: 1. The premium user opens the subscription page. 2. The premium user selects Cancel Subscription. 3. The system explains that premium access continues until the current expiry date and that the account will revert to the free tier afterwards. 4. The premium user confirms the cancellation. 5. The system cancels the automatic renewal and updates the subscription status to cancelled. 6. The system confirms the cancellation and displays the date on which premium access will end. 7. The system sends a confirmation to the registered email address.	
Sub-flow: S1. On the expiry date the system reverts the account to the free tier and applies the free allowance of one successful submission per day. S2. Saved results remain available to the user after the account reverts to the free tier.	
Alternative flow: 4a. If the premium user cancels the confirmation, the subscription remains unchanged. 5a. If the cancellation cannot be saved because of a network or database error, the system does not confirm the cancellation and offers a retry option. 6a. Cancelling the subscription does not delete the account. The user deletes the account through the separate account deletion function.	

Use Case 21 – Premium user uploads image with captions for analysis

Name: Premium user uploads image with captions for analysis	ID: 21
Stakeholders & Goals: Premium user – Understand the submitted image and caption and assess potentially misleading content.	
Description: A premium user uploads a supported static image with an accompanying English caption for combined content analysis and evidence-backed result generation.	
Actor(s): Premium user	
Trigger: The premium user selects Image and Caption Analysis from the submission screen.	
Pre-condition: The user is logged in with active premium access, has a remaining monthly submission allowance, has an internet connection and has a supported image they are permitted to process.	
Main Flow: 1. The user selects Image and Caption Analysis. 2. The system verifies premium entitlement and displays the upload form and remaining monthly allowance. 3. The user selects a supported static image. 4. The system displays a preview and a field for an accompanying caption. 5. The user enters or edits the English caption and selects Submit. 6. The system validates the image, caption and allowance, then uploads the submission securely. 7. The back end analyses the image and caption, identifies checkable content and retrieves relevant evidence. 8. When processing succeeds, the system records the result, deducts one monthly submission and displays the result page with explanation, evidence, uncertainty and next-action guidance.	
Sub-flow: S1. The user may replace the selected image or edit the caption before submitting. S2. From the preview, the user may choose the specialised context, OCR or image-authenticity function and continue with Use Case 22, 23 or 24.	
Alternative flow: 2a. If premium access is inactive, the system blocks the function and shows subscription guidance. 2b. If no monthly allowance remains, the system displays the next refresh information and does not accept the submission. 6a. If the image is missing, unsupported, corrupted or exceeds the permitted size, the system explains the problem and requests another image. 6b. If the caption is missing or cannot be processed as English text, the system asks the user to correct it. 7a. If processing fails, is cancelled or exceeds 20 minutes, the system records no successful use, preserves the allowance and offers a safe retry. 8a. If evidence is insufficient, the result uses Not Enough Information and explains the limitation.	

Use Case 22 – Premium user uploads image with captions for context analysis

Name: Premium user uploads image with captions for context analysis	ID: 22
Stakeholders & Goals: Premium user – Find out whether the image and caption are contextually consistent.	
Description: A premium user submits a static image and caption so the system can compare contextual cues such as date, location, event and description and identify a possible mismatch.	
Actor(s): Premium user	
Trigger: The premium user selects Caption-Image Context Analysis.	
Pre-condition: The user is logged in with active premium access, has a remaining monthly allowance, has an internet connection and has a supported image and English caption.	
Main Flow: 1. The user selects Caption-Image Context Analysis. 2. The system verifies premium entitlement and displays the upload form and allowance. 3. The user selects an image, enters or edits its caption and reviews the preview. 4. The user selects Analyse Context. 5. The system validates and securely uploads the submission. 6. The back end identifies contextual cues in the image and caption and searches for relevant corroborating evidence. 7. The system compares the available cues and generates a context-assessment result with uncertainty and limitations. 8. On success, the system records the result, deducts one monthly submission and displays the explanation, evidence links and recommended next action.	
Sub-flow: S1. If useful visible text is detected, the user may review OCR-extracted text before analysis by continuing with Use Case 23.	
Alternative flow: 2a. If premium access is inactive or no allowance remains, the system does not accept the submission and explains the available action. 5a. If an input is missing, unsupported or invalid, the system highlights the issue and does not start processing. 6a. If no reliable contextual evidence can be found, the system returns Not Enough Information rather than asserting a mismatch. 7a. If analysis fails, is cancelled or times out, the allowance is not deducted and the system offers a retry option.	

Use Case 23 – Premium user uploads image with captions for OCR analysis

Name: Premium user uploads image with captions for OCR analysis	ID: 23
Stakeholders & Goals: Premium user – Review visible text extracted from an image and assess it for misinformation.	
Description: A premium user uploads an image containing visible text, reviews and corrects the OCR output, and submits the corrected text for evidence-backed misinformation analysis.	
Actor(s): Premium user	
Trigger: The premium user selects OCR Image Analysis.	
Pre-condition: The user is logged in with active premium access, has a remaining monthly allowance, has an internet connection and has a supported image containing legible text.	
Main Flow: 1. The user selects OCR Image Analysis. 2. The system verifies premium entitlement and displays the image-upload form and allowance. 3. The user selects an image and chooses Extract Text. 4. The system validates the image, uploads it securely and performs OCR. 5. The system displays the extracted text in an editable field beside the image preview. 6. The user checks the extracted text, corrects any errors and selects Analyse Text. 7. The system validates the corrected text, identifies checkable claims and retrieves relevant evidence. 8. On successful analysis, the system records the result, deducts one monthly submission and displays the concern label, risk, uncertainty, explanation and citations.	
Sub-flow: S1. The user may re-run OCR after replacing or rotating the image before submitting the corrected text.	
Alternative flow: 2a. If premium access is inactive or no allowance remains, the system blocks the analysis and explains why. 4a. If the image is unsupported, corrupted or too unclear for usable OCR, the system requests a clearer supported image and does not deduct an allowance. 6a. If the corrected text is empty or contains no analysable content, the system requests a correction and does not start misinformation analysis. 7a. If no objectively checkable claim is identified, the system explains this and returns Not Enough Information where appropriate. 8a. If processing fails, is cancelled or times out, the system preserves the allowance and offers a retry option.	

Use Case 24 – Premium user uploads image for deepfake/AI analysis

Name: Premium user uploads image for deepfake/AI analysis	ID: 24
Stakeholders & Goals: Premium user – Assess whether an image may have been generated or manipulated using AI.	
Description: A premium user submits a supported static image to receive a probabilistic AI-generation or manipulation likelihood score and a plain-language media-forensics explanation.	
Actor(s): Premium user	
Trigger: The premium user selects Deepfake/AI Image Analysis.	
Pre-condition: The user is logged in with active premium access, has a remaining monthly allowance, has an internet connection and has a supported image they are permitted to process.	
Main Flow: 1. The user selects Deepfake/AI Image Analysis. 2. The system verifies premium entitlement and displays the upload form, limitations notice and remaining allowance. 3. The user selects an image and reviews the preview. 4. The user acknowledges that the output is a likelihood assessment rather than proof and selects Analyse Image. 5. The system validates and securely uploads the image. 6. The back end runs the supported image-authenticity detector and gathers the available signals. 7. The system converts the detector output into a likelihood score, uncertainty statement and plain-language explanation of the available signals and limitations, including suspicious-region indicators only when supported. 8. On success, the system records the result, deducts one monthly submission and displays the result and safe next-action guidance.	
Sub-flow: S1. The user may add an optional caption to provide context, but the image-authenticity score remains distinct from the misinformation-risk assessment.	
Alternative flow: 2a. If premium access is inactive or no allowance remains, the system blocks the function and explains the available action. 5a. If the image is unsupported, corrupted or too low-quality for the detector, the system asks for another image and does not deduct an allowance. 6a. If the detector is unavailable or cannot produce a reliable output, the system returns Not Enough Information rather than a conclusive label. 8a. If processing fails, is cancelled or exceeds 20 minutes, the system preserves the allowance and offers a retry option.	

3.3 Use Case Diagram – System Administrator

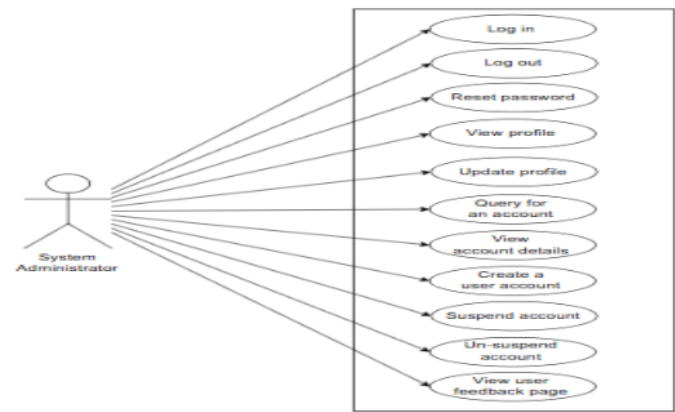


Figure 3.3. System Administrator Use Case Diagram

3.3.1 User Stories – System Administrator

- UC-25: As a system administrator, I want to log in to my account so that I can access the administrative functions of the app.
- UC-26: As a system administrator, I want to log out of my account so that I can maintain the security of my account.
- UC-27: As a system administrator, I want to reset my password so that I can improve account security or regain access when the password is forgotten.
- UC-28: As a system administrator, I want to view my profile so that I can check whether my information is up to date.
- UC-29: As a system administrator, I want to update my profile so that I can keep my information up to date.
- UC-30: As a system administrator, I want to query for an account so that I can find a specific account quickly.
- UC-31: As a system administrator, I want to view account details so that I can verify the account's information and access status.
- UC-32: As a system administrator, I want to create a user account so that I can create an account for a new system administrator or data engineer.
- UC-33: As a system administrator, I want to suspend an account so that I can revoke a specific account's access without deleting its record.
- UC-34: As a system administrator, I want to un-suspend an account so that I can restore a specific suspended account's access.
- UC-35: As a system administrator, I want to view the user feedback page so that I can review feedback and reviews provided by users.

3.3.2 Use Case Description – System Administrator

Use Case 25 – System administrator logs in to their account

Name: System administrator logs in to their account	ID: 25
Stakeholders & Goals: System administrator – Access the administrative functions of the app.	
Description: An authorised system administrator authenticates to access role-restricted uiabo functions and information.	
Actor(s): System administrator	
Trigger: The system administrator selects Log In on the login screen.	
Pre-condition: The system administrator has an active account created by an authorised system administrator, is not already logged in and has an internet connection.	
Main Flow: 1. The system administrator opens uiabo and selects Log In. 2. The system displays the login form. 3. The system administrator enters their registered email address and password. 4. The system administrator selects Log In. 5. The system validates the credentials through Firebase Authentication. 6. The system verifies that the account is active and has the system administrator role. 7. The system creates the authenticated session and displays the administrative home screen.	
Sub-flow: S1. If the system administrator has forgotten the password, they select Forgot Password and continue with Use Case 27.	
Alternative flow: 5a. If the credentials are invalid, the system denies access and displays a clear error message. 6a. If the account is suspended or has the wrong role, the system denies access without exposing protected information. 7a. If a network or service error occurs, the system keeps the user logged out and offers a retry option.	

Use Case 26 – System administrator logs out of their account

Name: System administrator logs out of their account	ID: 26
Stakeholders & Goals: System administrator – Maintain the security of their account.	
Description: An authenticated system administrator ends the current session so that protected operational functions are no longer accessible from the device.	
Actor(s): System administrator	
Trigger: The system administrator selects Log Out from the account or profile menu.	
Pre-condition: The system administrator is logged in to an active session.	
Main Flow: 1. The system administrator opens the account or profile menu. 2. The user selects Log Out. 3. The system asks for confirmation. 4. The user confirms the action. 5. The system ends the authenticated session and clears locally stored session credentials. 6. The system returns to the login screen and no protected operational information remains visible.	
Sub-flow: S1. After 30 minutes of inactivity, the system automatically expires the session and returns to the login screen.	
Alternative flow: 4a. If the user cancels, the system keeps the session active. 5a. If the remote sign-out request fails, the system clears the local session, returns to the login screen and reports that the service could not be reached.	

Use Case 27 – System administrator resets their password

Name: System administrator resets their password	ID: 27
Stakeholders & Goals: System administrator – Improve account security or regain access when the password is forgotten.	
Description: The system administrator requests a secure password-reset link and replaces the existing password with a valid new password.	
Actor(s): System administrator	
Trigger: The system administrator selects Forgot Password on the login screen or Reset Password in profile settings.	
Pre-condition: The system administrator has a registered email-and-password account, can access the registered inbox and has an internet connection.	
Main Flow: 1. The system administrator selects Forgot Password or Reset Password. 2. The system displays the password-reset form. 3. The user enters the registered email address and submits the request. 4. The system sends a time-limited password-reset link and displays a neutral confirmation. 5. The user opens the link and enters the new password twice. 6. The system validates the link, password match and password rules. 7. Firebase Authentication updates the password and invalidates the link. 8. The system confirms the change and directs the user to log in again.	
Sub-flow: S1. A logged-in user may start the same process from profile settings as a preventative security measure.	
Alternative flow: 3a. If the email format is invalid, the system highlights the field and does not submit the request. 4a. The system uses the same neutral confirmation whether or not an account can be disclosed. 6a. If the link is invalid, expired or used, the system rejects it and offers to send another link. 6b. If the passwords do not match or fail the password rules, the system explains the problem and requests a correction. 7a. If the network or authentication service is unavailable, the password remains unchanged and the system offers a retry option.	

Use Case 28 – System administrator views their profile

Name: System administrator views their profile	ID: 28
Stakeholders & Goals: System administrator – Check whether their information is up to date.	
Description: The system administrator views the profile information stored for their own operational account.	
Actor(s): System administrator	
Trigger: The system administrator selects Profile from the navigation or account menu.	
Pre-condition: The system administrator is logged in to an active account.	
Main Flow: 1. The user selects Profile. 2. The system verifies the current session and role. 3. The system retrieves only the authenticated user's profile information. 4. The system displays the profile information and operational role in a readable format. 5. The user reviews the displayed information.	
Sub-flow: S1. The user may select Edit Profile and continue with Use Case 29.	
Alternative flow: 2a. If the session has expired, the system returns to the login screen. 3a. If the profile cannot be retrieved, the system displays an error without exposing another account's information and offers a retry option.	

Use Case 29 – System administrator updates their profile

Name: System administrator updates their profile	ID: 29
Stakeholders & Goals: System administrator – Keep their information up to date.	
Description: The system administrator changes editable information in their own profile while protected role and access-control fields remain read-only.	
Actor(s): System administrator	
Trigger: The system administrator selects Edit Profile while viewing their profile.	
Pre-condition: The system administrator is logged in and the existing profile information is available.	
Main Flow: 1. The user views their profile and selects Edit Profile. 2. The system displays editable fields with their current values; the role is read-only. 3. The user changes one or more editable fields and selects Save Changes. 4. The system validates the information. 5. The system saves the valid changes to the authenticated user's profile. 6. The system displays the updated profile and confirms that the changes were saved.	
Sub-flow: None.	
Alternative flow: 3a. If the user cancels, the system discards unsaved changes. 4a. If required information is missing or invalid, the system highlights the relevant field and does not save the changes. 5a. If an authentication, network or database error occurs, the previous profile remains unchanged and the system offers a retry option.	

Use Case 30 – System administrator queries for an account

Name: System administrator queries for an account	ID: 30
Stakeholders & Goals: System administrator – Find a specific account quickly.	
Description: A system administrator searches authorised account records using one or more supported identifiers or filters.	
Actor(s): System administrator	
Trigger: The system administrator selects Account Management or uses the account search field.	
Pre-condition: The system administrator is logged in with an active administrative account.	
Main Flow: 1. The administrator opens Account Management. 2. The system displays the account search and permitted filters. 3. The administrator enters a supported query such as account email, identifier, role, tier or status. 4. The administrator selects Search. 5. The system validates the query and applies role-based access controls. 6. The system displays matching accounts with only the summary fields required to distinguish them. 7. The system records the administrative query in the audit log.	
Sub-flow: S1. The administrator may refine, sort or clear the query and search again. S2. The administrator may select a result and continue with Use Case 31.	
Alternative flow: 3a. If the query is empty, malformed or too broad, the system requests a more specific valid value. 6a. If no account matches, the system displays No Results without suggesting protected account information. 6b. If the database cannot be reached, the system displays an error and offers a retry option.	

Use Case 31 – System administrator views account details

Name: System administrator views account details	ID: 31
Stakeholders & Goals: System administrator – Verify the account's information and access status.	
Description: A system administrator views the authorised details, role, tier and status of a selected account before performing an account-management action.	
Actor(s): System administrator	
Trigger: The system administrator selects an account from the query results.	
Pre-condition: The administrator is logged in and the selected account is within the administrator's permitted scope.	
Main Flow: 1. The administrator queries for an account using Use Case 30. 2. The administrator selects a matching account. 3. The system verifies the session, administrative role and access scope. 4. The system retrieves the account's permitted profile summary, role, access tier, status and relevant timestamps. 5. The system displays the details while masking or excluding sensitive fields that are not required for administration. 6. The system records the view in the audit log.	
Sub-flow: S1. From the account-details screen, the administrator may start Use Case 33 or 34 when the selected account is eligible.	
Alternative flow: 3a. If the session has expired or access is not permitted, the system denies the request and displays no account details. 4a. If the account no longer exists or cannot be retrieved, the system displays an error and returns to the search results.	

Use Case 32 – System administrator creates a user account

Name: System administrator creates a user account	ID: 32
Stakeholders & Goals: System administrator – Create an account for a new system administrator or data engineer.	
Description: An authorised system administrator creates an operational account with the appropriate system-administrator or data-engineer role.	
Actor(s): System administrator	
Trigger: The system administrator selects Create Operational Account.	
Pre-condition: The administrator is logged in, has permission to create operational accounts and has the new user's required information.	
Main Flow: 1. The administrator opens Account Management and selects Create Operational Account. 2. The system displays the authorised account fields and role choices. 3. The administrator enters the required information and selects System Administrator or Data Engineer. 4. The administrator reviews the details and confirms creation. 5. The system validates the data, the administrator's permission and the uniqueness of the account identifier. 6. The system creates the authentication account and associated role/profile record in a consistent transaction or recoverable workflow. 7. The system sends the new user a secure account-activation or password-setup instruction. 8. The system confirms creation and records the action in the audit log.	
Sub-flow: None.	
Alternative flow: 4a. If the administrator cancels, the system creates no account. 5a. If required data is missing, invalid or already registered, the system highlights the problem and does not create the account. 5b. If the selected role is outside the administrator's authority, the system denies the request. 6a. If any required operation fails, the system does not report success, safely rolls back or flags incomplete creation and offers a retry or support path.	

Use Case 33 – System administrator suspends account

Name: System administrator suspends account	ID: 33
Stakeholders & Goals: System administrator – Revoke a specific account's access without deleting its record.	
Description: A system administrator changes an eligible active account to suspended status so that it cannot authenticate or use protected functions while its record remains available for authorised administration.	
Actor(s): System administrator	
Trigger: The system administrator selects Suspend Account on an eligible account-details screen.	
Pre-condition: The administrator is logged in, has suspension permission and has selected an active account that is eligible for suspension.	
Main Flow: 1. The administrator views the selected account using Use Case 31. 2. The administrator selects Suspend Account. 3. The system explains the effect, requests a reason and asks for confirmation. 4. The administrator enters the reason and confirms. 5. The system rechecks authority and safeguards against suspending the current session or removing the last active system administrator. 6. The system changes the account status to Suspended and revokes active sessions where supported. 7. The system confirms the action and records the administrator, account, reason and timestamp in the audit log.	
Sub-flow: None.	
Alternative flow: 4a. If the administrator cancels or provides no required reason, the account remains active. 5a. If the target is the current administrator or the last active system administrator, the system blocks the action and explains the safeguard. 6a. If the status update fails, the system leaves the account's previous status unchanged and offers a retry option.	

Use Case 34 – System administrator un-suspends account

Name: System administrator un-suspends account	ID: 34
Stakeholders & Goals: System administrator – Restore a specific suspended account's access.	
Description: A system administrator restores an eligible suspended account to active status without creating a new account record.	
Actor(s): System administrator	
Trigger: The system administrator selects Unsuspend Account on a suspended account-details screen.	
Pre-condition: The administrator is logged in, has unsuspension permission and has selected a suspended account.	
Main Flow: 1. The administrator views the suspended account using Use Case 31. 2. The administrator selects Unsuspend Account. 3. The system displays the current suspension information and asks for confirmation. 4. The administrator confirms the action. 5. The system verifies the administrator's authority and the target account's current status. 6. The system changes the account status to Active. 7. The system confirms the action and records it in the audit log.	
Sub-flow: None.	
Alternative flow: 4a. If the administrator cancels, the account remains suspended. 5a. If the account is already active or no longer exists, the system does not apply a duplicate update and refreshes the details. 6a. If the update fails, the account remains suspended and the system offers a retry option.	

Use Case 35 – System administrator views user feedback page

Name: System administrator views user feedback page	ID: 35
Stakeholders & Goals: System administrator – Review feedback and reviews provided by users.	
Description: A system administrator views and filters user reviews, feedback and reported service problems so recurring issues can be identified and followed up.	
Actor(s): System administrator	
Trigger: The system administrator selects User Feedback from the administrative navigation.	
Pre-condition: The administrator is logged in with permission to review feedback.	
Main Flow: 1. The administrator selects User Feedback. 2. The system verifies the session and administrative role. 3. The system retrieves the feedback records within the administrator's permitted scope. 4. The system displays a summary and a list containing the feedback category, status, submission date and only the user information required for follow-up. 5. The administrator filters, sorts or selects an item. 6. The system displays the selected feedback details and records the administrative access.	
Sub-flow: S1. A reported incorrect analysis may link to the data-engineering workflow in Use Cases 41 and 42 while access remains role-controlled.	
Alternative flow: 3a. If no feedback is available, the system displays an empty state. 3b. If feedback cannot be retrieved, the system displays an error and offers a retry option. 6a. If the item was removed or access is no longer permitted, the system displays no details and refreshes the list.	

3.4 Use Case Diagram – Data Engineer

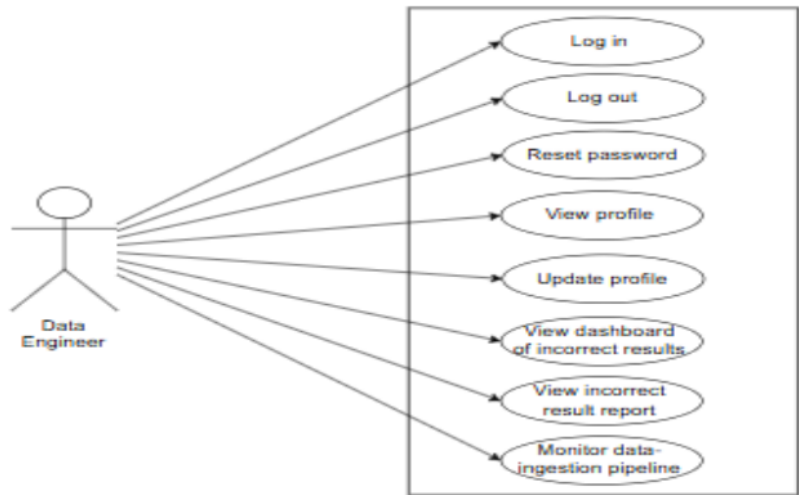


Figure 3.4. Data Engineer Use Case Diagram

3.4.1 User Stories – Data Engineer

- UC-36: As a data engineer, I want to log in to my account so that I can access the data-engineering functions of the app.
- UC-37: As a data engineer, I want to log out of my account so that I can maintain the security of my account.
- UC-38: As a data engineer, I want to reset my password so that I can improve account security or regain access when the password is forgotten.
- UC-39: As a data engineer, I want to view my profile so that I can check whether my information is up to date.
- UC-40: As a data engineer, I want to update my profile so that I can keep my information up to date.
- UC-41: As a data engineer, I want to view the dashboard of incorrect results so that I can analyse reported incorrect results and identify possible flaws in the AI models or data workflow.
- UC-42: As a data engineer, I want to view incorrect-result reports so that I can view the details of a reported incorrect result for investigation.
- UC-43: As a data engineer, I want to monitor data-ingestion pipelines so that I can identify failed, delayed or incomplete data updates.

3.4.2 Use Case Description – Data Engineer

Use Case 36 – Data engineer logs in to their account

Name: Data engineer logs in to their account	ID: 36
Stakeholders & Goals: Data engineer – Access the data-engineering functions of the app.	
Description: An authorised data engineer authenticates to access role-restricted uiabo functions and information.	
Actor(s): Data engineer	
Trigger: The data engineer selects Log In on the login screen.	
Pre-condition: The data engineer has an active account created by an authorised system administrator, is not already logged in and has an internet connection.	
Main Flow: 1. The data engineer opens uiabo and selects Log In. 2. The system displays the login form. 3. The data engineer enters their registered email address and password. 4. The data engineer selects Log In. 5. The system validates the credentials through Firebase Authentication. 6. The system verifies that the account is active and has the data engineer role. 7. The system creates the authenticated session and displays the data-engineering dashboard.	
Sub-flow: S1. If the data engineer has forgotten the password, they select Forgot Password and continue with Use Case 38.	
Alternative flow: 5a. If the credentials are invalid, the system denies access and displays a clear error message. 6a. If the account is suspended or has the wrong role, the system denies access without exposing protected information. 7a. If a network or service error occurs, the system keeps the user logged out and offers a retry option.	

Use Case 37 – Data engineer logs out of their account

Name: Data engineer logs out of their account	ID: 37
Stakeholders & Goals: Data engineer – Maintain the security of their account.	
Description: An authenticated data engineer ends the current session so that protected operational functions are no longer accessible from the device.	
Actor(s): Data engineer	
Trigger: The data engineer selects Log Out from the account or profile menu.	
Pre-condition: The data engineer is logged in to an active session.	
Main Flow: 1. The data engineer opens the account or profile menu. 2. The user selects Log Out. 3. The system asks for confirmation. 4. The user confirms the action. 5. The system ends the authenticated session and clears locally stored session credentials. 6. The system returns to the login screen and no protected operational information remains visible.	
Sub-flow: S1. After 30 minutes of inactivity, the system automatically expires the session and returns to the login screen.	
Alternative flow: 4a. If the user cancels, the system keeps the session active. 5a. If the remote sign-out request fails, the system clears the local session, returns to the login screen and reports that the service could not be reached.	

Use Case 38 – Data engineer resets their password

Name: Data engineer resets their password	ID: 38
Stakeholders & Goals: Data engineer – Improve account security or regain access when the password is forgotten.	
Description: The data engineer requests a secure password-reset link and replaces the existing password with a valid new password.	
Actor(s): Data engineer	
Trigger: The data engineer selects Forgot Password on the login screen or Reset Password in profile settings.	
Pre-condition: The data engineer has a registered email-and-password account, can access the registered inbox and has an internet connection.	
Main Flow: 1. The data engineer selects Forgot Password or Reset Password. 2. The system displays the password-reset form. 3. The user enters the registered email address and submits the request. 4. The system sends a time-limited password-reset link and displays a neutral confirmation. 5. The user opens the link and enters the new password twice. 6. The system validates the link, password match and password rules. 7. Firebase Authentication updates the password and invalidates the link. 8. The system confirms the change and directs the user to log in again.	
Sub-flow: S1. A logged-in user may start the same process from profile settings as a preventative security measure.	
Alternative flow: 3a. If the email format is invalid, the system highlights the field and does not submit the request. 4a. The system uses the same neutral confirmation whether or not an account can be disclosed. 6a. If the link is invalid, expired or used, the system rejects it and offers to send another link. 6b. If the passwords do not match or fail the password rules, the system explains the problem and requests a correction. 7a. If the network or authentication service is unavailable, the password remains unchanged and the system offers a retry option.	

Use Case 39 – Data engineer views their profile

Name: Data engineer views their profile	ID: 39
Stakeholders & Goals: Data engineer – Check whether their information is up to date.	
Description: The data engineer views the profile information stored for their own operational account.	
Actor(s): Data engineer	
Trigger: The data engineer selects Profile from the navigation or account menu.	
Pre-condition: The data engineer is logged in to an active account.	
Main Flow: 1. The user selects Profile. 2. The system verifies the current session and role. 3. The system retrieves only the authenticated user's profile information. 4. The system displays the profile information and operational role in a readable format. 5. The user reviews the displayed information.	
Sub-flow: S1. The user may select Edit Profile and continue with Use Case 40.	
Alternative flow: 2a. If the session has expired, the system returns to the login screen. 3a. If the profile cannot be retrieved, the system displays an error without exposing another account's information and offers a retry option.	

Use Case 40 – Data engineer updates their profile

Name: Data engineer updates their profile	ID: 40
Stakeholders & Goals: Data engineer – Keep their information up to date.	
Description: The data engineer changes editable information in their own profile while protected role and access-control fields remain read-only.	
Actor(s): Data engineer	
Trigger: The data engineer selects Edit Profile while viewing their profile.	
Pre-condition: The data engineer is logged in and the existing profile information is available.	
Main Flow: 1. The user views their profile and selects Edit Profile. 2. The system displays editable fields with their current values; the role is read-only. 3. The user changes one or more editable fields and selects Save Changes. 4. The system validates the information. 5. The system saves the valid changes to the authenticated user's profile. 6. The system displays the updated profile and confirms that the changes were saved.	
Sub-flow: None.	
Alternative flow: 3a. If the user cancels, the system discards unsaved changes. 4a. If required information is missing or invalid, the system highlights the relevant field and does not save the changes. 5a. If an authentication, network or database error occurs, the previous profile remains unchanged and the system offers a retry option.	

Use Case 41 – Data engineer views dashboard of incorrect results

Name: Data engineer views dashboard of incorrect results	ID: 41
Stakeholders & Goals: Data engineer – Analyse reported incorrect results and identify possible flaws in the AI models or data workflow.	
Description: A data engineer views aggregated information about user-reported incorrect results, including counts, trends and supported categories, to prioritise investigation.	
Actor(s): Data engineer	
Trigger: The data engineer selects Incorrect Results Dashboard.	
Pre-condition: The data engineer is logged in with an active data-engineer account and authorised report data is available.	
Main Flow: 1. The data engineer selects Incorrect Results Dashboard. 2. The system verifies the session and data-engineer role. 3. The system retrieves authorised report and evaluation summary data. 4. The system displays dashboard metrics and trends such as report volume, input type, concern label, model or pipeline component and report status where those fields are available. 5. The data engineer changes the time range or applies supported filters. 6. The system refreshes the dashboard using the selected criteria. 7. The data engineer selects a report or group requiring investigation.	
Sub-flow: S1. The data engineer may continue from a selected report to Use Case 42.	
Alternative flow: 3a. If no reports match, the system displays an empty state rather than misleading zeroed charts. 3b. If data is incomplete, the system identifies the affected metric or time range. 4a. If the dashboard cannot be generated, the system displays an error and offers a retry option without exposing partial protected data.	

Use Case 42 – Data engineer views incorrect results reports

Name: Data engineer views incorrect results reports	ID: 42
Stakeholders & Goals: Data engineer – View the details of a reported incorrect result for investigation.	
Description: A data engineer inspects a selected user report together with the associated submission, generated result, model metadata, evidence and provenance information that the role is authorised to view.	
Actor(s): Data engineer	
Trigger: The data engineer selects an incorrect-result report from the dashboard or report list.	
Pre-condition: The data engineer is logged in, the report exists and the associated data is within the role's permitted scope.	
Main Flow: 1. The data engineer opens a report from the dashboard or report list. 2. The system verifies the session, role and access scope. 3. The system retrieves the user's report reason, relevant submitted content, generated assessment, evidence references, timestamps and available model or pipeline metadata. 4. The system displays the information in sections and masks personal information that is not required for investigation. 5. The data engineer reviews the discrepancy and supporting provenance. 6. The data engineer records an investigation note or permitted status update. 7. The system validates and saves the update and records it in the audit history.	
Sub-flow: S1. The data engineer may open a related pipeline record and continue with Use Case 43.	
Alternative flow: 2a. If access is not permitted or the session has expired, the system displays no report information. 3a. If the report or associated result is unavailable, the system identifies the missing record and returns to the list. 6a. If the user cancels an edit, the system discards the unsaved note or status change. 7a. If saving fails, the previous report state remains unchanged and the system offers a retry option.	

Use Case 43 – Data engineer monitors data-ingestion pipelines

Name: Data engineer monitors data-ingestion pipelines	ID: 43
Stakeholders & Goals: Data engineer – Identify failed, delayed or incomplete data updates.	
Description: A data engineer monitors authorised data-ingestion pipelines, source provenance and run status so failures or incomplete updates that may affect the verification workflow can be investigated.	
Actor(s): Data engineer	
Trigger: The data engineer selects Data-Ingestion Pipelines from the data-engineering navigation.	
Pre-condition: The data engineer is logged in and pipeline monitoring data is available.	
Main Flow: 1. The data engineer selects Data-Ingestion Pipelines. 2. The system verifies the session and data-engineer role. 3. The system retrieves the latest authorised pipeline runs, status, source, timestamps and data-quality indicators. 4. The system displays pipeline health and clearly marks successful, running, delayed, incomplete and failed runs. 5. The data engineer filters the list or selects a pipeline run. 6. The system displays the run details, available error information, provenance and affected data range. 7. The data engineer records the issue for follow-up or uses an authorised recovery action where one is implemented. 8. The system logs the monitoring or recovery action and refreshes the displayed status.	
Sub-flow: S1. The data engineer may follow a related incorrect-result report by continuing with Use Case 42.	
Alternative flow: 3a. If monitoring data is stale or incomplete, the system labels its last update time and does not present it as current. 3b. If the monitoring service cannot be reached, the system displays an error and offers a retry option. 6a. If detailed logs are unavailable, the system displays the available summary and identifies the missing information. 7a. If the user lacks authority for a recovery action, the system provides read-only information and records no change. 8a. If an authorised recovery action fails, the system preserves the failure state, records the attempt and provides the available error information.	

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5.2 Non-Functional Requirements

The following numbers are requirements/targets, not achieved measurements. In particular, concurrent capacity, requests per minute, UI load time, inactivity expiry and storage warnings remain unverified. Existing source diagrams may retain earlier wording; current authentication uses email/password.

5.2.1 User Capacity (Scalability)

Concurrent usage: The system must be able to support at least 100 concurrent users.

Peak Traffic Load: The system must be able to function during peak traffic times, being able to handle 100 requests per minute.

Scalability: The system and database architecture must be designed to handle a 20% increase in user base per year.

5.2.2 Load Time

Initial UI Load: The initial UI of the app (homepage/login/register) must load within 2 seconds over a stable mobile connection.

Interaction Latency: UI elements must respond to user input (clicks/taps) within 1 second.

Image/Asset Optimisation: The app should lazy-load images and compress assets to ensure they load in no longer than 2.5 seconds.

5.2.3 Database Storage Space

Limit: The system should provide a notification to the user if they exceed 80% of their allocated personal storage quota.

5.2.4 Result Generation Time

Short-form text-based content: The system should generate results in under 5 minutes.

Links: The system should generate results in under 5 minutes.

Image related functions: The system should generate results in under 10 minutes.

Timeout: The system should have a timeout mechanism in place, canceling processes that exceed 20 minutes to prevent resource exhaustion.

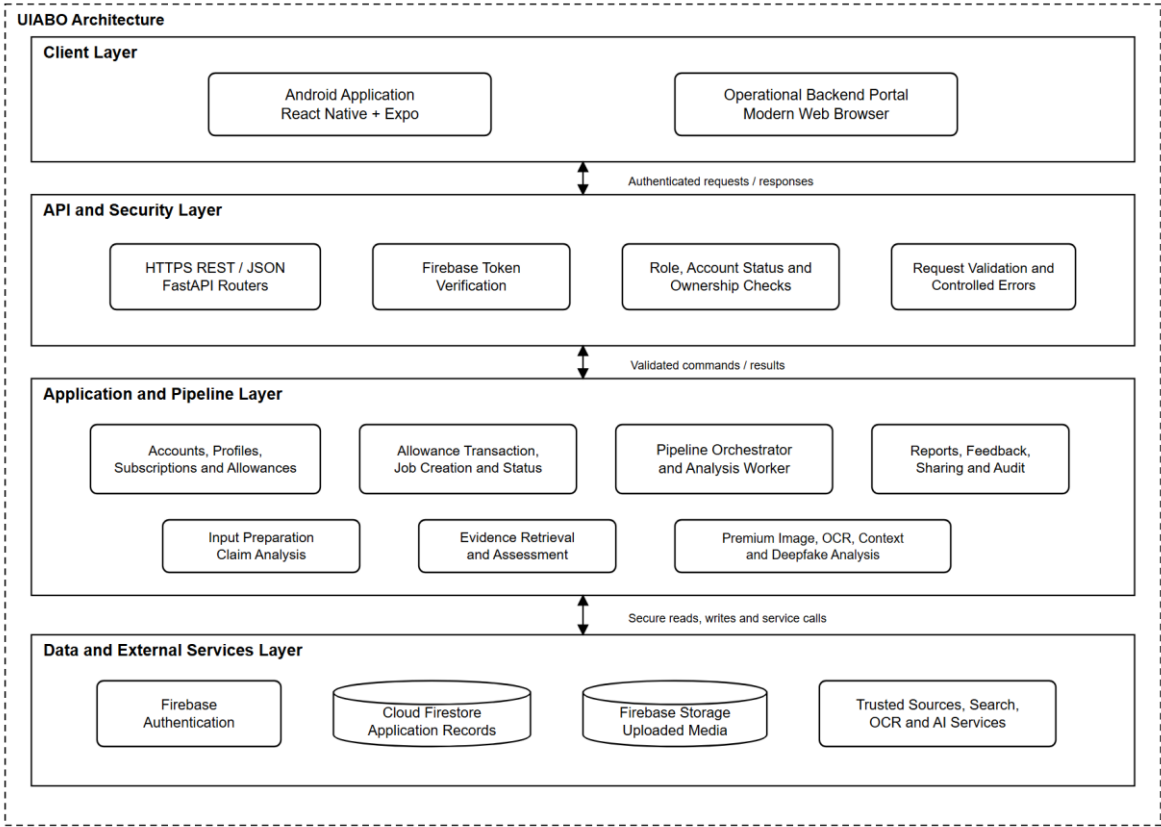
5.2.5 System Security

Data in database/app: All sensitive user data stored in the database must be encrypted. All sensitive user data available to system administrators must also be encrypted.

Authentication: individual accounts use an email address and password through Firebase Authentication. The backend verifies identity and account access for protected operations.

Session Management: User sessions should automatically expire after 30 minutes of inactivity.

5.3 Architectural Design

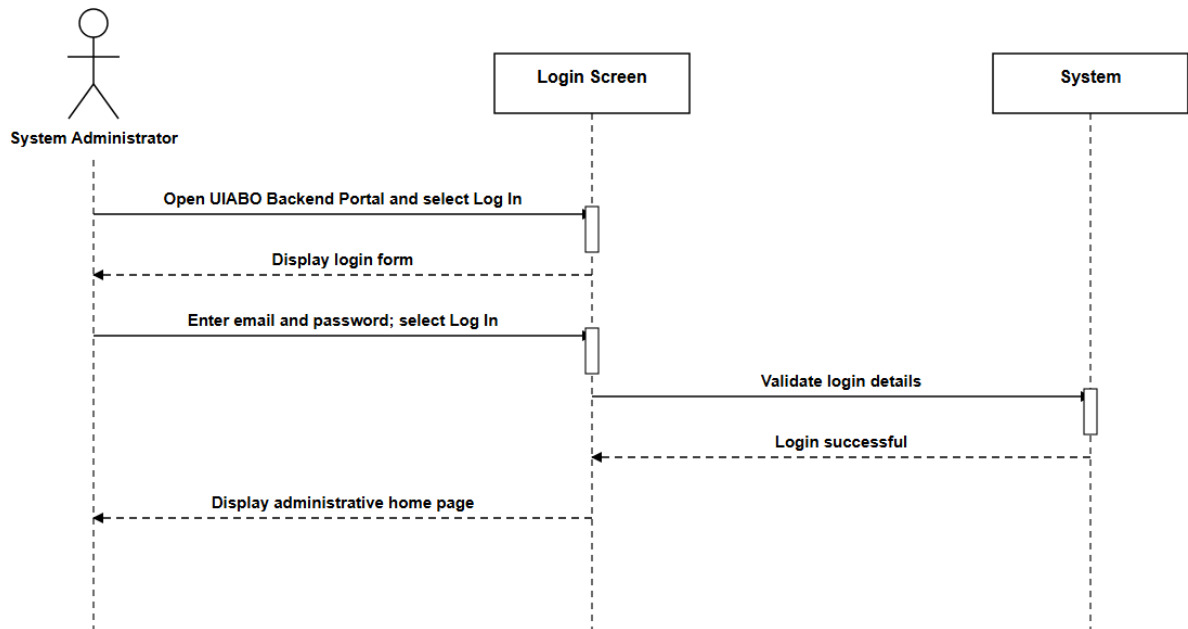


5.4 System Design

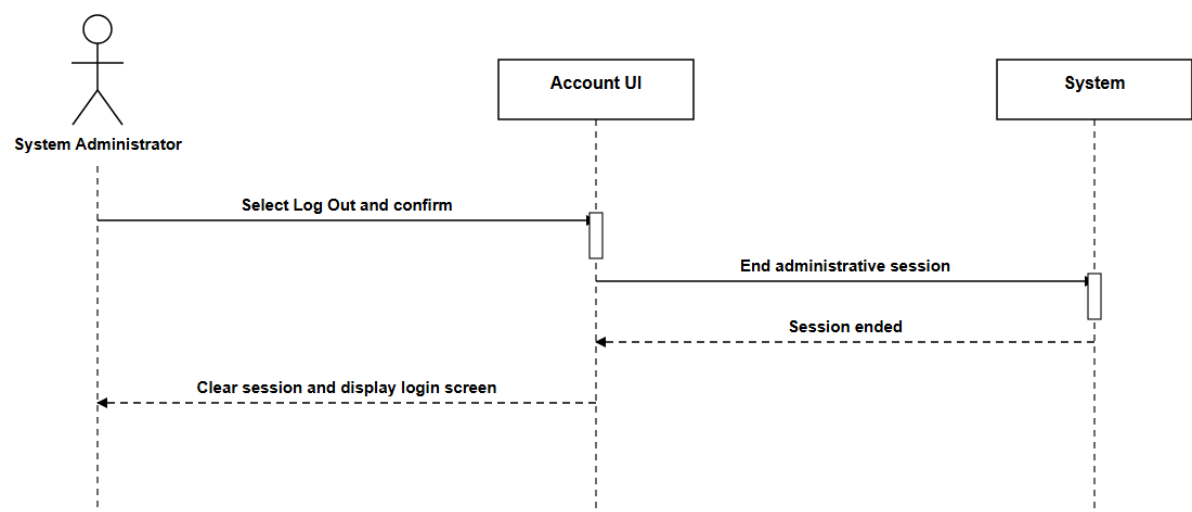
5.4.1 Sequence Diagrams

5.4.1.1 System Administrator

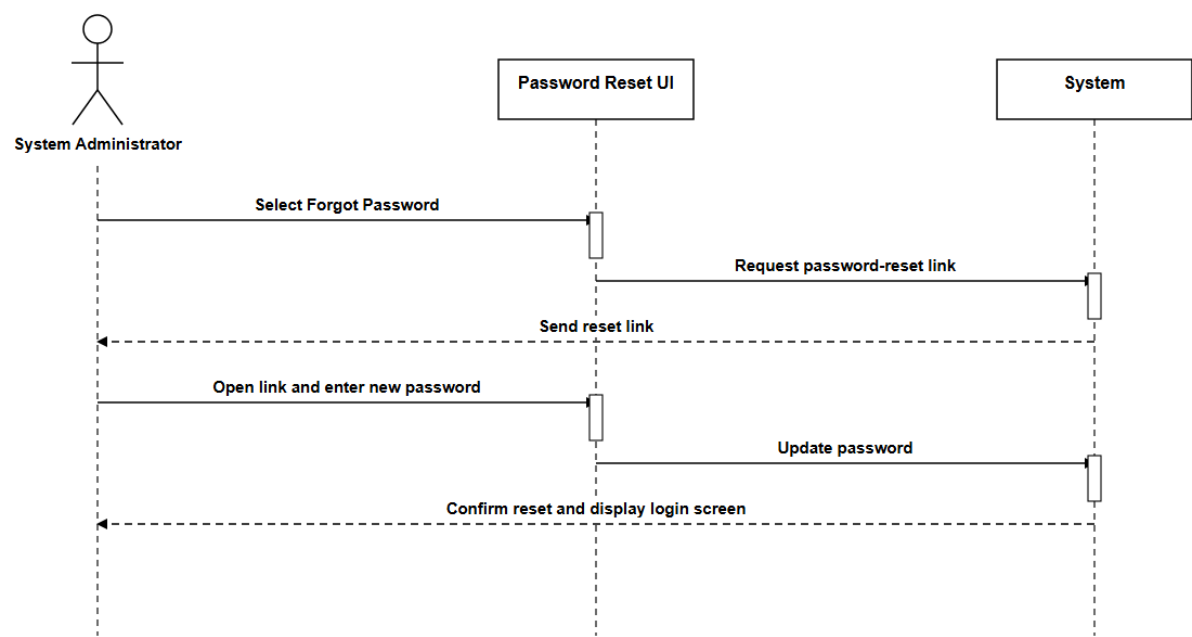
Login (System Administrator)



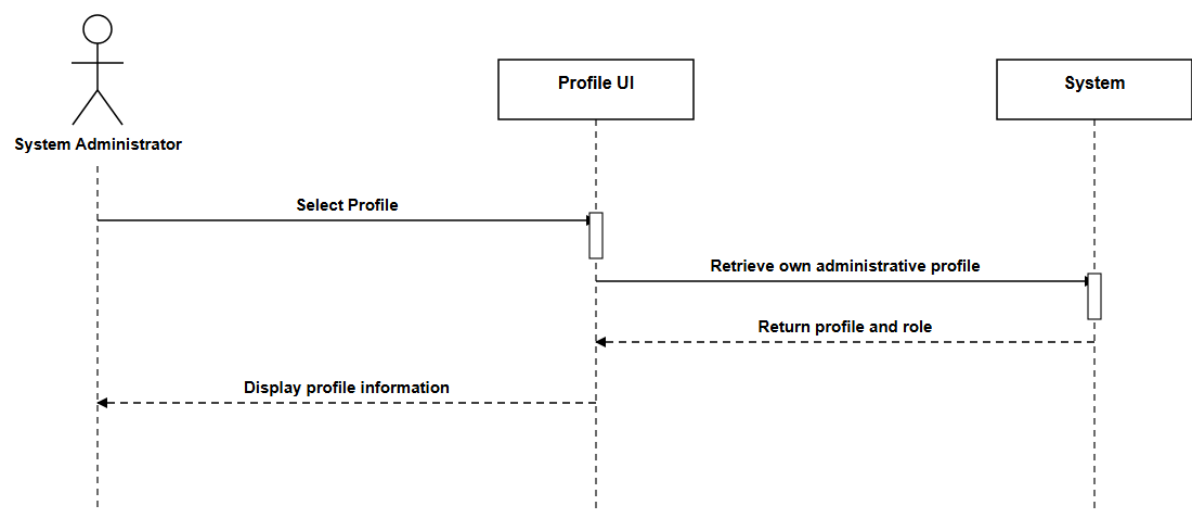
Logout (System Administrator)



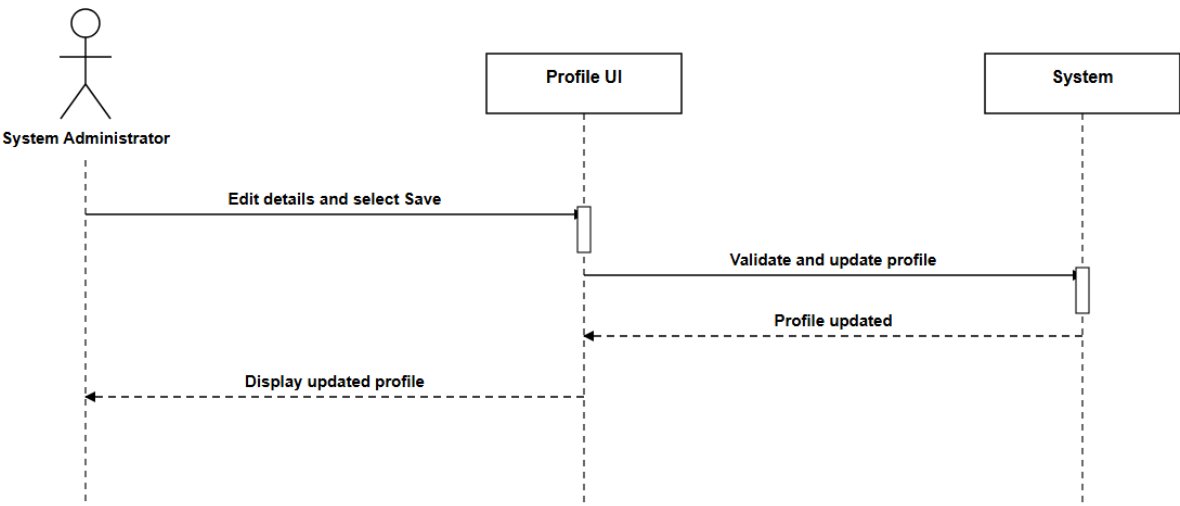
Reset Password (System Administrator)



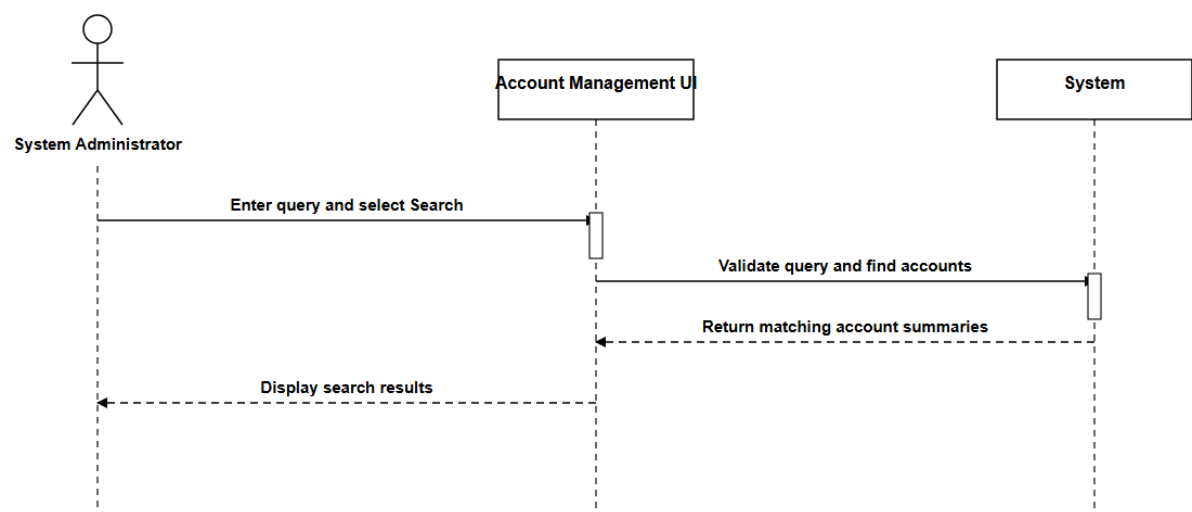
View Profile (System Administrator)



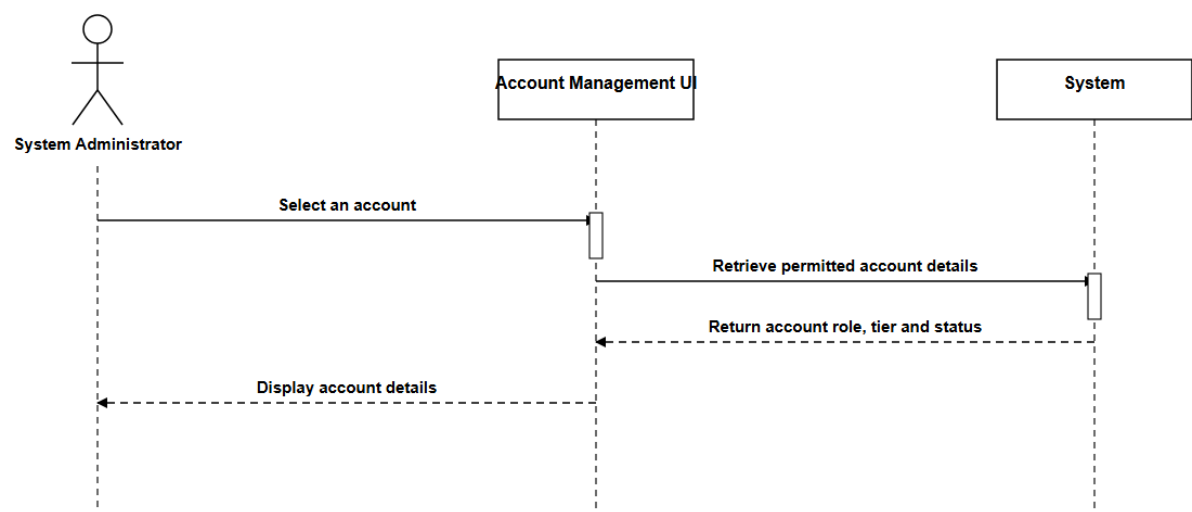
Update Profile (System Administrator)



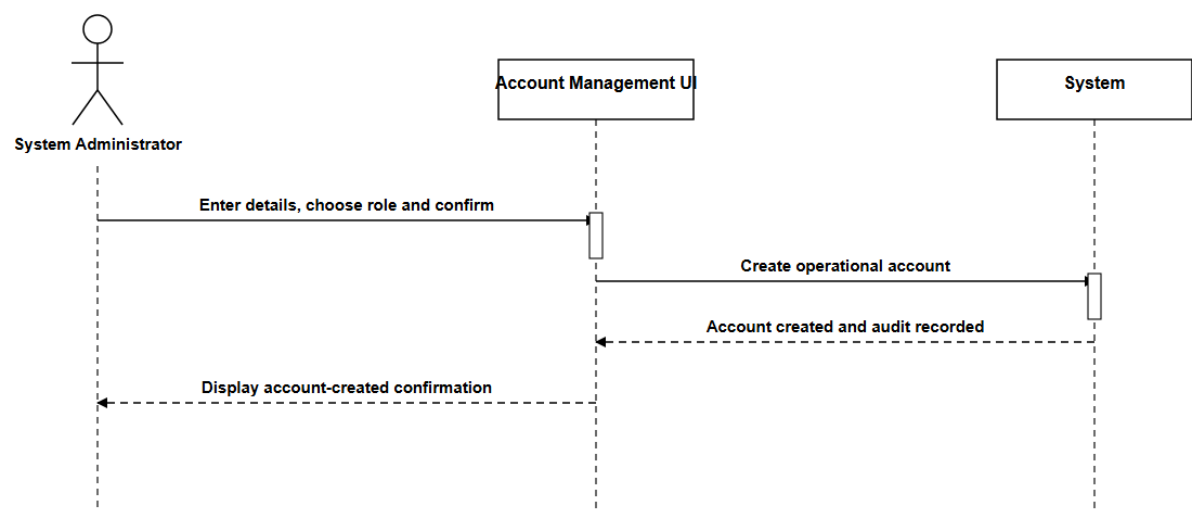
Query Account (System Administrator)



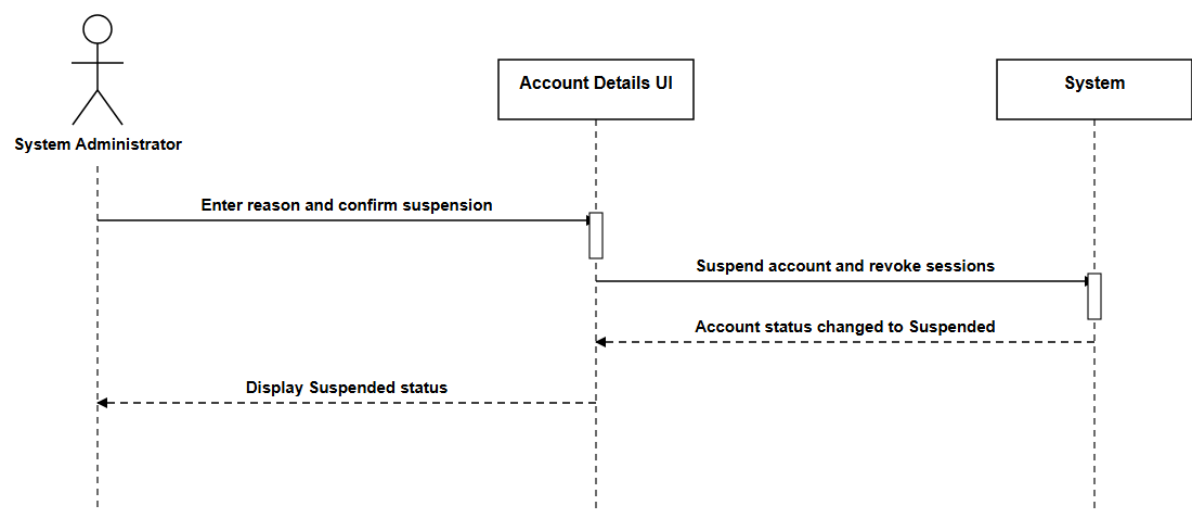
View Account Details (System Administrator)



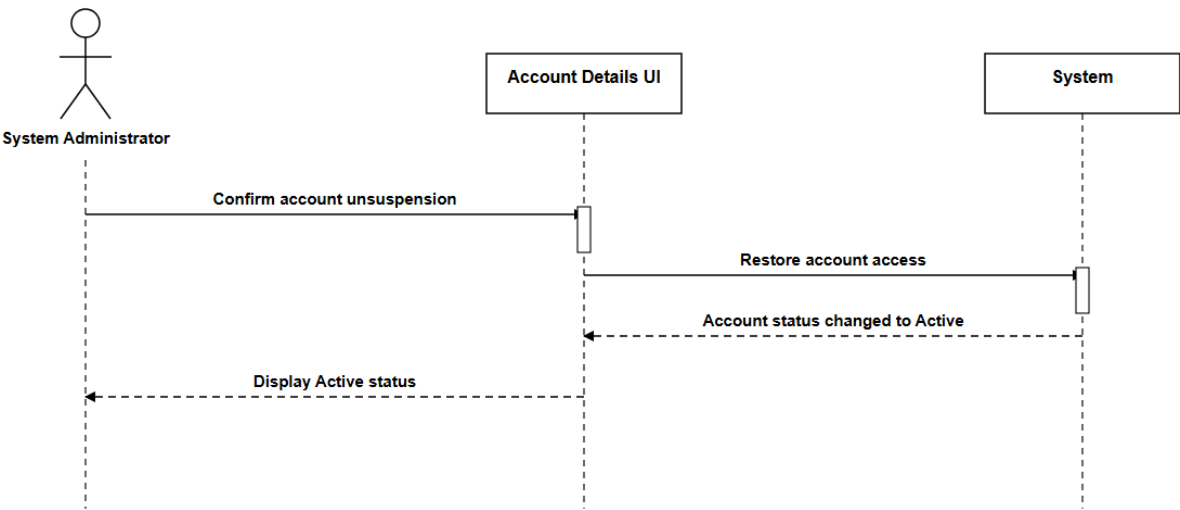
Create Operational Account (System Administrator)



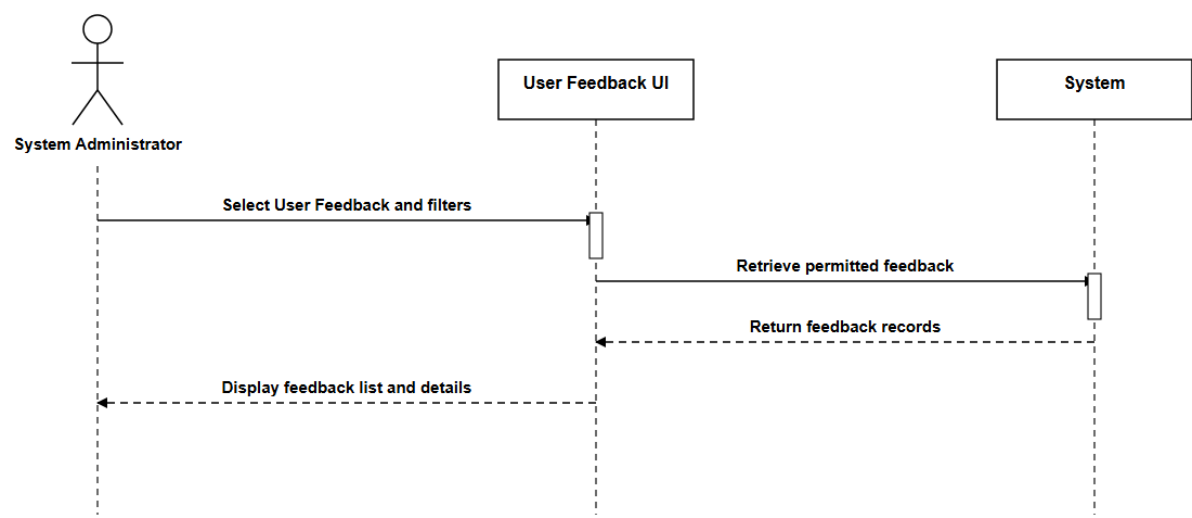
Suspend Account (System Administrator)



Unsuspend Account (System Administrator)

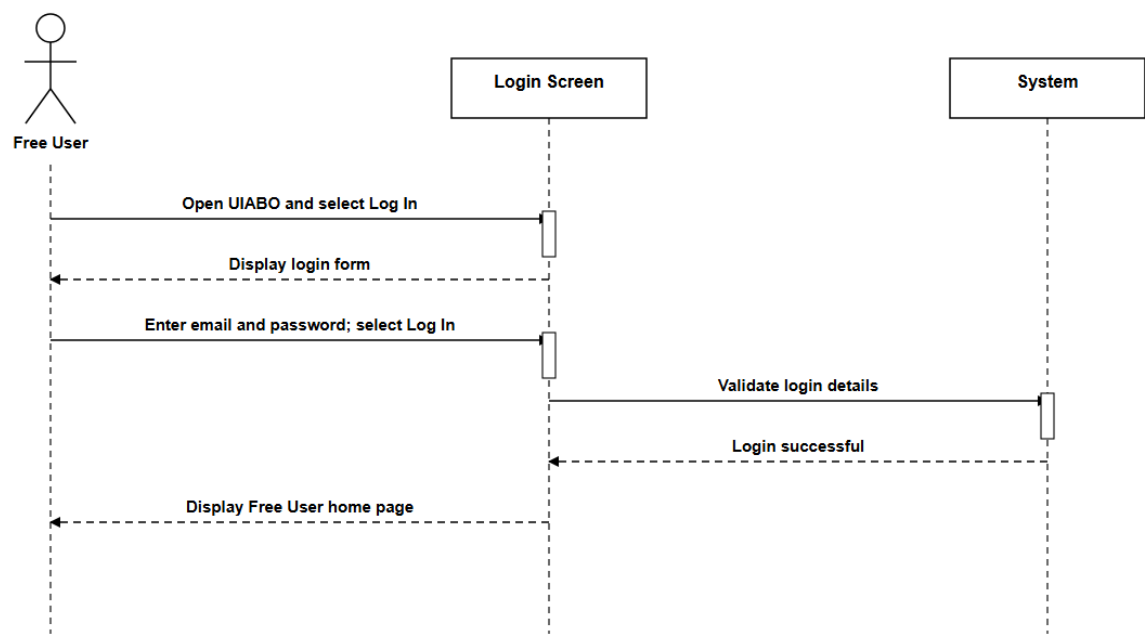


View User Feedback (System Administrator)

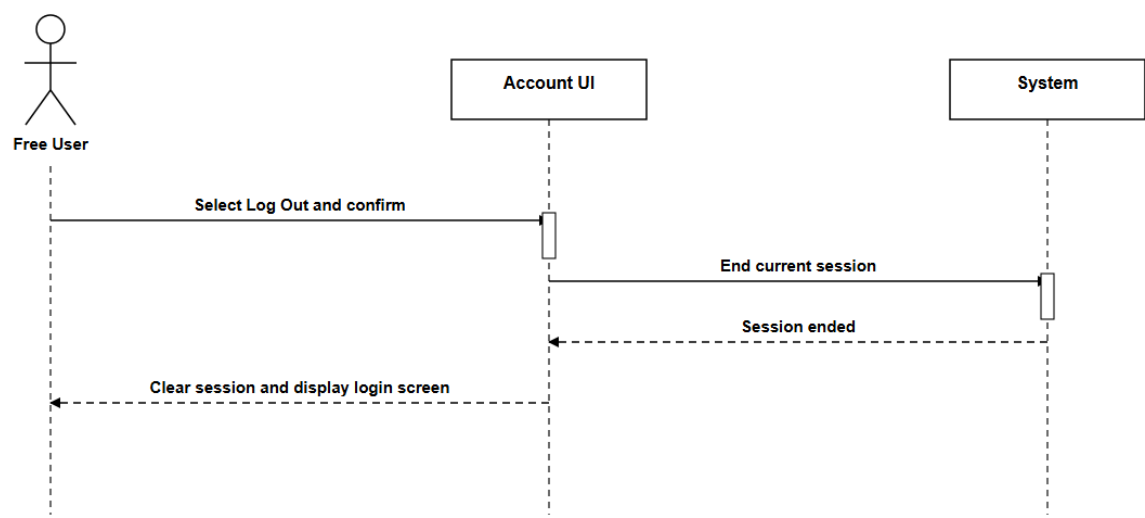


5.4.1.2 Free User

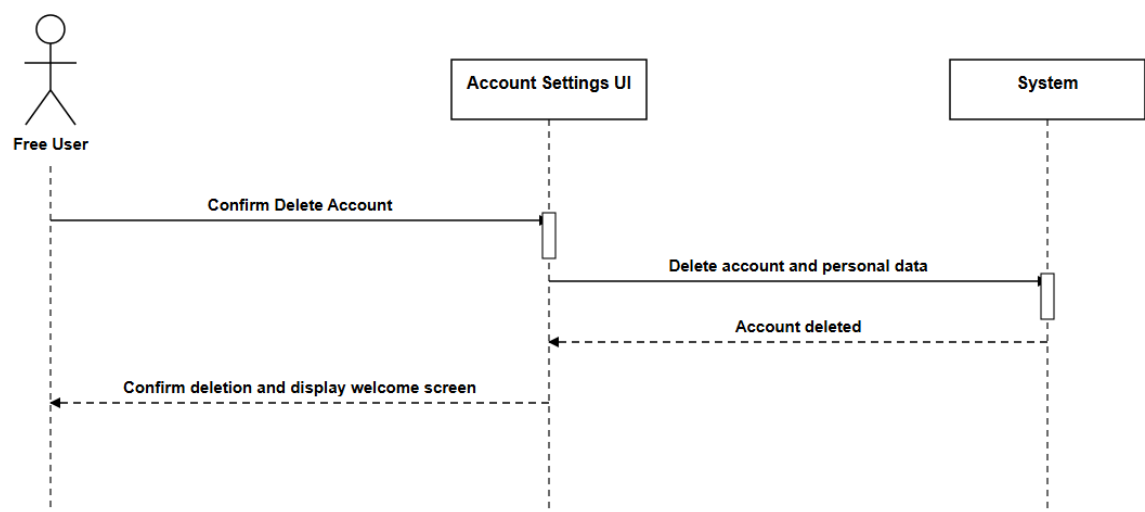
Login (Free User)



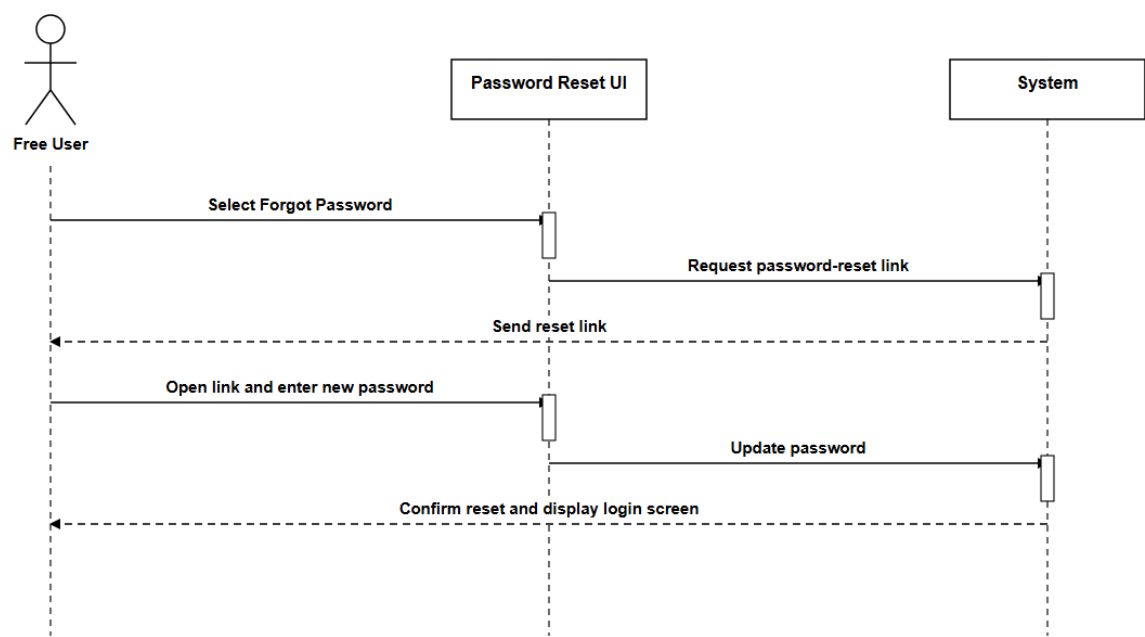
Logout (Free User)



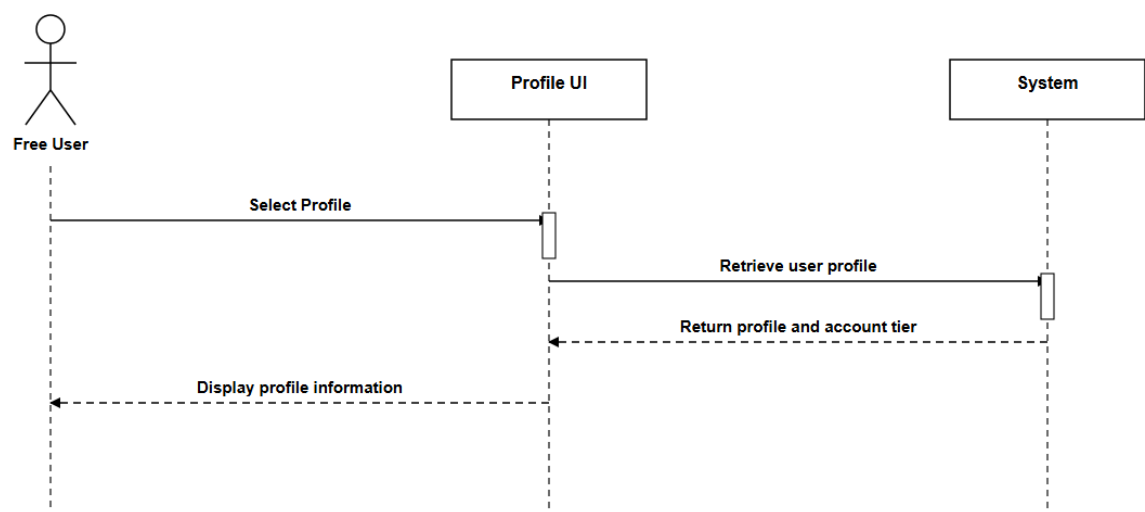
Delete Account (Free User)



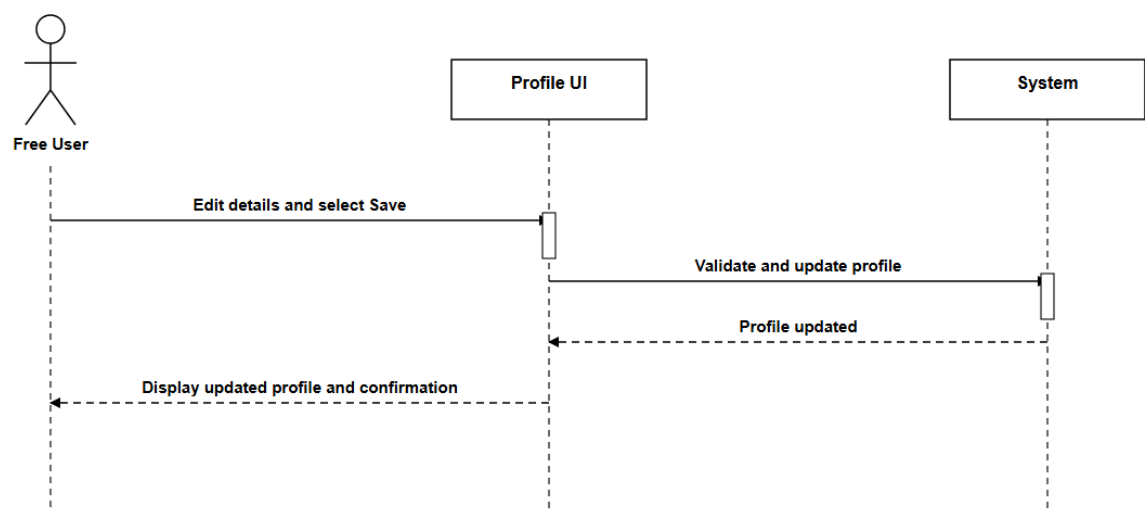
Reset Password (Free User)



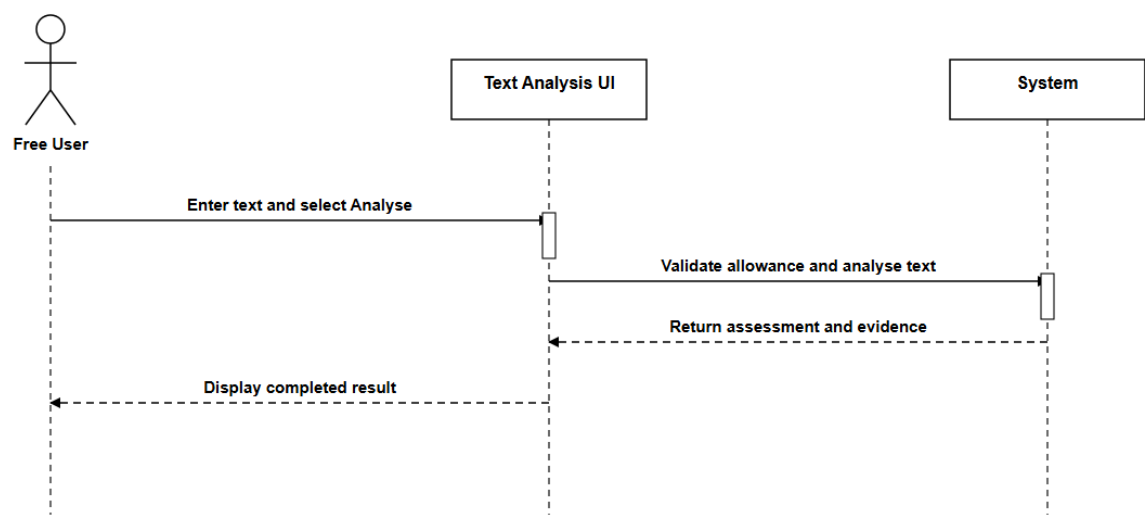
View Profile (Free User)



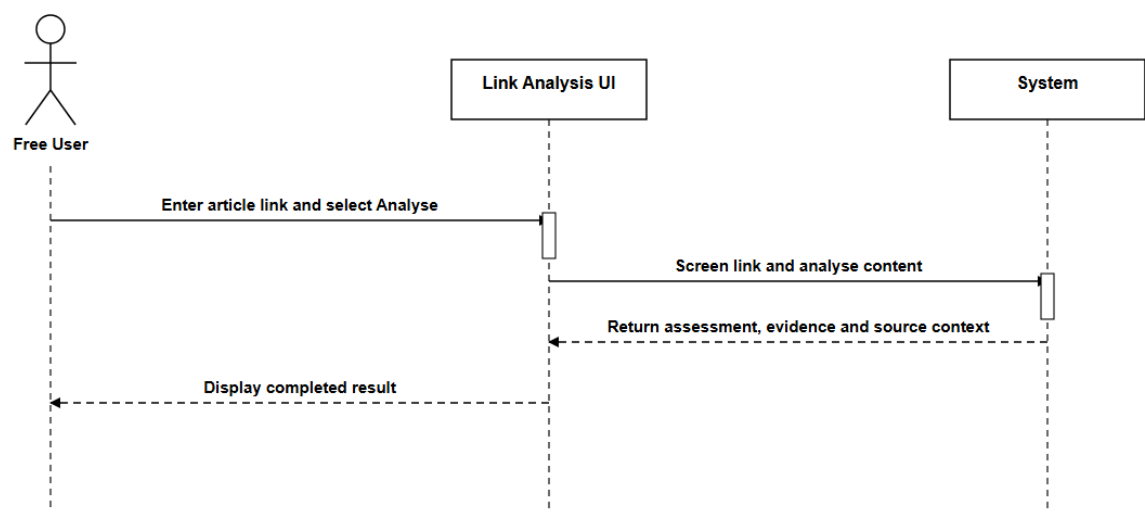
Update Profile (Free User)



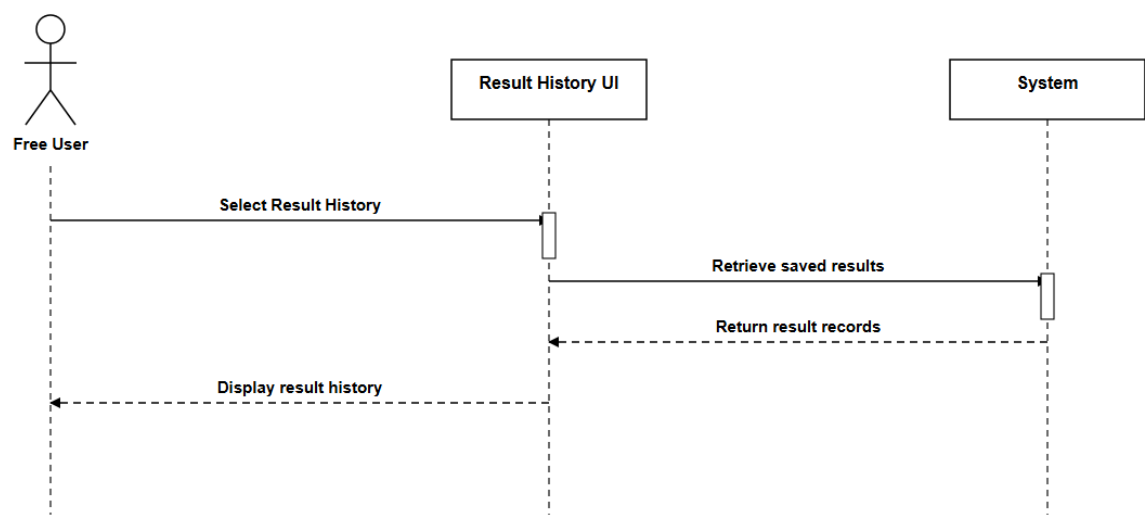
Analyse Short-form Text (Free User)



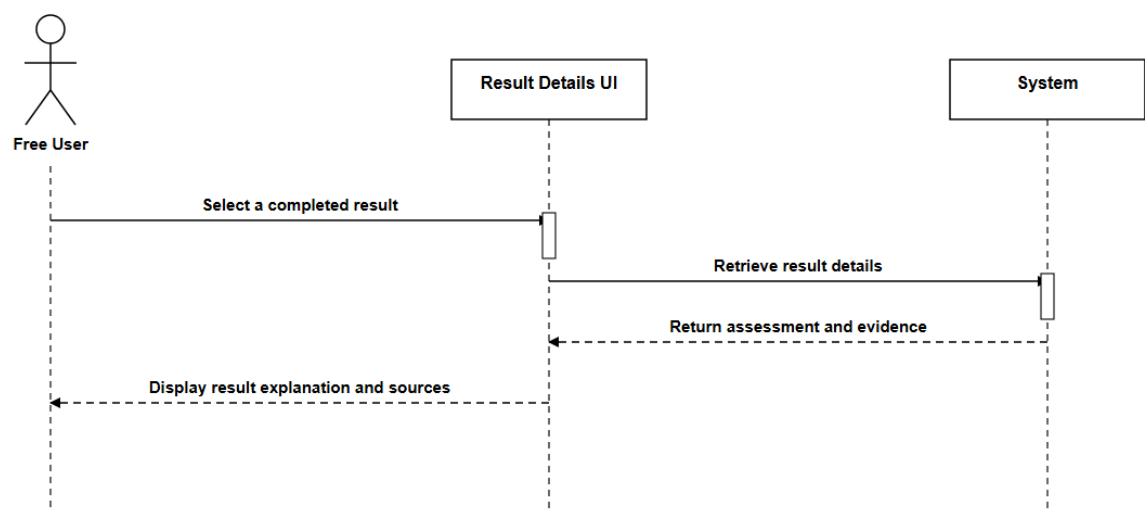
Analyse Public Webpage Link (Free User)



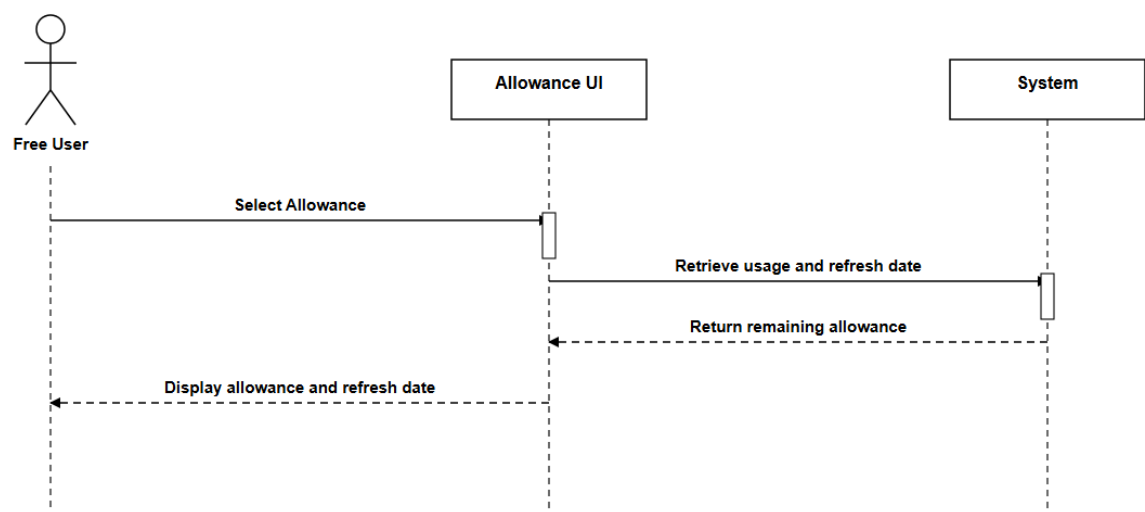
View Result History (Free User)



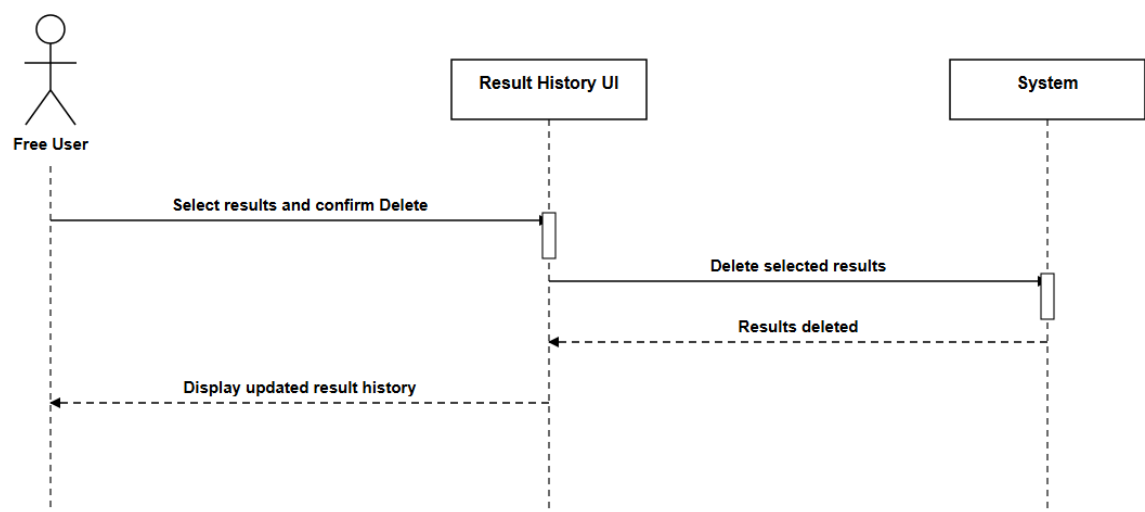
View Result Details (Free User)



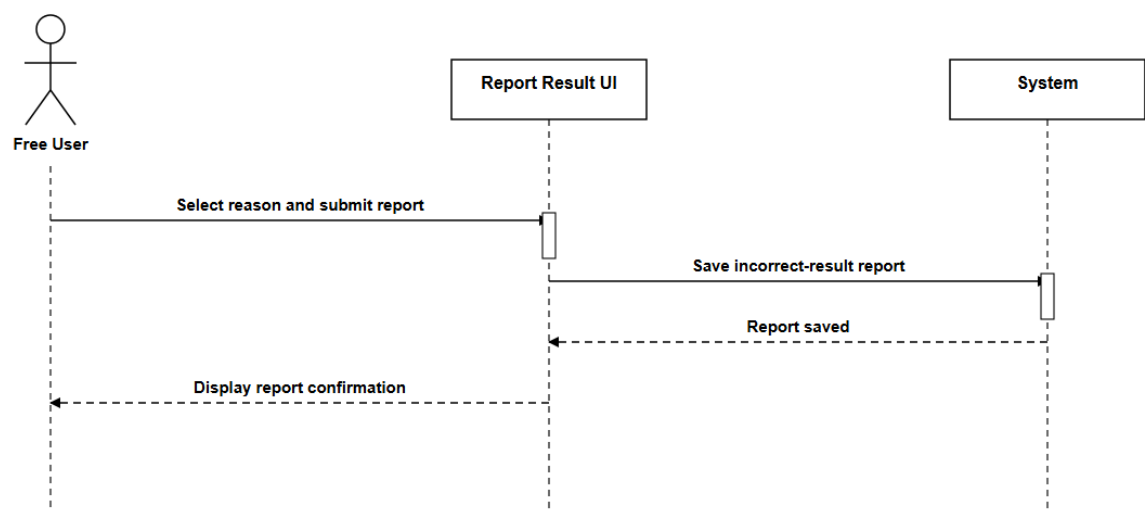
View Remaining Allowance (Free User)



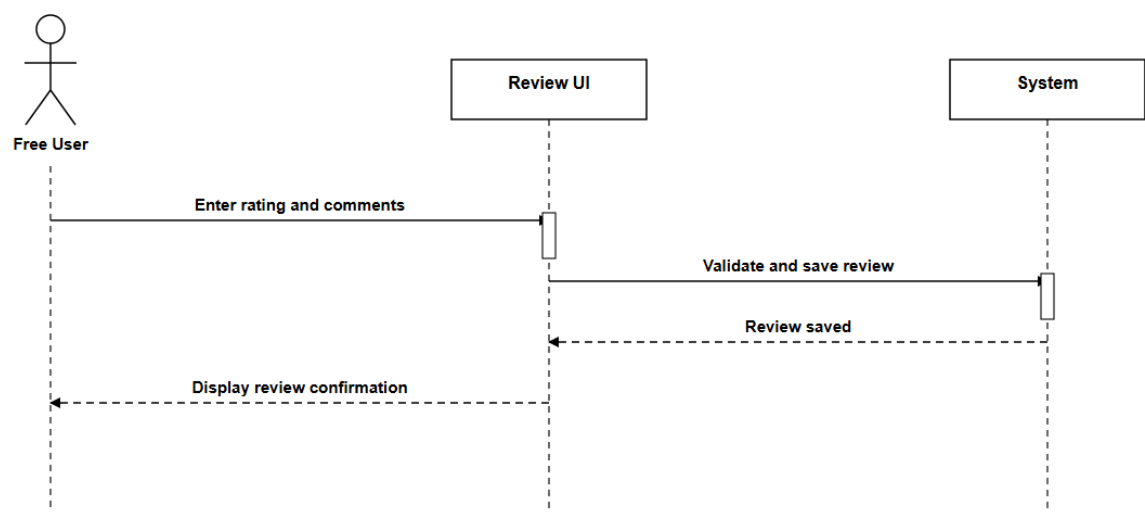
Delete Results (Free User)



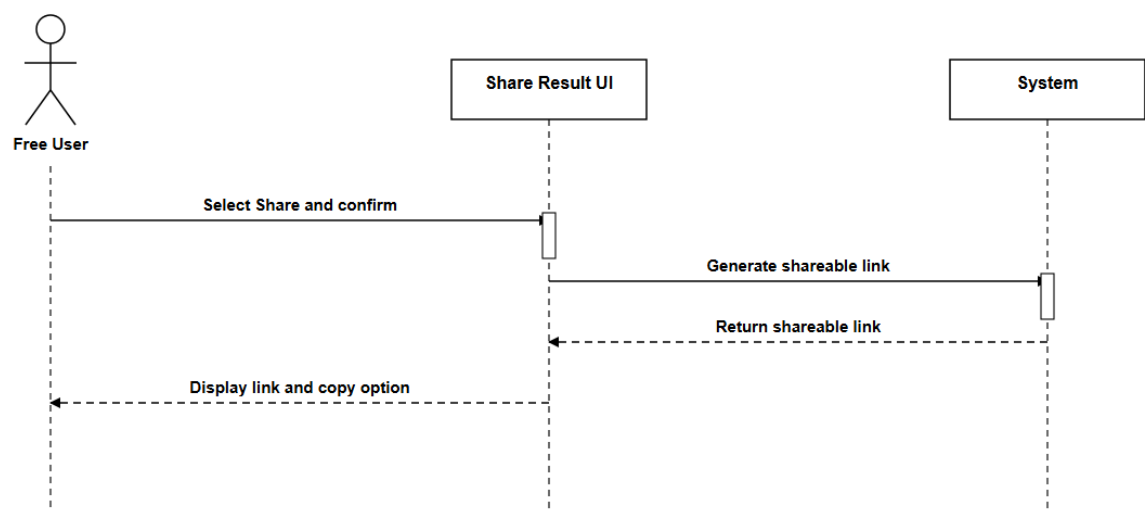
Report Incorrect Result (Free User)



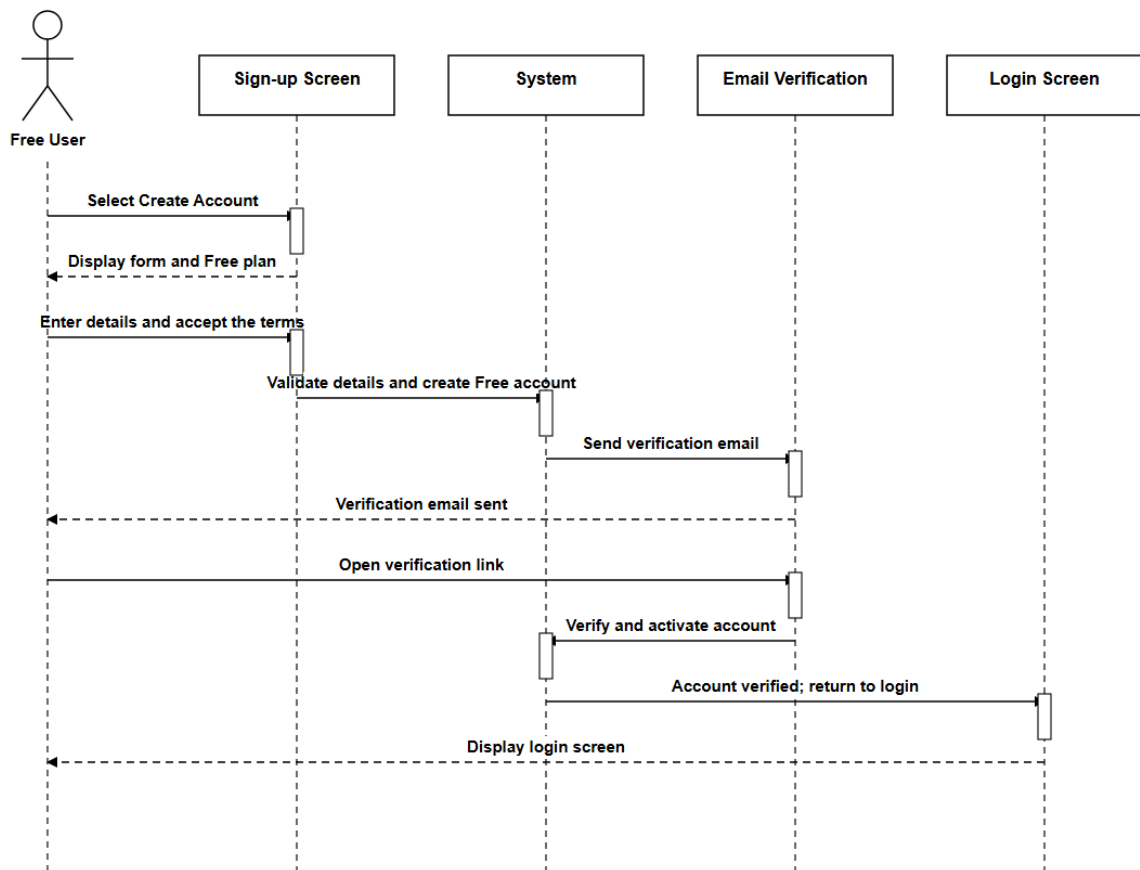
Leave a Review (Free User)



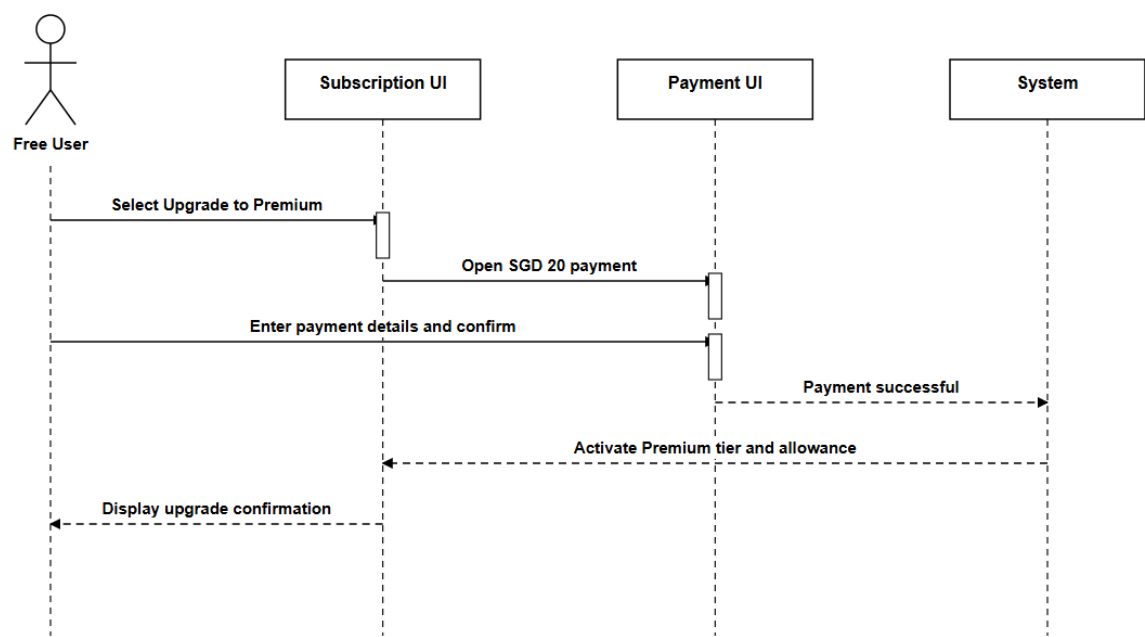
Share Result (Free User)



Create Account (Free User)

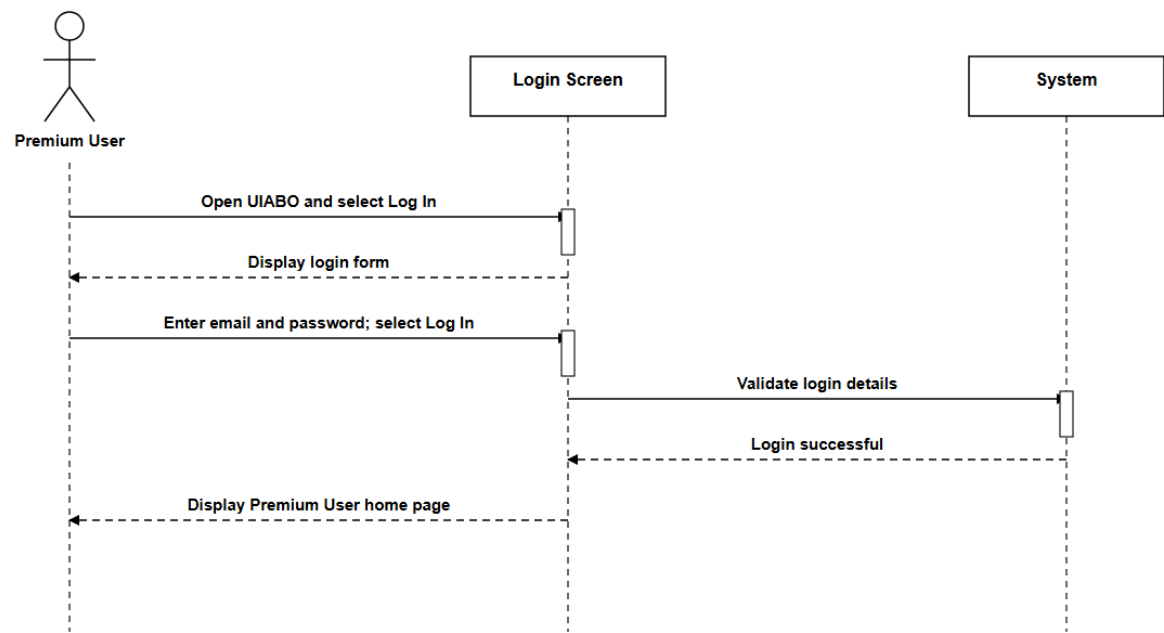


Upgrade to Premium (Free User)

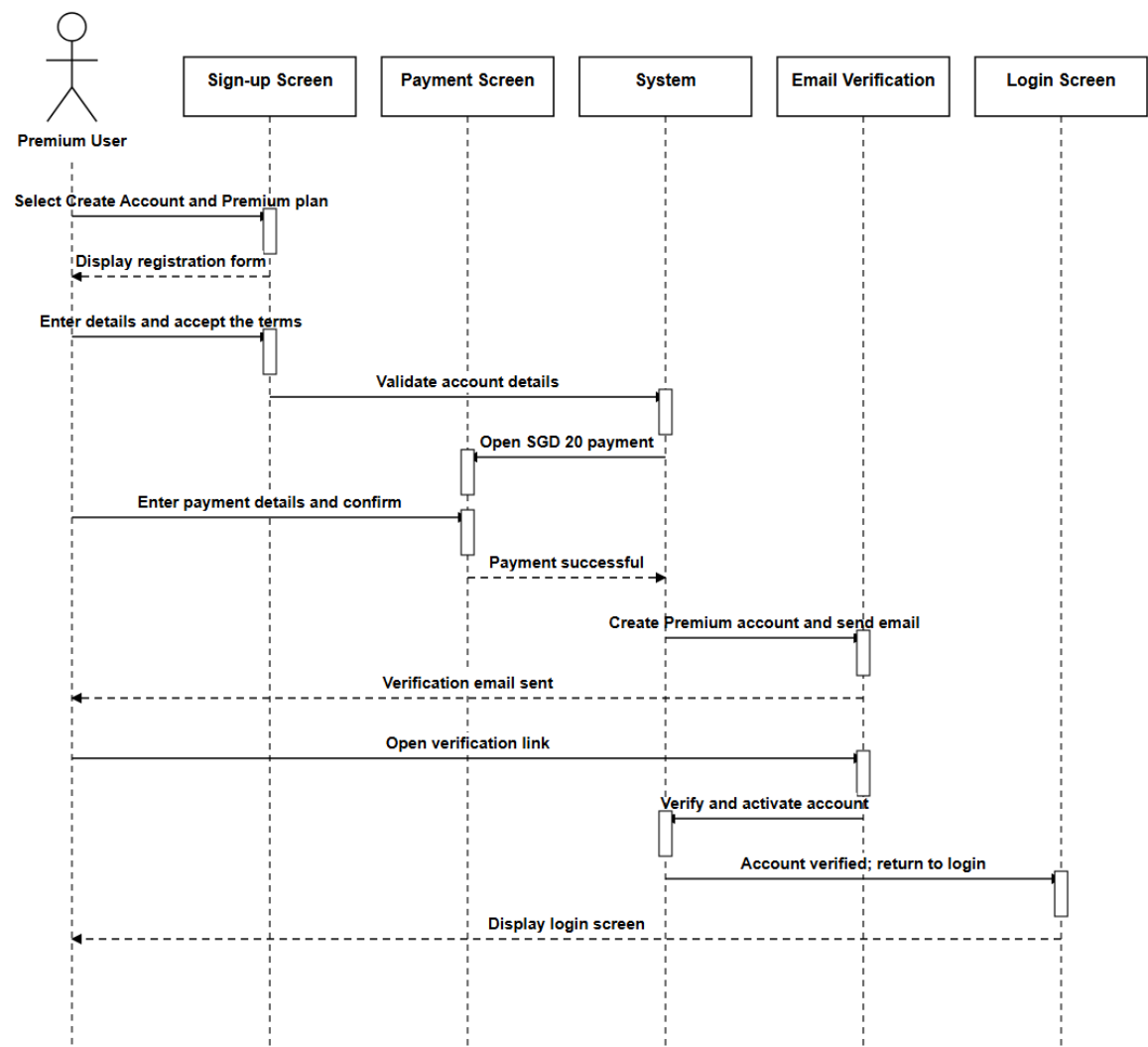


5.4.1.3 Premium User

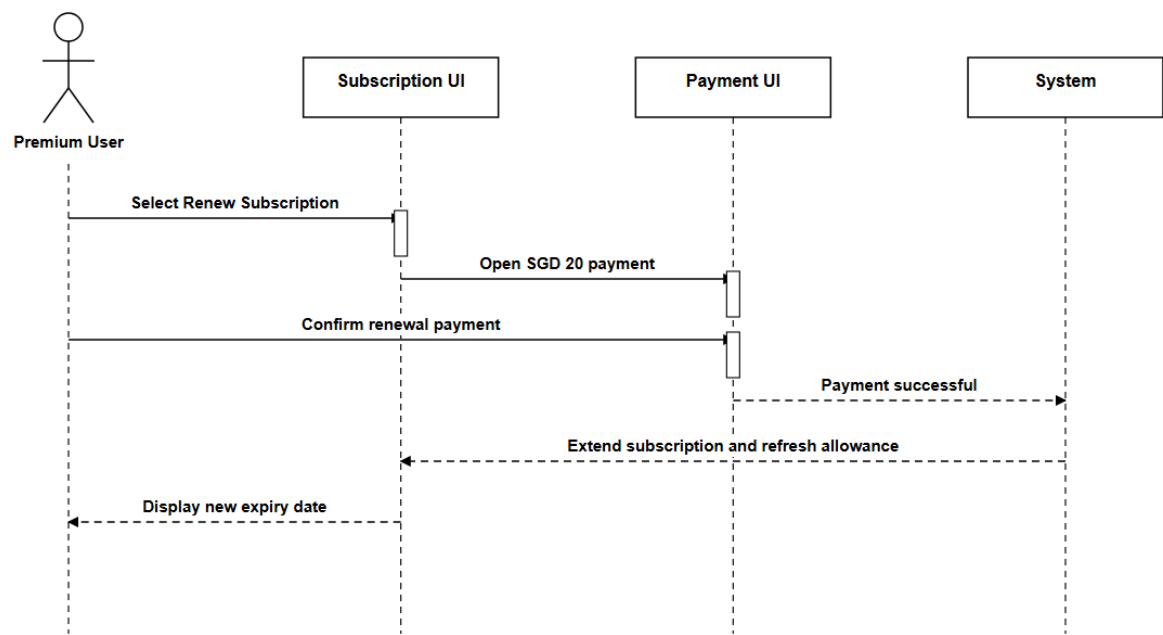
Login (Premium User)



Create Account (Premium User)



Renew Subscription (Premium User)



Cancel Subscription (Premium User)

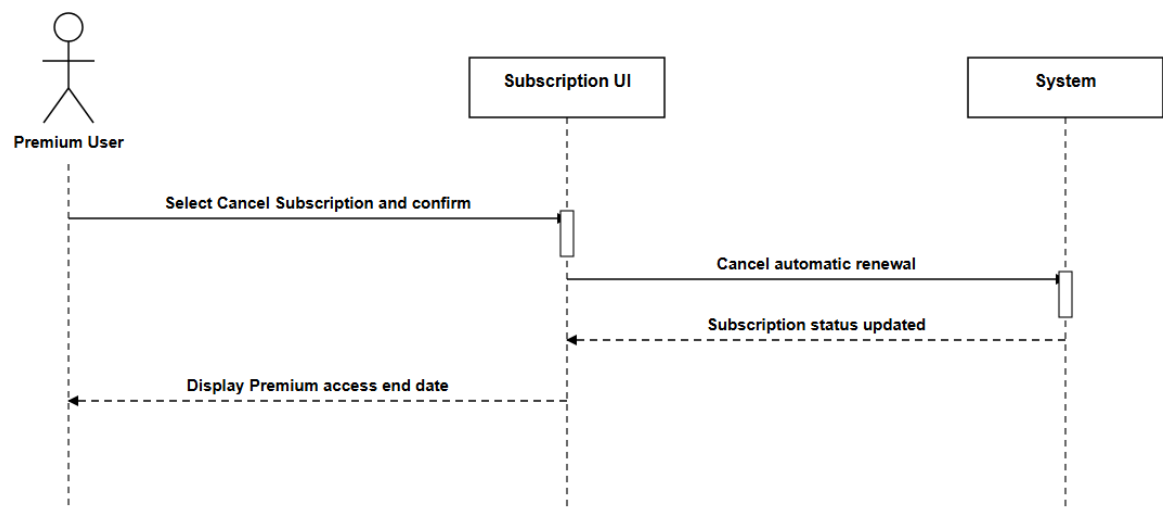
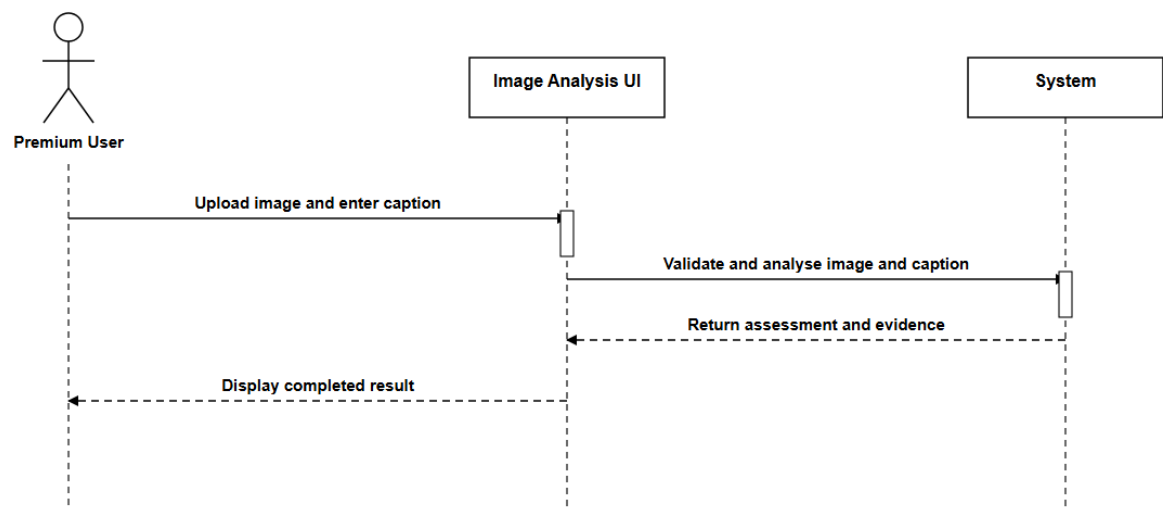
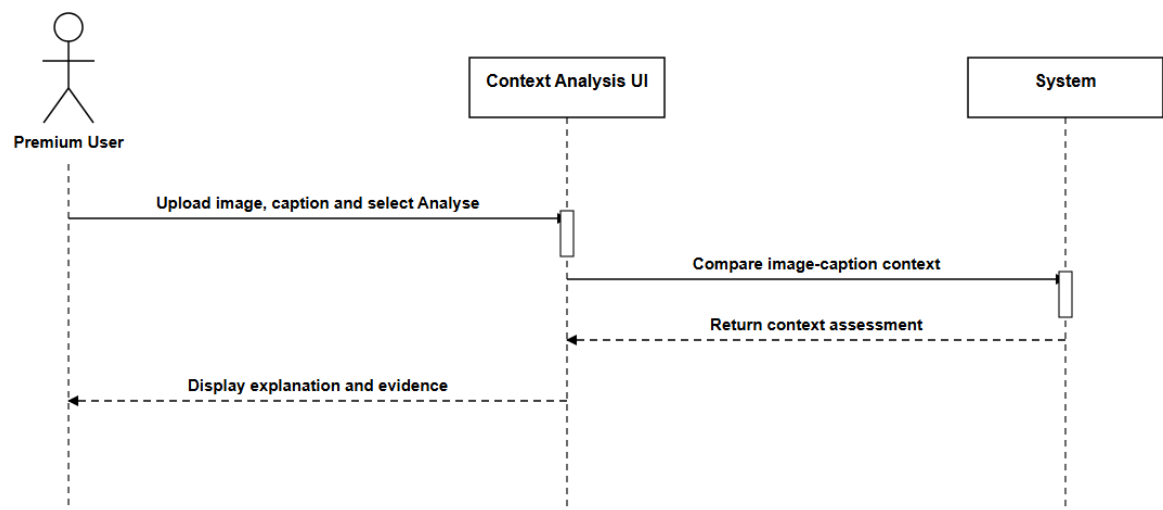


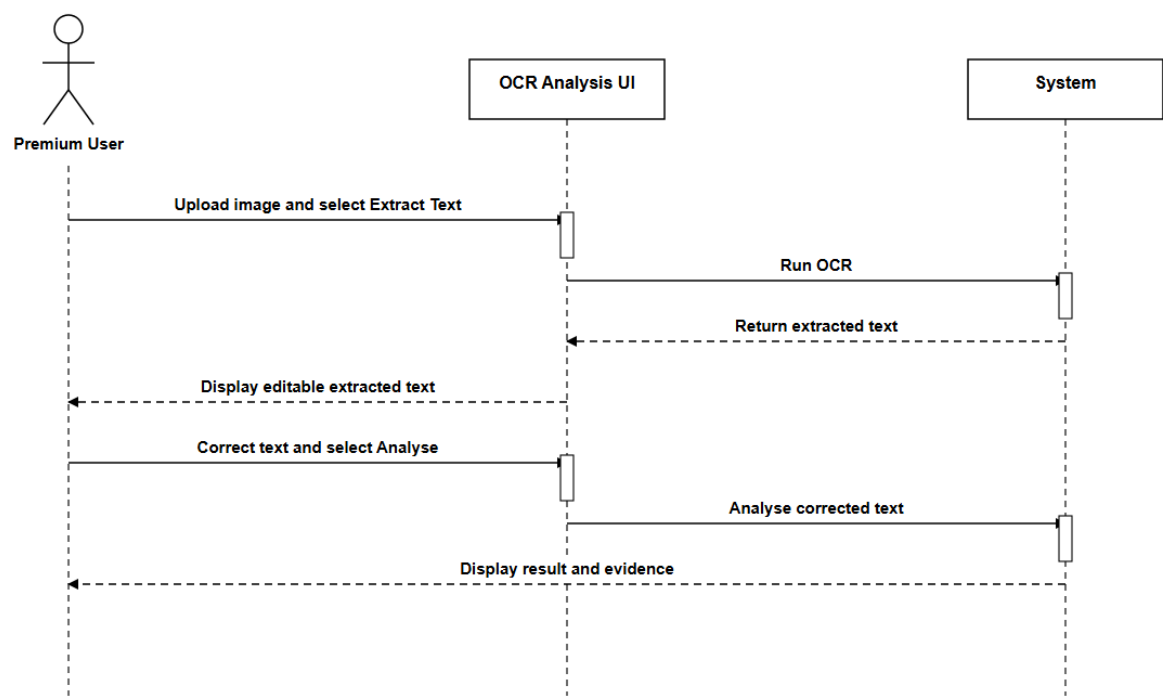
Image and Caption Analysis (Premium User)



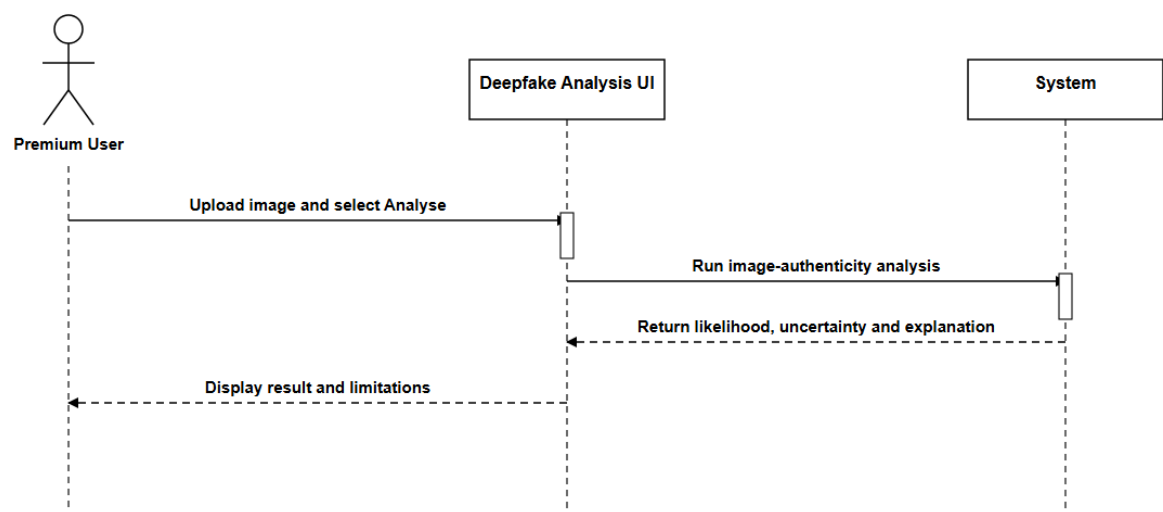
Caption-Image Context Analysis (Premium User)



OCR Image Analysis (Premium User)

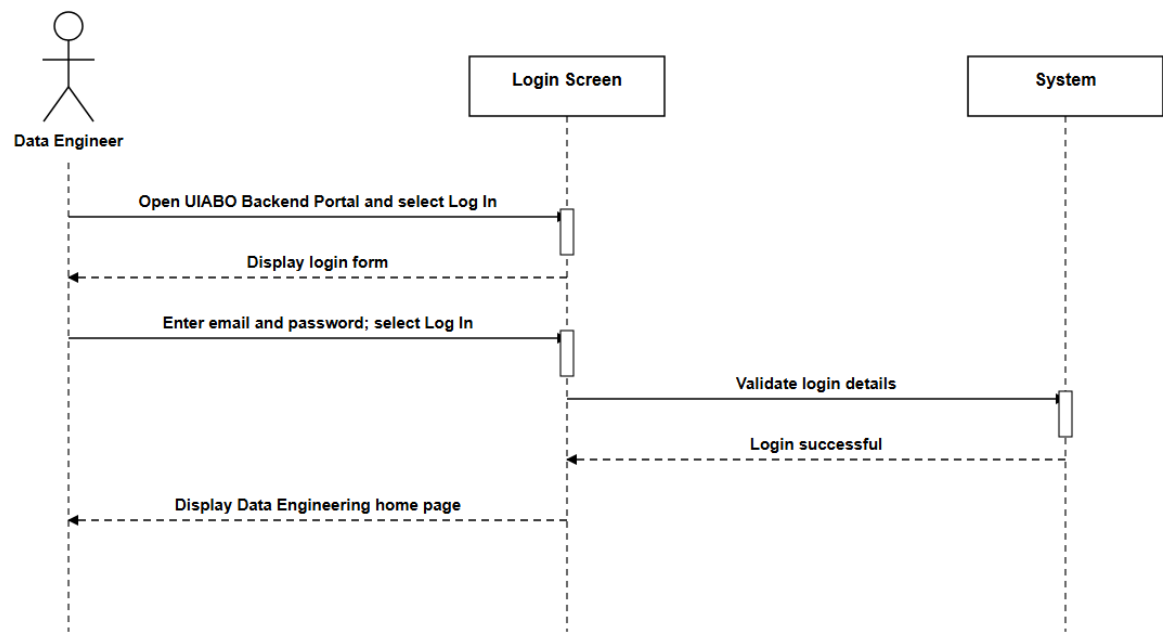


Deepfake/AI Image Analysis (Premium User)

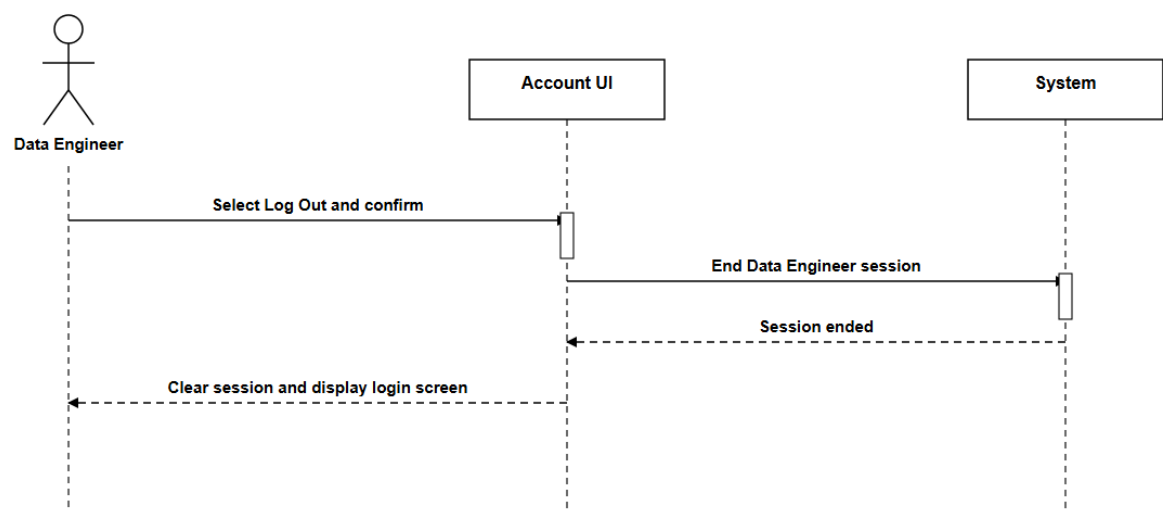


5.4.1.4 Data Engineer

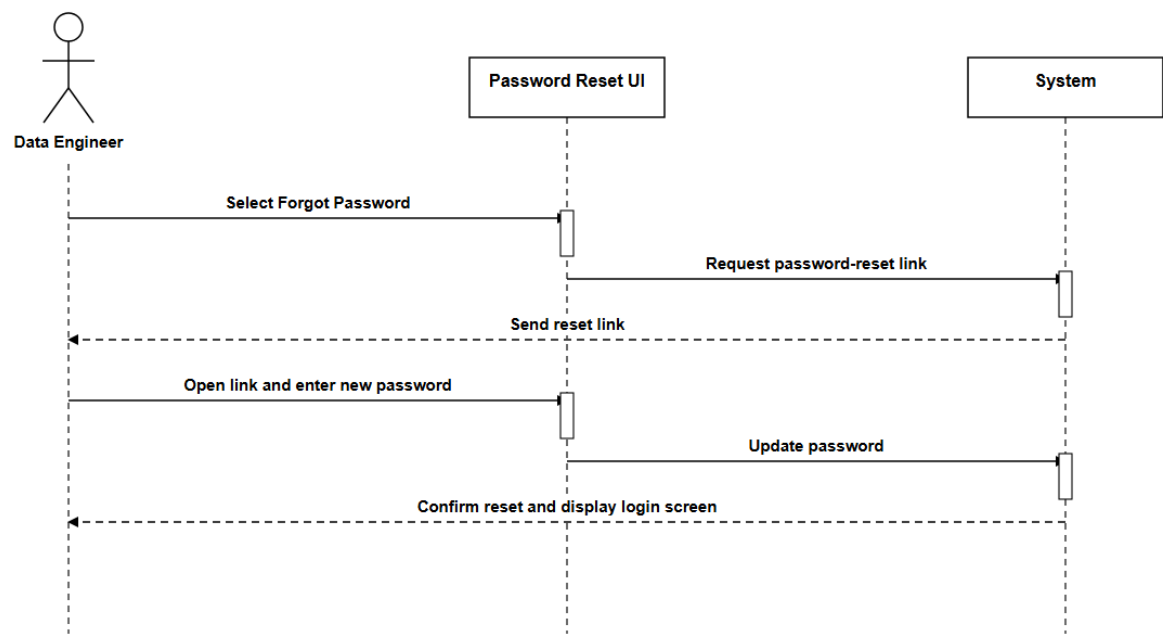
Login (Data Engineer)



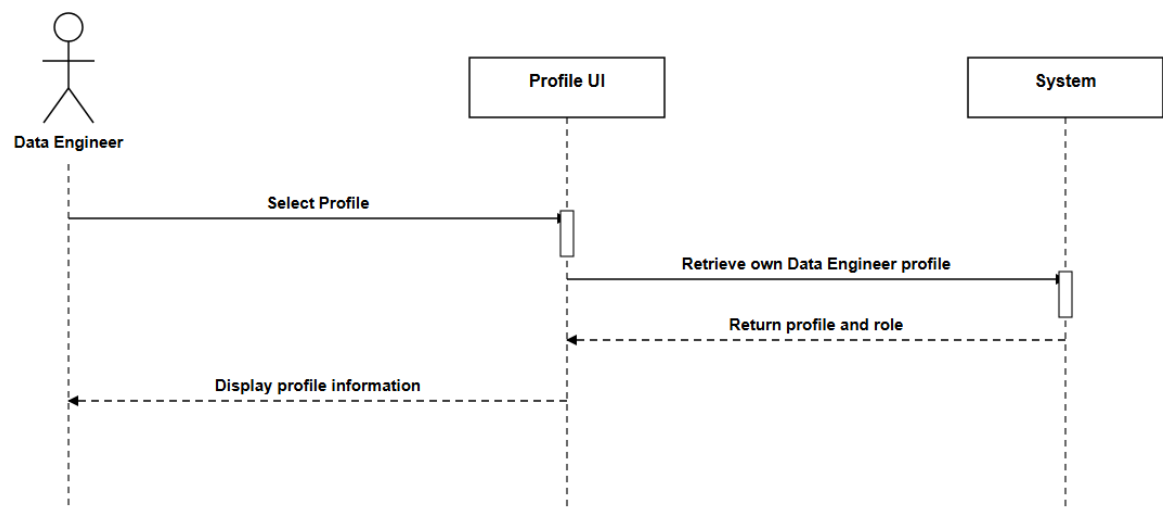
Logout (Data Engineer)



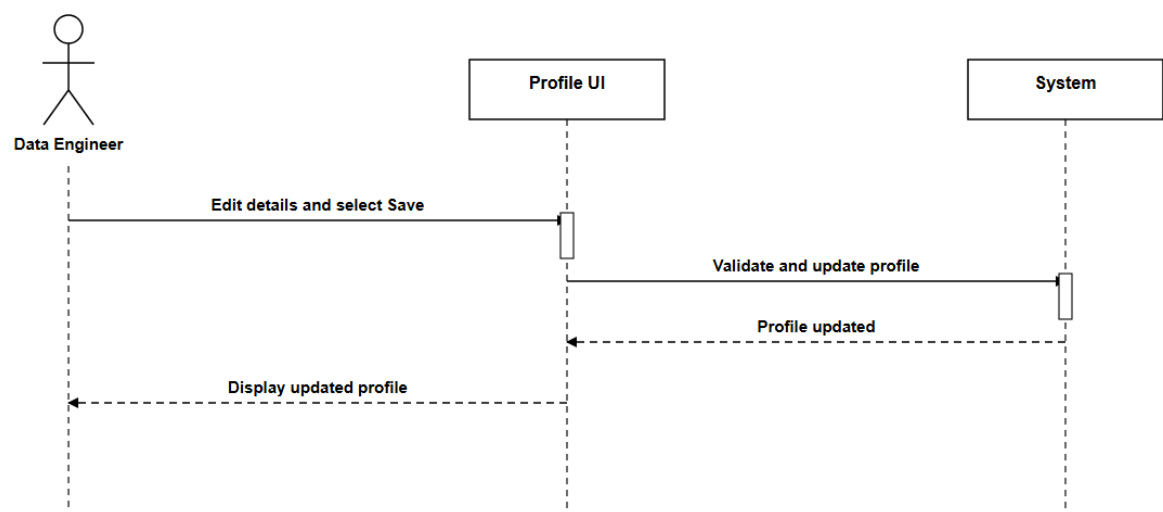
Reset Password (Data Engineer)



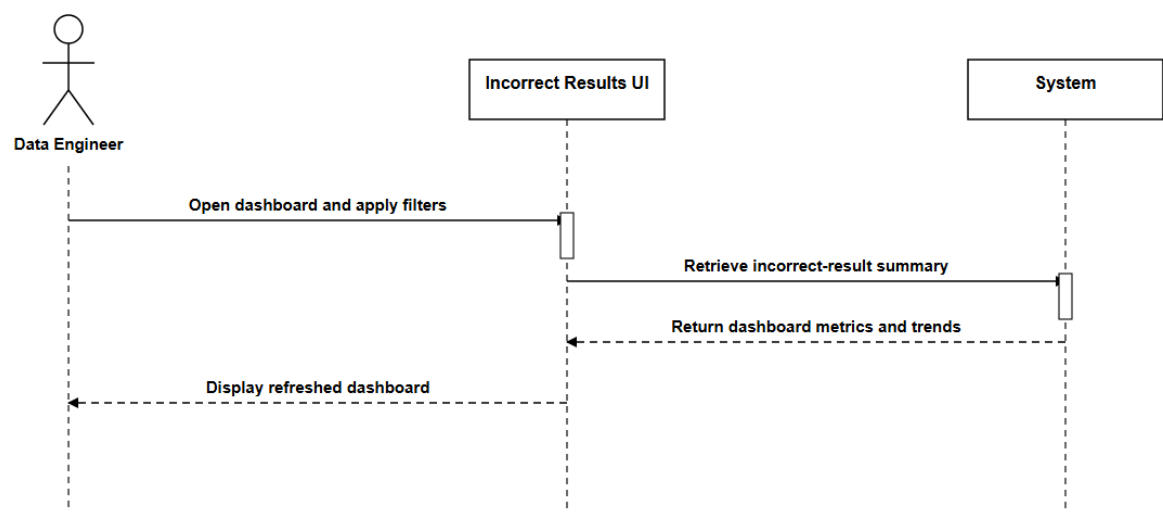
View Profile (Data Engineer)



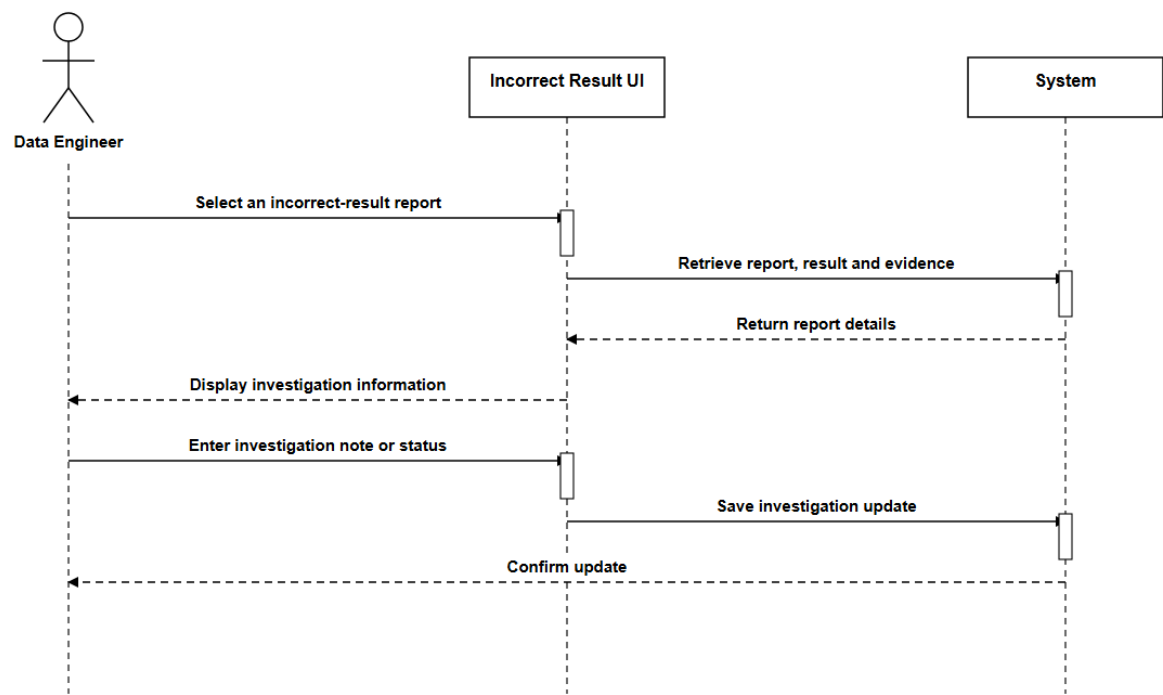
Update Profile (Data Engineer)



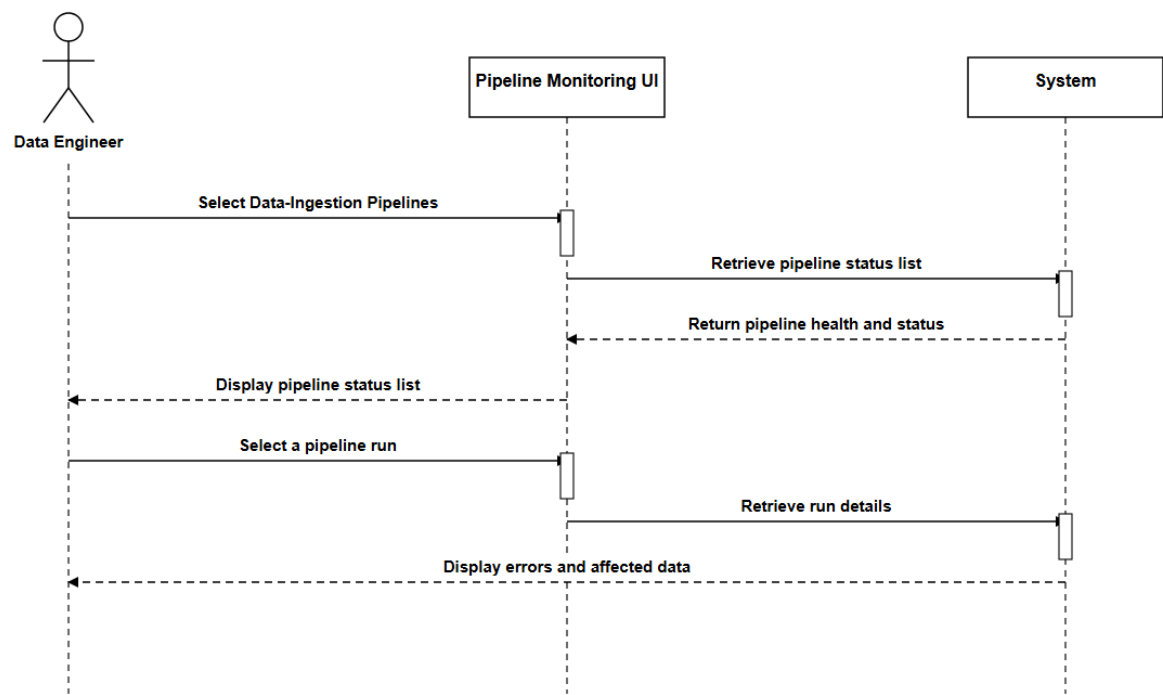
View Incorrect Results Dashboard (Data Engineer)



View Incorrect Result Report (Data Engineer)

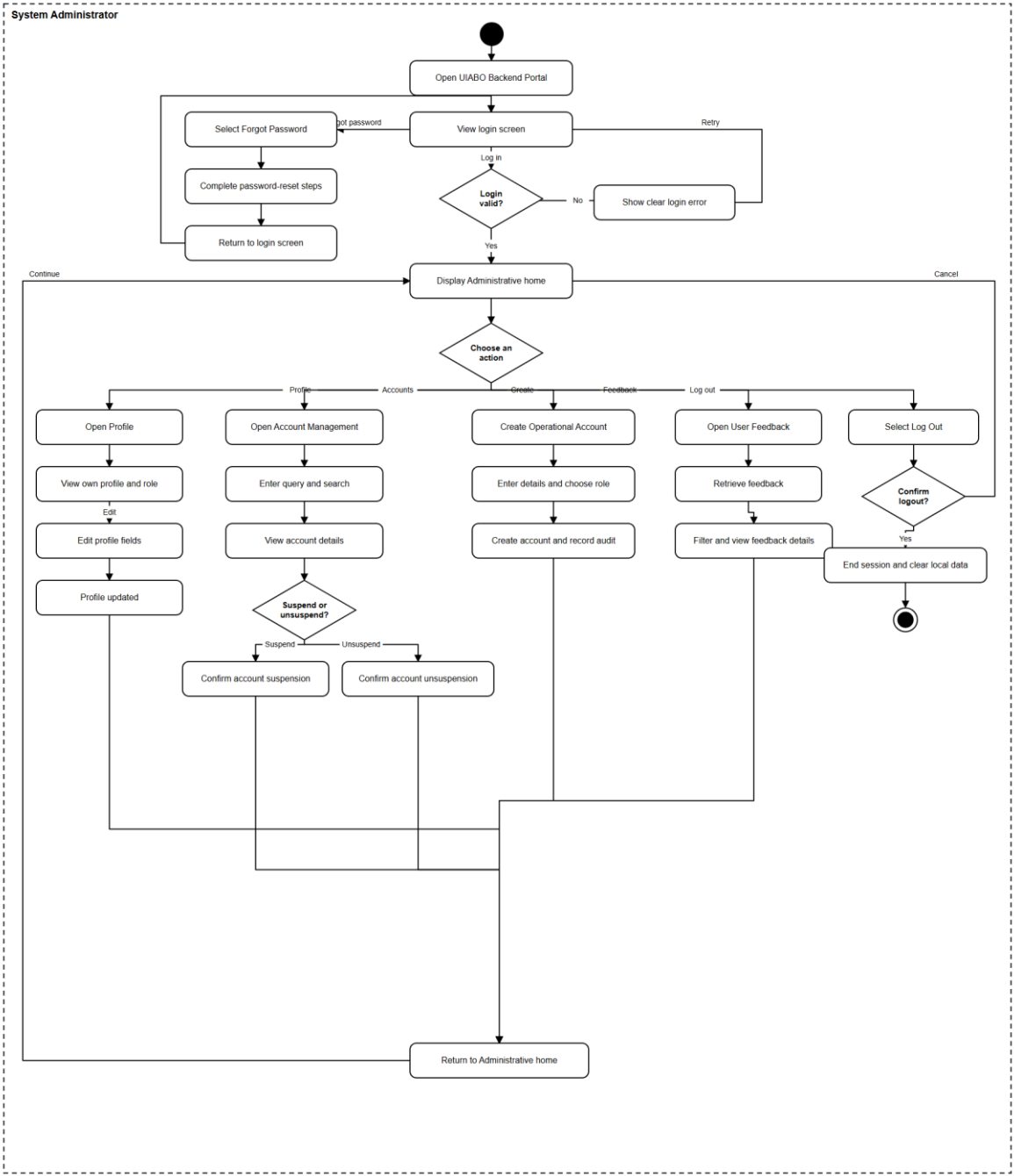


Monitor Data-Ingestion Pipelines (Data Engineer)

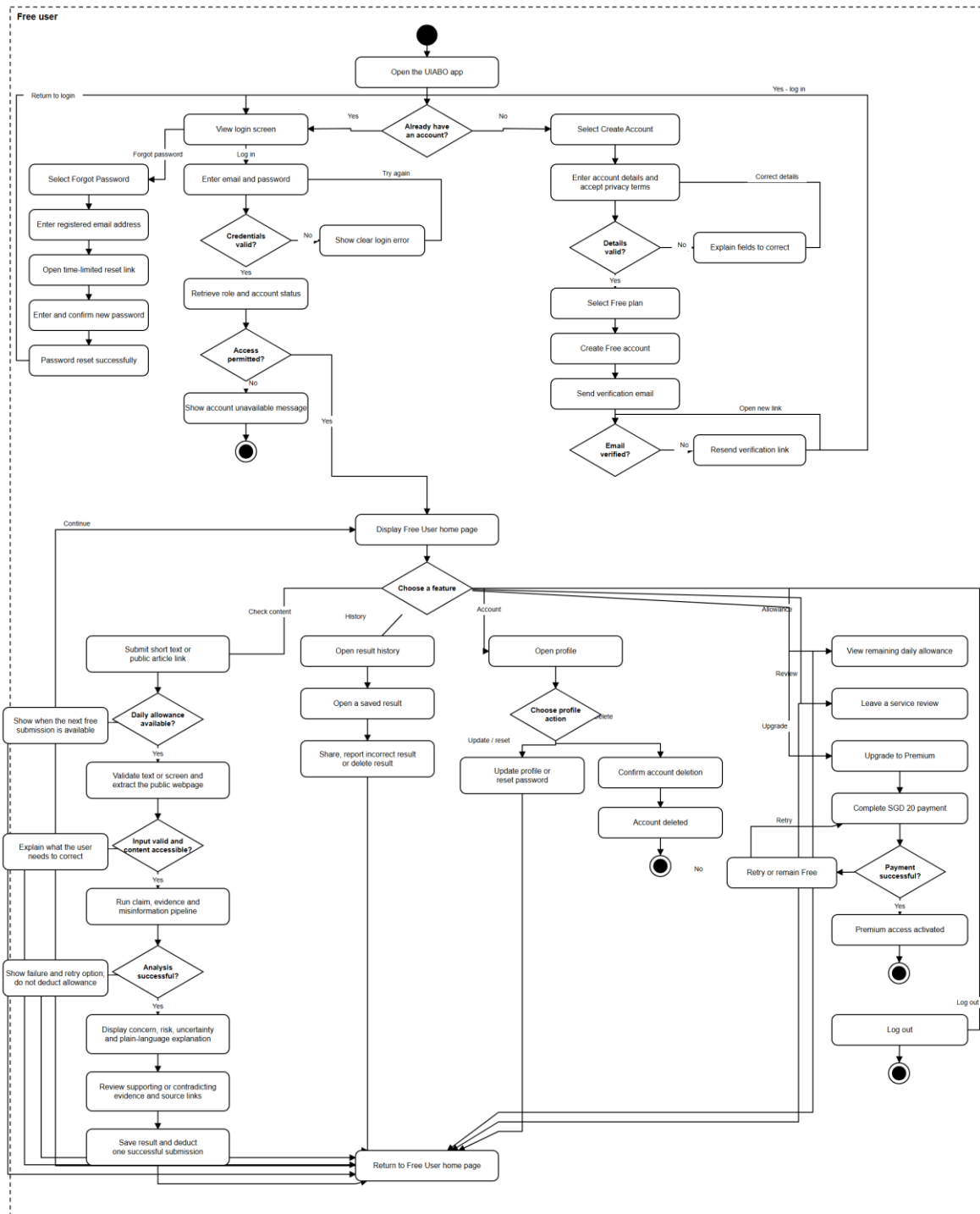


5.4.2 Activity Diagrams

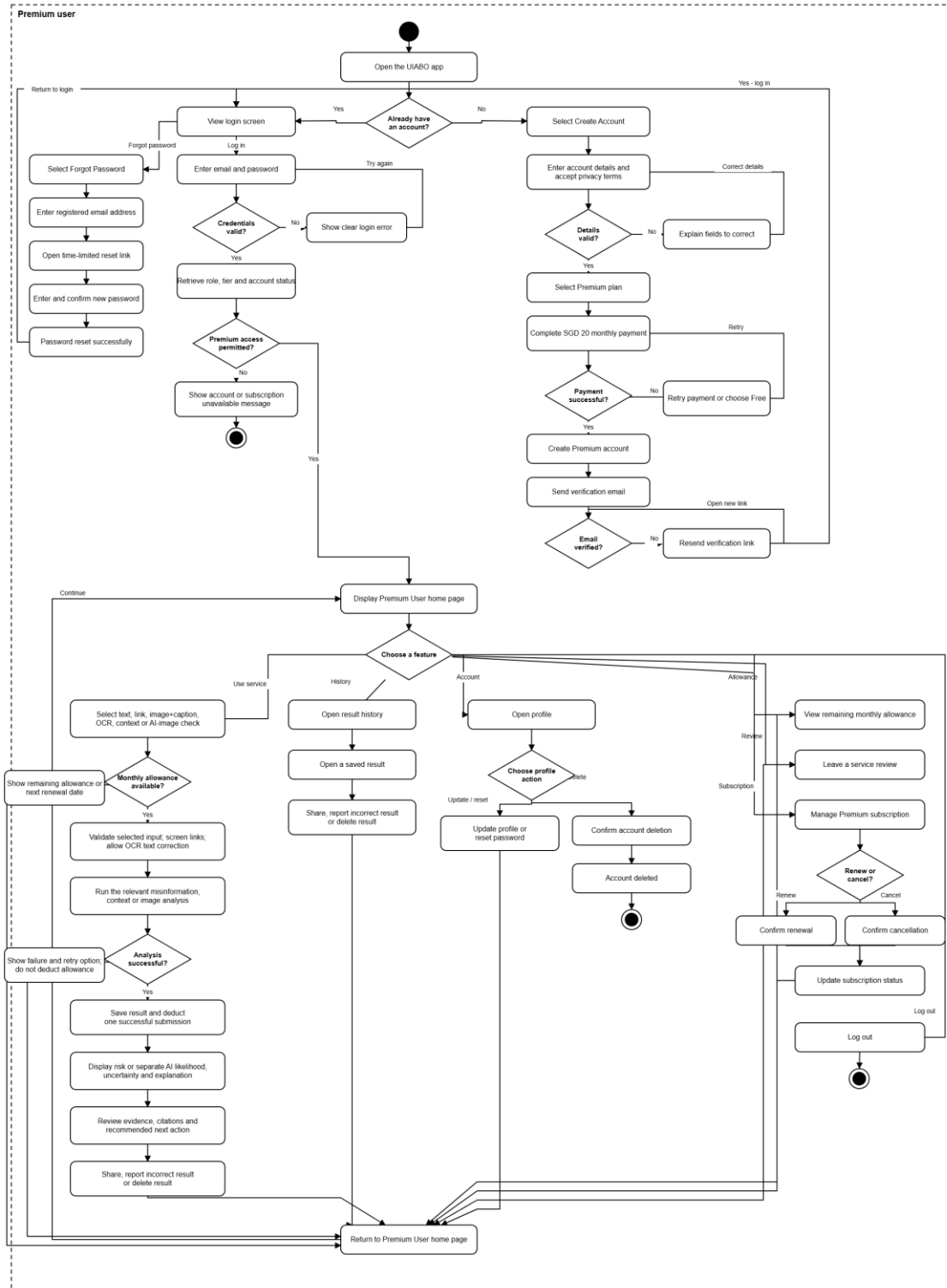
5.4.2.1 System Administrator



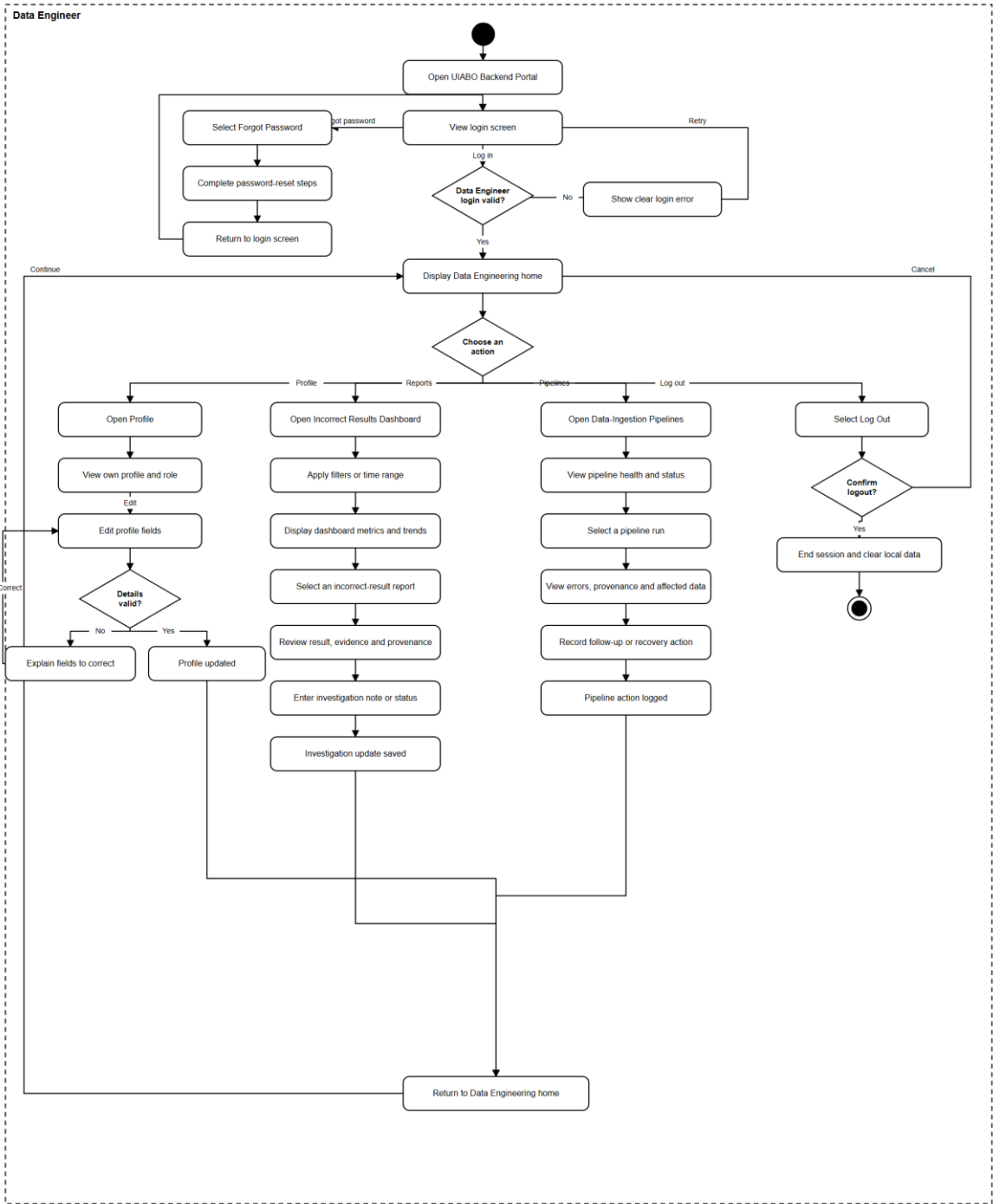
5.4.2.2 Free User



5.4.2.3 Premium User



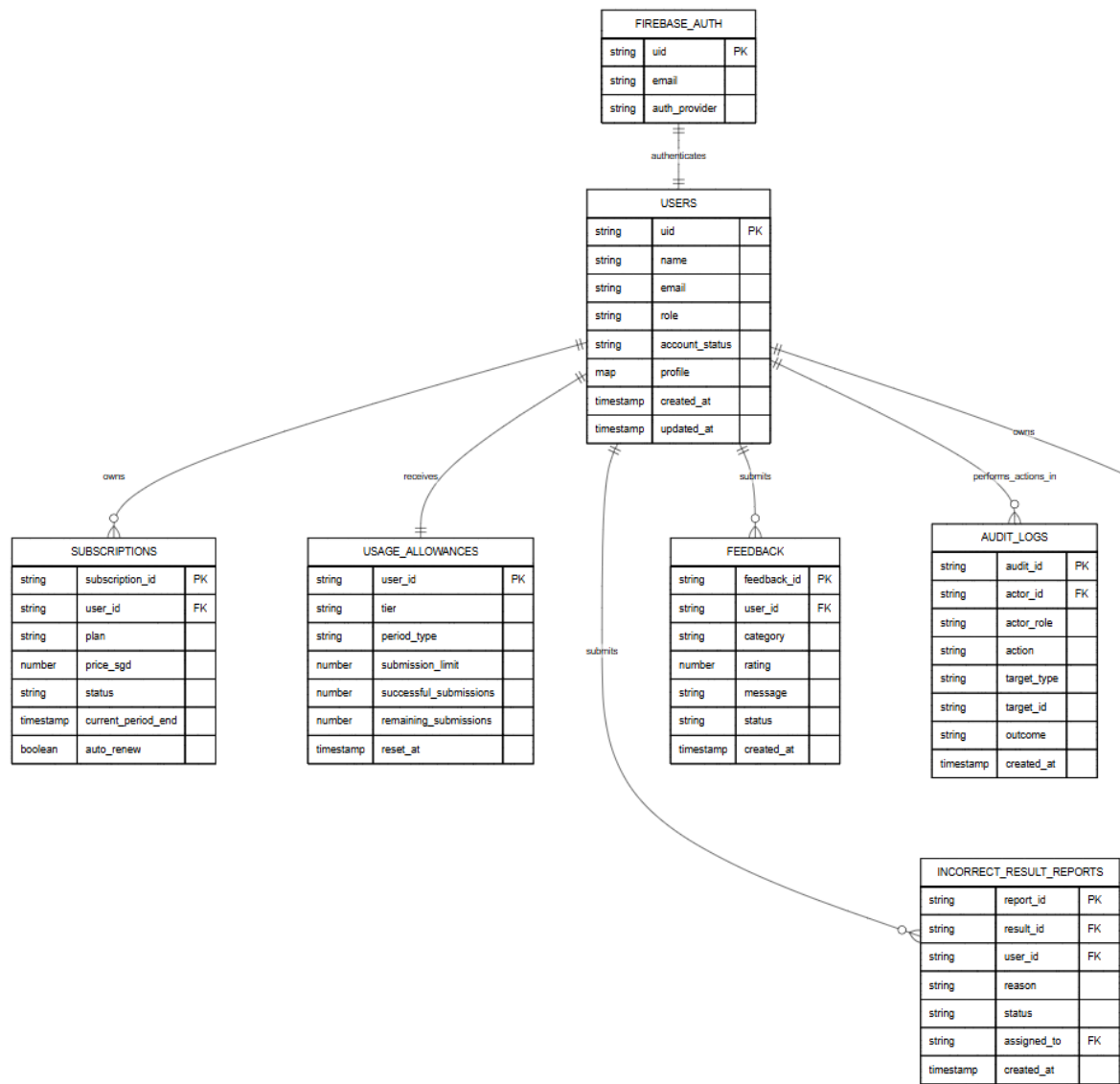
5.4.2.4 Data Engineer

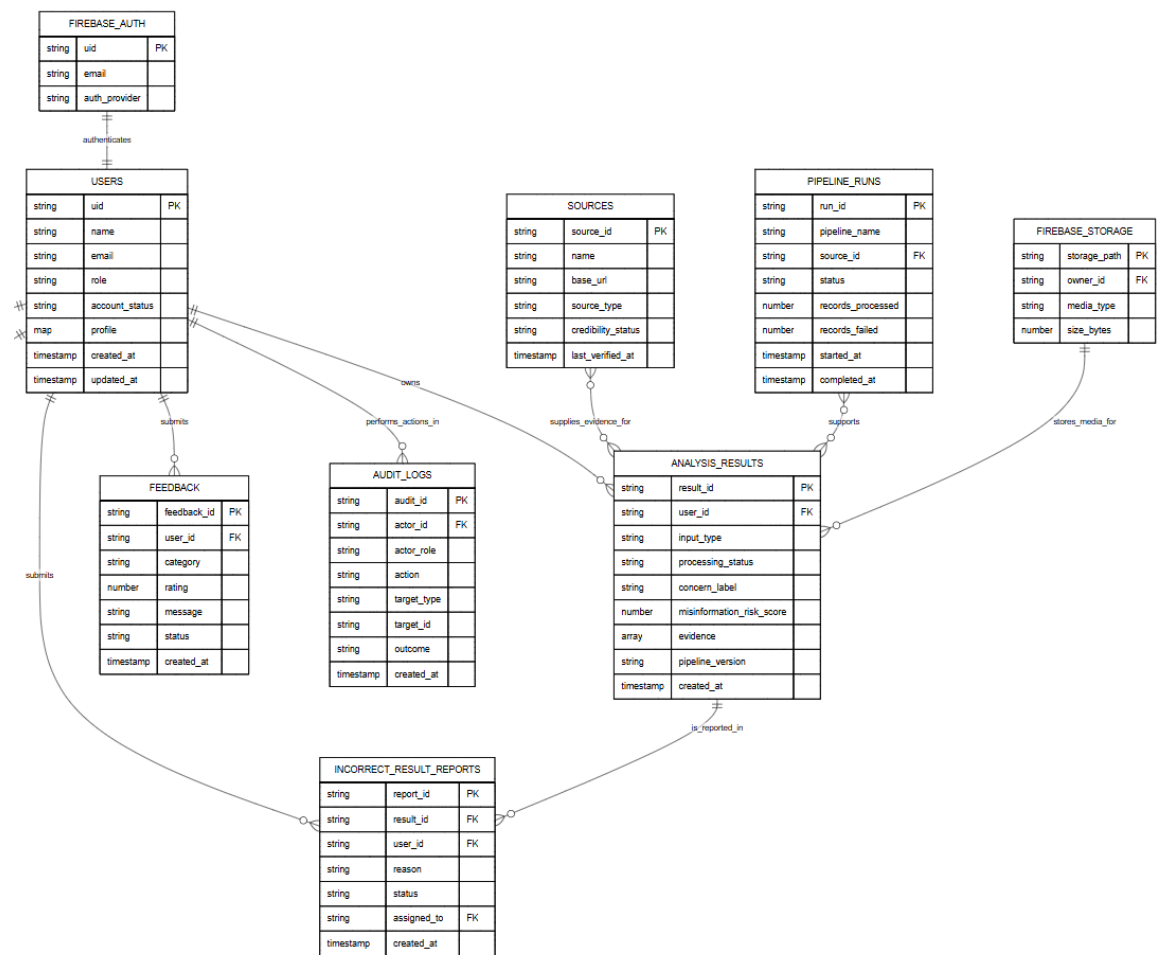


5.5 Database Design

Current consumer persistence uses users, usage_allowances, analysis_results and an implementation-level analysis_locks collection. Subscription, report, feedback, source-catalogue, ingestion-run and operational-audit schemas below remain proposed. The runtime source catalogue is currently a code allowlist, not an operational Firestore sources editor.

5.5.1 Firestore Database Diagram





Cloud Firestore is a NoSQL document database. The lines in the diagram show document-ID references rather than SQL joins. Evidence snapshots are embedded inside analysis_results to match the current pipeline implementation and preserve the evidence used for each result.

5.5.2. Users Collection

Document path: users/{uid}

Stores the Firestore profile for every Free User, Premium User, System Administrator and Data Engineer. The document ID is the Firebase Authentication UID. Passwords are not stored here.

Field	Type	Required	Purpose
uid	String	Yes	Firebase Authentication UID and Firestore document ID.
name	String	Yes	User's display name.
email	String	Yes	Email copied from Firebase Authentication for profile display and authorised account queries.
role	String	Yes	Access role: free, premium, system_admin or data_engineer.
account_status	String	Yes	Account state: active, suspended or deactivated.
profile	Map	Yes	Editable profile information. Only data needed by the application should be stored.
storage_used_bytes	Number	Yes	Current personal storage usage used for the 80% warning.
storage_quota_bytes	Number	Yes	Maximum personal storage allocation.
last_login_at	Timestamp	No	Most recent successful login time.

Field	Type	Required	Purpose
created_at	Timestamp	Yes	When the profile record was created.
updated_at	Timestamp	Yes	When the profile record was last changed.

5.5.3. Subscriptions Collection

Document path: subscriptions/{subscription_id}

Stores premium subscription periods and renewal state. Payment-card details must remain with the payment provider and must not be stored in Firestore.

Field	Type	Required	Purpose
subscription_id	String	Yes	Unique subscription document ID.
user_id	String	Yes	UID of the user who owns the subscription.
plan	String	Yes	Plan identifier, such as premium_monthly.
price_sgd	Number	Yes	Recorded monthly price; SGD 20 for the prototype.
status	String	Yes	pending, active, cancelled, expired or payment_failed.
current_period_start	Timestamp	Yes	Start of the current paid period.
current_period_end	Timestamp	Yes	End of the current paid period.
auto_renew	Boolean	Yes	Whether automatic renewal remains enabled.
payment_provider	String	Yes	Provider or app-store mechanism used for payment.
provider_reference	String	No	Non-sensitive provider transaction or subscription reference.
created_at	Timestamp	Yes	When the subscription record was created.
updated_at	Timestamp	Yes	When the subscription record was last updated.

5.5.4. Usage Allowances Collection

Document path: usage_allowances/{user_id}

Stores the current free daily or premium monthly allowance. One successful Free User submission is allowed per day; Premium Users receive up to 60 successful submissions per month.

Field	Type	Required	Purpose
user_id	String	Yes	User UID and document ID.
tier	String	Yes	free or premium.
period_type	String	Yes	daily for free access or monthly for premium access.
period_start	Timestamp	Yes	Beginning of the current allowance period.
period_end	Timestamp	Yes	End of the current allowance period.
submission_limit	Number	Yes	Maximum successful submissions for this period: 1 or 60.
successful_submissions	Number	Yes	Number of completed analyses charged during the period.
remaining_submissions	Number	Yes	Remaining amount displayed in the application.
reset_at	Timestamp	Yes	When the allowance will reset.
updated_at	Timestamp	Yes	When the counter was last changed.

5.5.5. Analysis Results Collection

Document path: analysis_results/{result_id}

Stores completed and failed pipeline records. Result history is retrieved by querying user_id and created_at. Evidence is embedded as an array so the explanation preserves the evidence that was used when the result was generated.

Field	Type	Required	Purpose
result_id	String	Yes	Unique analysis result and document ID.
user_id	String	Yes	UID of the user who owns the result.
input_type	String	Yes	text, link, image_caption, ocr, context or deepfake.
original_text	String	No	Submitted text, caption or corrected OCR text.
submitted_url	String	No	Public webpage URL for a link analysis.
media_storage_path	String	No	Firebase Storage path for an authorised image submission.
processing_status	String	Yes	completed or failed.
extracted_claim	String	No	Factual claim identified by the pipeline.
claim_category	String	Yes	factual, opinion, satire, prediction, personal experience or unverifiable.
checkable	Boolean	Yes	Whether the identified claim can be fact-checked.
concern_label	String	No	Low Concern, Needs Caution, High Concern or Not Enough Information.
misinformation_risk_score	Number	No	Risk indicator from 0 to 100; null when information is insufficient.
ai_manipulation_likelihood	Number	No	Premium image AI-generation or manipulation likelihood from 0 to 100.
uncertainty	String	No	Low, Medium or High uncertainty.
uncertainty_reasons	Array	No	Plain-language reasons for uncertainty.
explanation	String	No	Plain-language result explanation.
recommended_action	String	No	Suggested next action for the user.
evidence	Array<Map>	No	Evidence snapshots containing source, URL, passage, stance and quality score.
warnings	Array	No	Non-fatal preparation, retrieval or assessment warnings.
pipeline_version	String	Yes	Version of the pipeline that generated the record.
failure	Map	No	Failure stage, error code, safe message and retryable flag for failed runs.
share_token	String	No	Random token used when result sharing is enabled.
share_enabled	Boolean	Yes	Whether a shareable result link is active.
created_at	Timestamp	Yes	When analysis processing finished or failed.
saved_at	Timestamp	Yes	Server timestamp written by Firestore.

5.5.6. Incorrect Result Reports Collection

Document path: incorrect_result_reports/{report_id}

Stores reports submitted from an analysis result and the Data Engineer's investigation state.

Field	Type	Required	Purpose
report_id	String	Yes	Unique report document ID.
result_id	String	Yes	Analysis result being reported.
user_id	String	Yes	UID of the reporting user.
reason	String	Yes	Selected reason or report category.
user_comment	String	No	Optional explanation supplied by the user.
status	String	Yes	new, investigating, resolved or rejected.
priority	String	Yes	low, medium or high investigation priority.
assigned_to	String	No	UID of the assigned Data Engineer.
investigation_notes	Array<Map>	No	Timestamped investigation notes and permitted status updates.
resolution_summary	String	No	Final investigation outcome without unnecessary personal information.
created_at	Timestamp	Yes	When the user submitted the report.
updated_at	Timestamp	Yes	When the report was last changed.
resolved_at	Timestamp	No	When the investigation was closed.

5.5.7. Feedback Collection

Document path: feedback/{feedback_id}

Stores application reviews, accessibility feedback, bug reports and general comments for System Administrator review.

Field	Type	Required	Purpose
feedback_id	String	Yes	Unique feedback document ID.
user_id	String	Yes	UID of the user who submitted the feedback.
category	String	Yes	review, accessibility, bug or general.
rating	Number	No	Optional service rating, normally from 1 to 5.
message	String	Yes	Feedback text entered by the user.
status	String	Yes	new, reviewed or resolved.
reviewed_by	String	No	UID of the reviewing System Administrator.
admin_note	String	No	Operational follow-up note.
created_at	Timestamp	Yes	When the feedback was submitted.
updated_at	Timestamp	Yes	When the feedback record was last changed.

5.5.8. Sources Collection

Document path: sources/{source_id}

Stores trusted-source and provenance metadata used by evidence retrieval and maintained by authorised Data Engineers.

Field	Type	Required	Purpose
source_id	String	Yes	Unique source document ID.
name	String	Yes	Publisher or organisation name.
base_url	String	Yes	Canonical source website URL.

Field	Type	Required	Purpose
source_type	String	Yes	fact_check, government, academic, news or other.
credibility_status	String	Yes	trusted, review_required or blocked.
country	String	No	Country or region associated with the source.
provenance_notes	String	No	Reason for its status and relevant validation notes.
active	Boolean	Yes	Whether the retrieval pipeline may currently use the source.
last_verified_at	Timestamp	Yes	Most recent source validation time.
created_at	Timestamp	Yes	When the source record was created.
updated_at	Timestamp	Yes	When source metadata was last changed.

5.5.9. Pipeline Runs Collection

Document path: pipeline_runs/{run_id}

Stores the monitoring data needed to identify running, delayed, incomplete and failed ingestion updates.

Field	Type	Required	Purpose
run_id	String	Yes	Unique pipeline run document ID.
pipeline_name	String	Yes	Name of the ingestion or update pipeline.
source_id	String	No	Related source record when the run imports source data.
status	String	Yes	running, completed, delayed, incomplete or failed.
records_expected	Number	No	Expected number of records, when known.
records_processed	Number	Yes	Number of successfully handled records.
records_failed	Number	Yes	Number of failed records.
data_range	Map	No	Affected dates, record range or dataset version.
error_summary	String	No	Safe summary of the run failure or delay.
triggered_by	String	Yes	system or authorised operational-user UID.
started_at	Timestamp	Yes	When the run started.
completed_at	Timestamp	No	When the run ended.
updated_at	Timestamp	Yes	Most recent monitoring update.

5.5.10. Audit Logs Collection

Document path: audit_logs/{audit_id}

Stores immutable records of sensitive operational actions such as account queries, account views, account creation, suspension, unsuspension and investigation updates.

Field	Type	Required	Purpose
audit_id	String	Yes	Unique audit record ID.
actor_id	String	Yes	UID of the operational user or system process.
actor_role	String	Yes	system_admin, data_engineer or system.
action	String	Yes	Action that was attempted.
target_type	String	Yes	Collection or resource type affected.

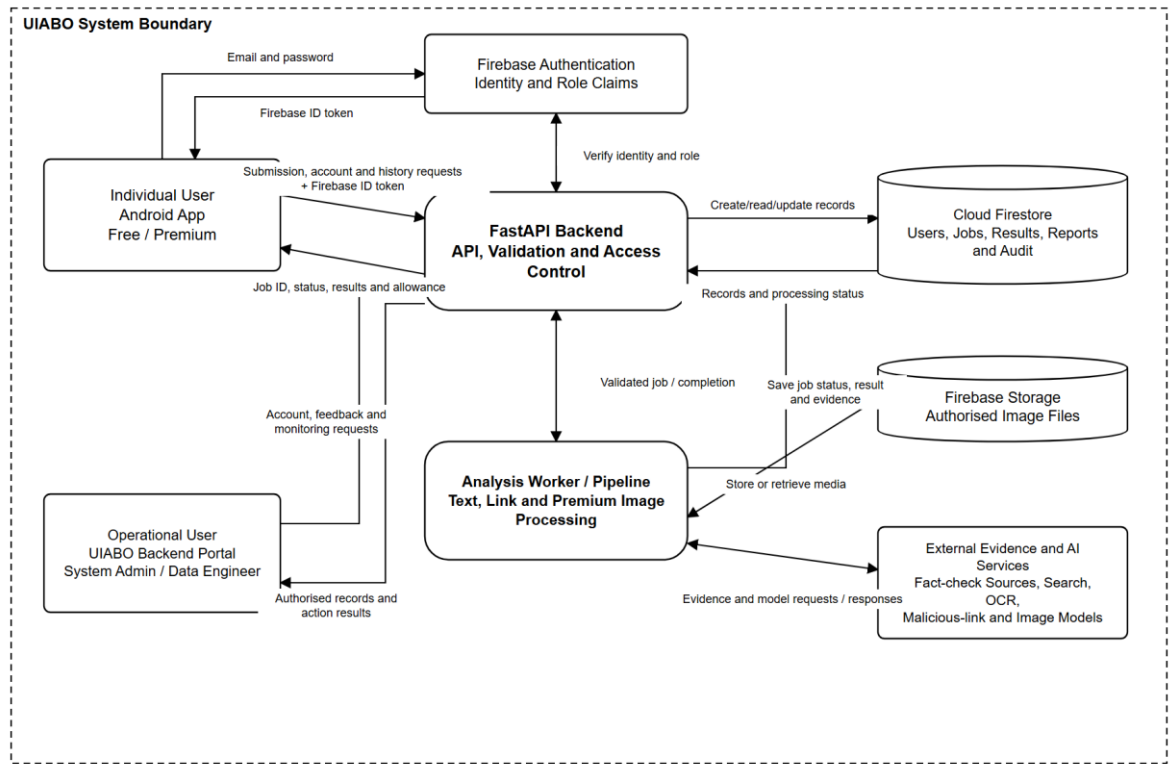
Field	Type	Required	Purpose
target_id	String	No	Identifier of the affected record.
reason	String	No	Required reason for sensitive changes where applicable.
outcome	String	Yes	success, denied or failed.
metadata	Map	No	Non-sensitive context needed for security or investigation.
created_at	Timestamp	Yes	Server timestamp. Audit records are not edited after creation.

5.5.11. Supporting Firebase Services

These services support the database but are not Firestore collections.

Service	Responsibility
Firestore Authentication	Stores email-and-password credentials, creates the UID and manages authenticated sessions. Password hashes must not be copied into Firestore.
Firestore Storage	Stores uploaded images and screenshots. Firestore keeps only authorised storage paths and media metadata.
Cloud Firestore	Stores the document collections described above. Security rules and backend role checks restrict every read and write.

5.6 Data Flow Diagram



5.7 User Interface Design

These figures are the submitted TDM design mock-ups. Section 5.8 shows the available Android captures; unfinished image, payment and portal screens must not be described as deployed.

5.7.1 Login and Account Creation

Login Page

9:41

uiabo.

Welcome back

Log in to check suspicious content and review your saved results.

Email address

alex.tan@example.com

Password

.....

Forgot password?

Log in

New to UIABO?

Create an account

One free check every day.

Free accounts can analyse text and supported public links.

By continuing, you agree to the Terms of Use and Privacy Notice.

Sign Up Page

9:41

<

Create account

CHOOSE YOUR ACCESS

Start checking with UIABO

You can change your plan later.

Free

1 text or link check daily

Premium • SGD 20

60 checks monthly + images

Full name

Enter your name

Email address

name@example.com

Create password

At least 8 characters

Use at least 8 characters with a number.

☒ I agree to the Terms of Use and Privacy Notice.

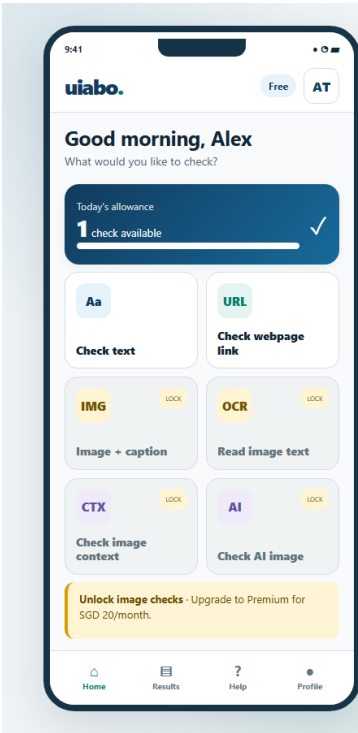
Continue to payment

Premium is SGD 20 per month. This interface is a visual sample.

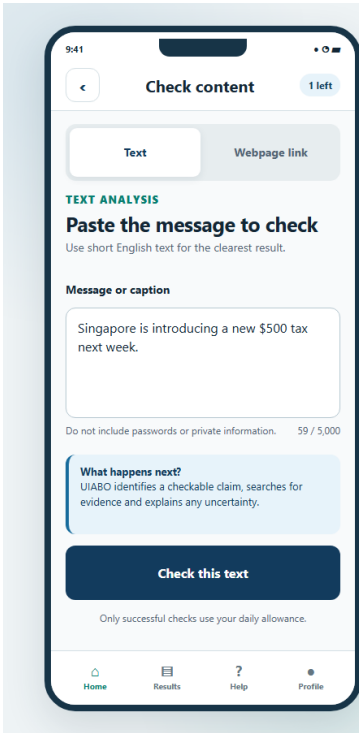
Working draft 0.2 | Page 134

5.7.2 Free User

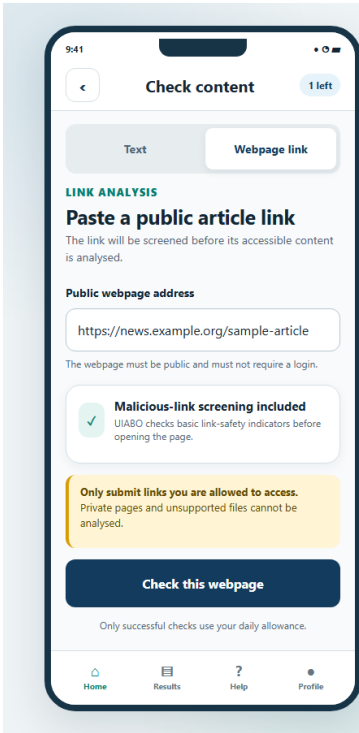
Home Page
(Premium features locked)



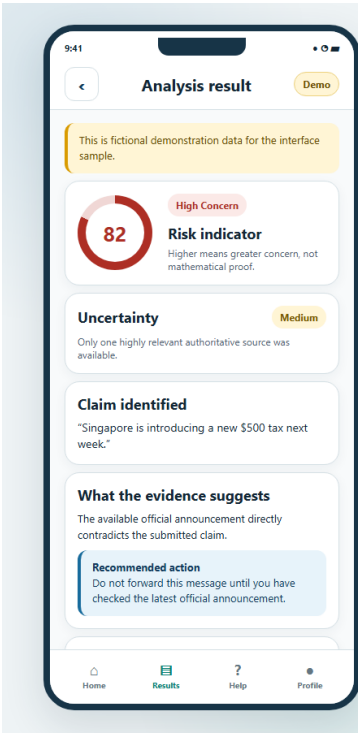
Text Analysis



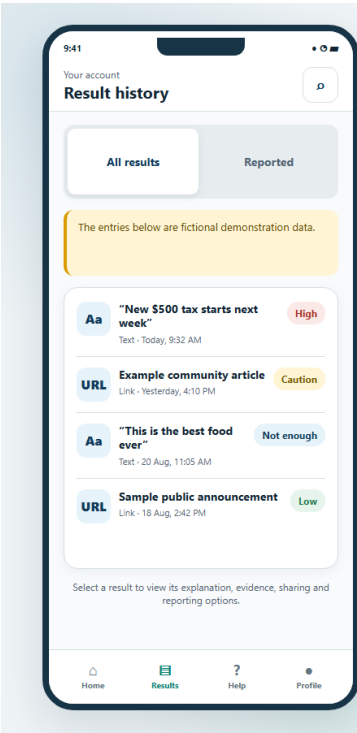
Link Analysis



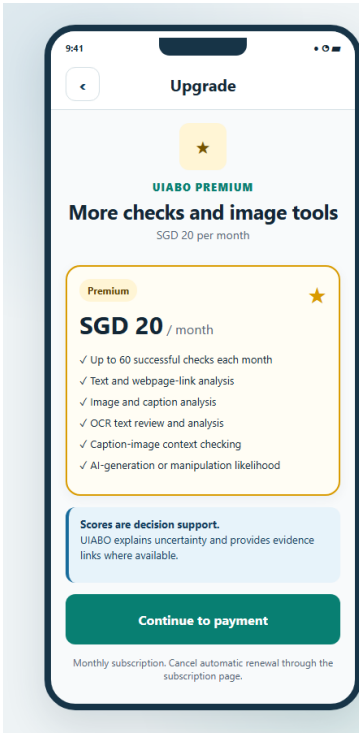
Result Page



Result History



Upgrade to Premium



5.7.3 Premium User

Home Page

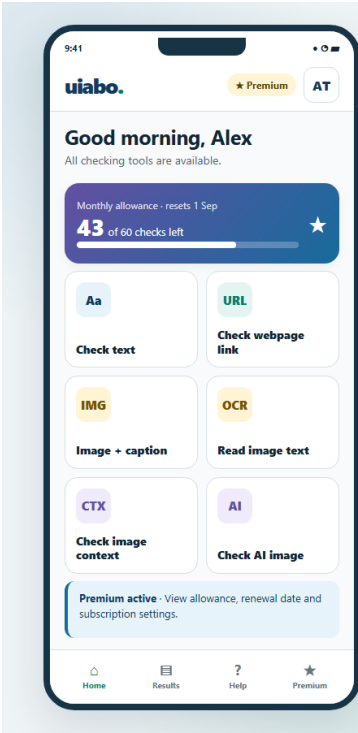
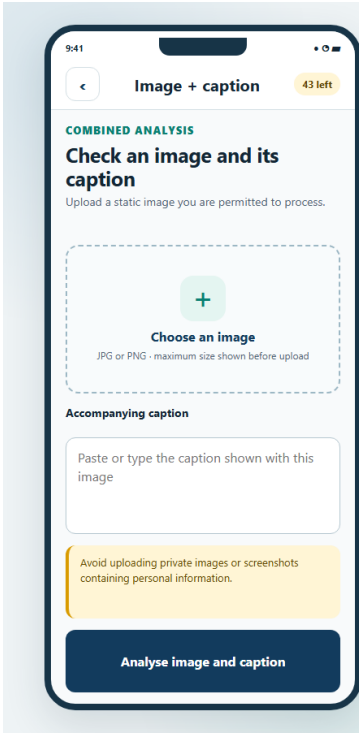
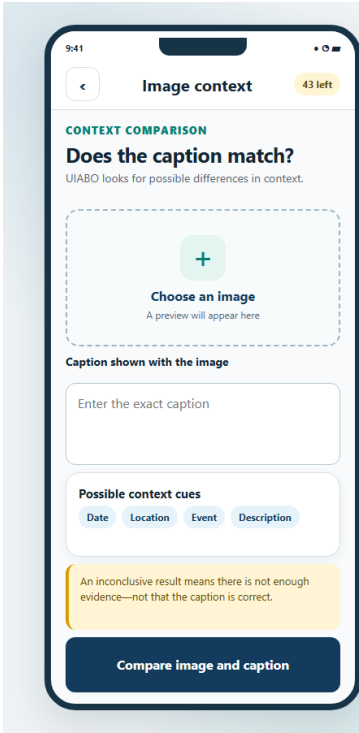


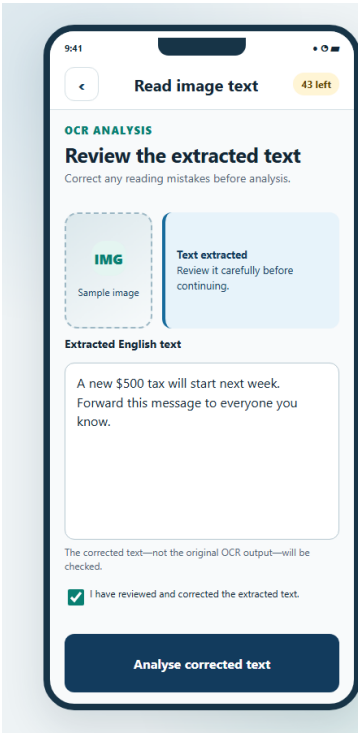
Image Caption Analysis



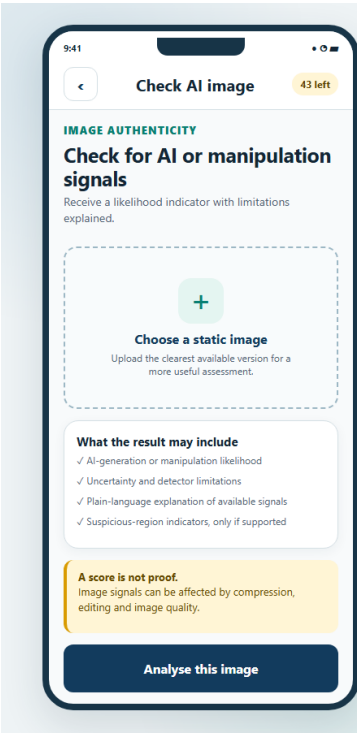
Context Analysis



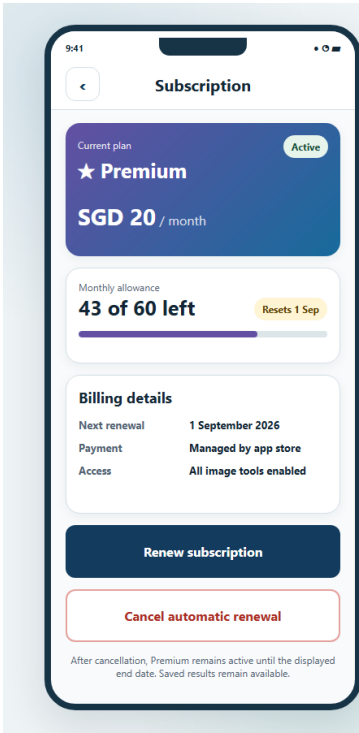
OCR Analysis



Deepfake Analysis



Subscription



5.7.4 System Administrator

System Administrator Dashboard

uiabo.

Operational Portal

Dashboard

USR Accounts

+ Create account

MSG User feedback

My profile

Sam Lee

System Administrator

System Administrator

Interface sampleSL

ACCOUNT OPERATIONS

Good morning, Sam

Review access, operational accounts and user feedback.

Total accounts

1,284

+ 34 this month

Premium accounts

318

24.8% of users

Suspended accounts

7

2 require review

Unread feedback

12

Oldest: 2 days

Recently updated accounts

View all

USER	ROLE / TIER	STATUS	LAST UPDATED
Alex Tan alex.tan@example.com	Free user	Active	Today, 9:18 AM
Mei Ling mei.ling@example.com	Premium user	Active	Yesterday
Jordan Lim jordan.lim@example.com	Free user	Suspended	22 Aug 2026

Feedback requiring review

12 unread

ATDifficulty reading evidence text

Accessibility - 18 minutes ago

New

MLLink check was easy to follow

Review - Yesterday

4/5

RKRequest for larger buttons

Accessibility - 2 days ago

New

Open feedback page

Account Management

uiabo.

Operational Portal

Dashboard

USR Accounts

+ Create account

MSG User feedback

My profile

Sam Lee

System Administrator

Account management

Interface sampleSL

AUTHORISED ACCESS ONLY

Find and manage accounts

+ Create operational account

View permitted account details and change access status.

Search by name, email or user ID

All roles

All statuses

Search

Account results

1,284 accounts

NAME	USER ID	ROLE / TIER	STATUS	LAST LOGIN	ACTION
Alex Tan alex.tan@example.com	usr_1028	Free user	Active	Today, 9:18 AM	View
Mei Ling mei.ling@example.com	usr_1027	Premium user	Active	Yesterday	View
Jordan Lim jordan.lim@example.com	usr_0994	Free user	Suspended	20 Aug 2026	View
Nora Ong nora.ong@uiabo.example	opi_0032	Data engineer	Active	Today, 8:02 AM	View
Ben Goh ben.goh@example.com	usr_0988	Premium user	Review	18 Aug 2026	View

Account access: suspending an account blocks access without deleting the account record. Every status change should be audited.

Create Operational Account

uiabo.

Operational Portal

Dashboard

USR Accounts

Create account

MSG User feedback

My profile

Sam Lee
System Administrator

Create operational account

Interface sampleSL

RESTRICTED ADMINISTRATOR FUNCTION

New operational user

Create an authorised system-administrator or data-engineer account.

Account information

Approval required

Full name

Enter legal name

Work email

name@uiabo.example

Operational role

Data Engineer

Initial status

Pending activation

Business reason

Explain why this operational access is required

☐ I have verified that this person is authorised to receive operational access.

Cancel

Create and send invite

!

Access is role-specific

System administrators manage accounts and feedback. Data engineers investigate incorrect results and monitor data-ingestion pipelines.

Security reminder

Use the minimum necessary role. Operational accounts must not be shared.

After creation

1. The user receives an invitation.
2. Email ownership is verified.
3. The account remains auditable.

User Feedback

uiabo.

Operational Portal

Dashboard

USR Accounts

Create account

User feedback

My profile

Sam Lee
System Administrator

User feedback

Interface sampleSL

REVIEWS AND EXPERIENCE

Feedback from UIABO users

Review comments about accessibility, usability and the service experience.

Export summary

Average review

4.2/5

Based on demo data

Unread

12

Requires review

Accessibility

8

This month

Resolved

37

Last 30 days

Latest feedback

All categories

AT

Some evidence text is difficult to read

Accessibility - "Please make the source summary larger."

New

ML

The link checker was easy to follow

Review - Rated 4 out of 5

Review

RK

Request for clearer locked-feature labels

Usability - "Tell me why I cannot open the image tool."

New

JL

Result explanation was useful

Review - Rated 5 out of 5

Reviewed

Selected feedback

Some evidence text is difficult to read

"The main result is clear, but the evidence summary and source name are smaller than I expected. Please make them easier to read."

Category

Accessibility

Submitted

24 Aug 2026, 9:12 AM

User tier

Free

Status

New

Mark reviewed

Add note

5.7.5 Data Engineer

Data Engineer Dashboard

uiabo.
Operational Portal

Dashboard

RPT Incorrect results

PLN Pipeline monitoring

My profile

Nora Ong
Data Engineer

Data Engineer

Interface sampleNO

DATA QUALITY OPERATIONS

Pipeline and report overview

Investigate reported results and monitor evidence-data updates.

Open incorrect-result reports

18

5 high priority

Reports reviewed today

7

+2 from yesterday

Healthy pipelines

5/6

1 delayed

Evidence freshness

96%

Within target

View pipeline health

Incorrect-result reports

View report

REPORT	RESULT TYPE	CONCERN	REPORTED	STATUS
RPT-1048	Text	High	24 Aug. 8:48 AM	New
RPT-1047	Link	Medium	23 Aug. 4:20 PM	Reviewing
RPT-1046	Text	Low	23 Aug. 2:03 PM	Resolved

Pipeline health

1 delayed

FC

Fact-check ingestion

Last update: 9:05 AM

Healthy

TS

Trusted-source updates

Expected 8:30 AM

Delayed

EV

Evaluation records

Last update: Yesterday

Healthy

Open monitoring

Incorrect Result Report

uiabo.
Operational Portal

Dashboard

RPT Incorrect results

PLN Pipeline monitoring

My profile

Nora Ong
Data Engineer

Incorrect-result report

Demo dataNO

RPT-1048

Review reported analysis result

Investigate the result, evidence and recorded pipeline information.

Assign to me

Update status

User report

New · High priority

"I believe the announcement used in this result may be outdated. A newer official page appears to say something different."

Reported

24 Aug 2026, 8:48 AM

Reason

Evidence may be outdated

Result ID

res_demo_1048

Input type

Short-form text

Original result

High Concern - 82/100

Identified claim

"Singapore is introducing a new \$500 tax next week."

Original explanation

The available official announcement directly contradicted the submitted claim.

Evidence and provenance

1 source

SOURCE	PUBLISHED	RETRIEVED	STANCE
Example official announcement example.gov/announcement	10 Aug 2026	20 Aug 2026	Contradicting

Investigation status

New

Investigation notes

Record findings and follow-up actions

Save notes

Pipeline record

Version

sprint-1-v1

Completed

20 Aug. 10:00 AM

Uncertainty

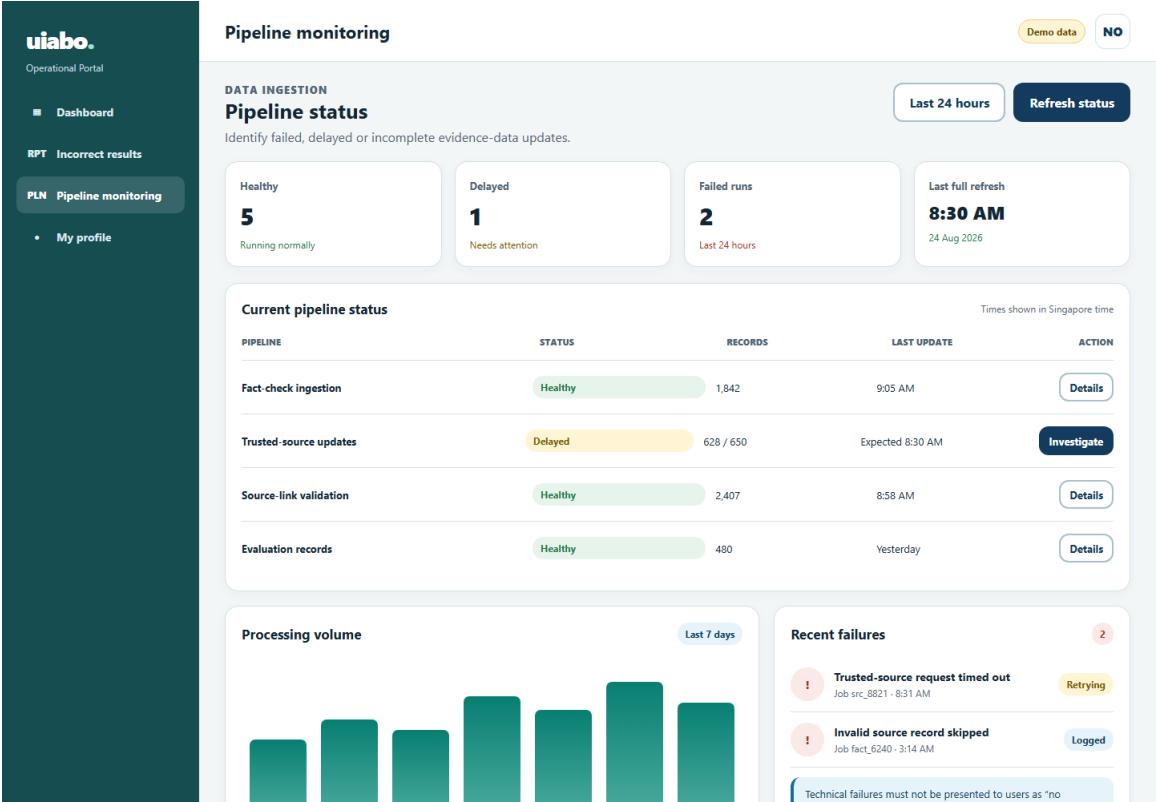
Medium

Warnings

One source found

Privacy: view only the information required to investigate this report.

Pipeline Monitoring



5.8 Current Android implementation

The current Home follows the TDM grid: Check text, Check webpage link, Image + caption, Read image text, Check image context and Check AI image. There is no audio placeholder. Only text analysis is connected; the link tab offers copied text and the image cards remain unavailable.

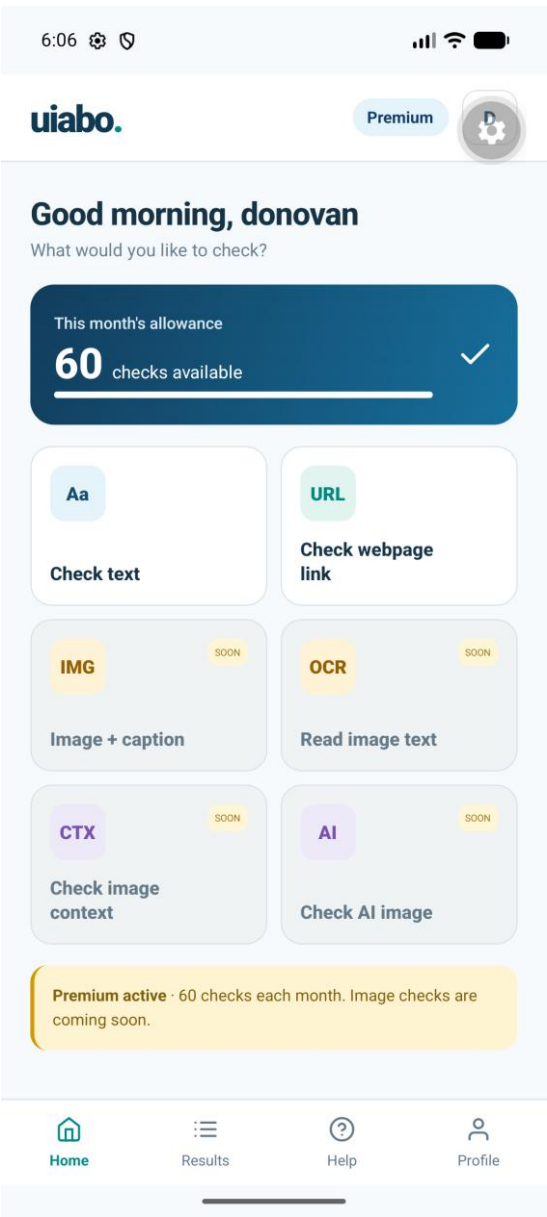


Figure 5.8a. Premium Home and monthly allowance.

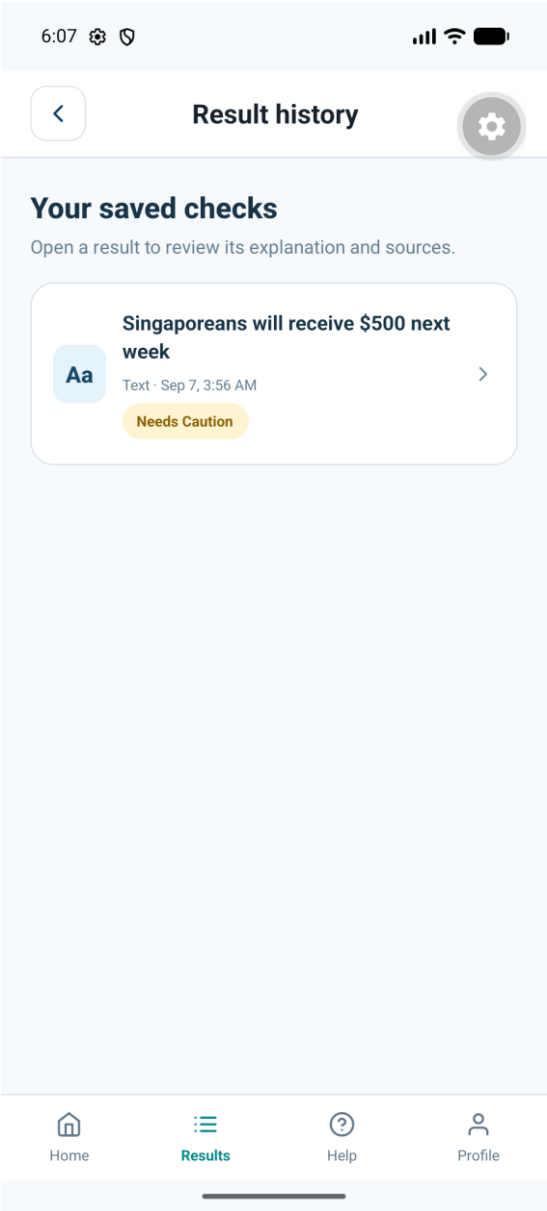


Figure 5.8b. Private result history.

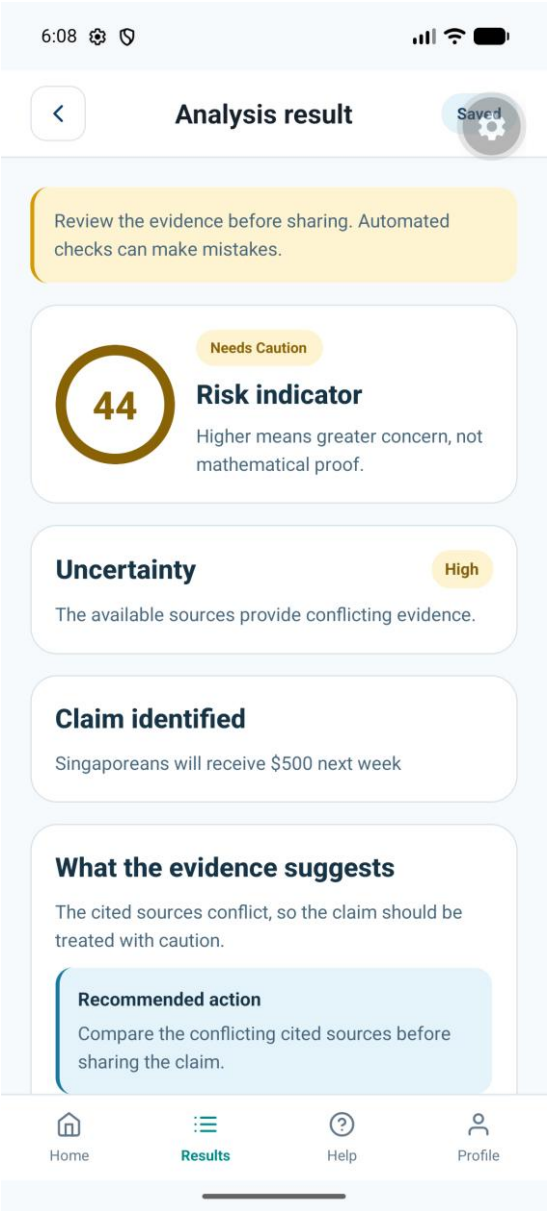


Figure 5.8c. Saved assessment view; this is a UI example, not a validated verdict.

Evidence / source: Existing Android emulator captures from 7 September 2026. Development overlay and account values are snapshot details; Premium grant did not create billing.

5.9 Current authentication, persistence and retry design

The Android client obtains a Firebase ID token and sends it to FastAPI. The backend verifies the token, email-verification state and active account, then derives the owner and tier from server-side records. It does not accept a client-selected user_id, role or allowance. Result/history reads use authenticated API routes.

Step	Current behavior
Reserve	An idempotency key and text fingerprint identify the submission. A bounded per-user lease prevents concurrent requests from consuming the same allowance.
Analyse	The pipeline runs with request-local intermediate storage and validated component handoffs.
Commit	A transaction saves the authoritative owned result and decrements allowance for a completed analysis together. Failed analyses are saved without a charge.
Retry	A successful repeated request with the same key/text returns its existing result. Reusing a key for different text is rejected.
Read history	Query records belonging to the user, sort by creation time and result ID, then paginate. Other users cannot retrieve the record through the API.

Free counters reset each Singapore calendar day; premium counters reset each Singapore calendar month. A completed Not Enough Information result consumes one check. A client connection timeout may happen after server completion, so the app directs users to inspect saved history before starting a new request.

Current Firestore additions include analysis_locks/{uid}, request_fingerprint and owner metadata on analysis_results. The history implementation reads the user’s matching records and sorts them in the backend. A proposed composite index exists in firestore.indexes.json but was not deployed because the current service account lacks index-administration permission; large histories need query optimization.

Evidence / source: backend/app/analyses/store.py; accounts/repository.py; routers/analysis.py; Firebase ID-token verification and transaction documentation listed in Section 13.

6. Project Implementation

6.1 Software Development Methodology

Waterfall Model

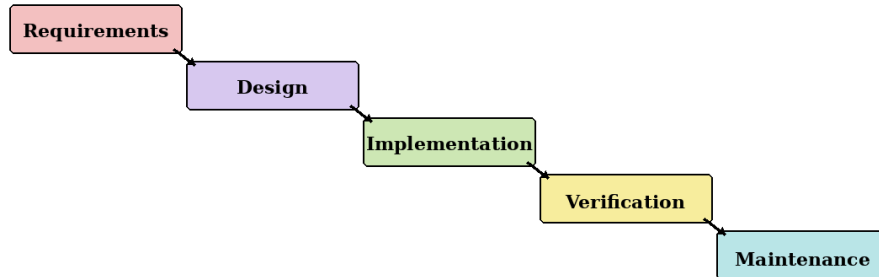


Figure 4.1: Waterfall model.

Waterfall follows a fixed sequence from requirements to maintenance. Although it offers clear stages, it is unsuitable as the main approach because AI models and data sources may require repeated changes. It will only be used for stable tasks such as final documentation and submission checks.

Prototyping Model

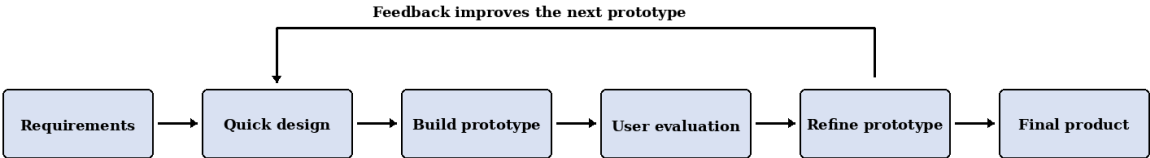


Figure 4.2: Prototyping model.

Prototyping allows the team to test claim extraction, evidence retrieval, scam detection and AI-authorship scoring before integrating them. To avoid endless changes, all prototypes will be compared using the same test dataset and only useful improvements will be retained.

Kanban Model

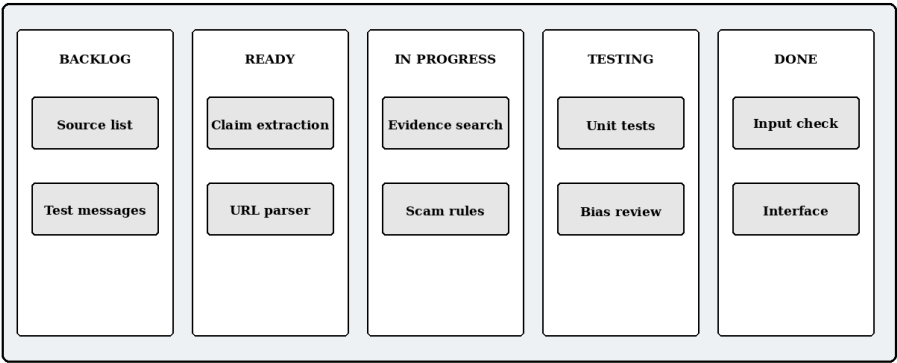


Figure 4.3: Example Kanban task board.

Kanban provides a visual view of tasks in Backlog, Ready, In Progress, Testing and Done. It supports daily coordination and limits unfinished work, but it lacks fixed review dates. Therefore, it will support Scrum rather than replace it.

Scrum

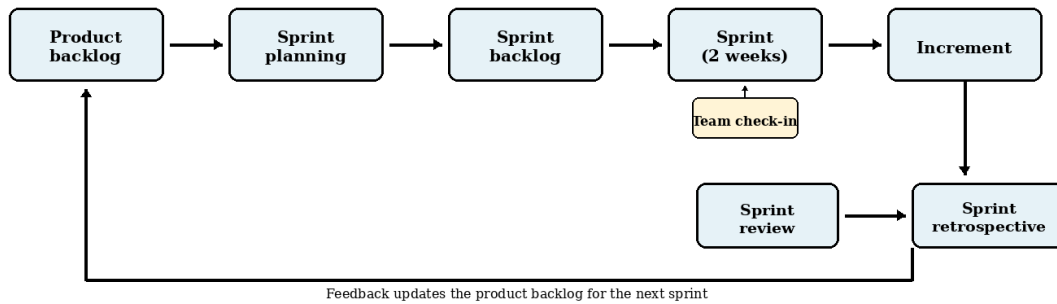


Figure 4.4: Scrum development cycle.

Scrum divides the project into short sprints that each produce a tested increment. The main practices used will be:

- Two-week sprints with planning, development, testing and documentation.
- User stories and a prioritised product backlog for project requirements and technical tasks.
- Sprint reviews to evaluate results, collect supervisor feedback and revise the backlog.
- Sprint retrospectives to identify delays and improve the next sprint.

Scrum is suitable because the team can respond when a model performs poorly, a Singapore source is unavailable or a feature exceeds the available time.

6.1.1 Development Methodology Chosen: Scrum

Scrum is selected as the main methodology because it combines structure with flexibility. This is important for an AI-based misinformation detection project, where models, datasets, trusted sources and scoring methods must be evaluated before the final approach is confirmed.

Justification for Choosing Scrum

1. Adaptability: Requirements can be revised when testing or supervisor feedback reveals problems.
2. Iterative progress: Short sprints provide regular working results within the six-month schedule.
3. Early evaluation: AI detection, factual verification and scam analysis can be tested separately.
4. Scope control: Essential functions are prioritised, while unrealistic features are moved to future work.
5. Explainability and ethics: Each sprint can review false positives, data quality, robustness and user-facing explanations.

How Scrum Will Be Used

At the start of each sprint, the team will select a small number of high-priority tasks. A task is complete only when it works, has passed basic tests and has updated documentation. The sprint review and retrospective will then guide the next sprint.

Conclusion

A lightweight Scrum process, supported by prototyping and Kanban, is the most realistic approach for this six-month, zero-budget student project.

6.2 Development Tools

Current implementation clarification: JavaScript/React Native with Expo 57, Python/FastAPI/Uvicorn, Firebase Authentication and Cloud Firestore. The mobile API wrapper uses fetch rather than the Axios library originally proposed. Database descriptions below retain the wider planned storage scope; see Section 5.5 for implemented collections.

The development stack has been selected for an Android-based AI-assisted misinformation-checking prototype. The selected tools provide native Android support, straightforward integration with Python-based processing, and a manageable architecture for the FYP team.

6.2.1 Front-End

Front-End	Selected Technology
Language	JavaScript
Framework	React Native with Expo
Development Tool	Visual Studio Code, Expo and Android Studio Emulator
Description	JavaScript will be used to implement the Android application with React Native. React Native will provide reusable components for the interface, navigation and application state, while Expo will simplify development, testing and Android builds. Visual Studio Code will be used to write the application code, and the Android Studio Emulator will be used for device testing.
Why We Chose Them	JavaScript and React Native were selected because they allow the team to develop the Android interface using familiar web-style component concepts. Expo supports rapid prototyping and reduces the initial Android project configuration required. This combination is appropriate for the FYP because it supports faster interface development while still producing an Android application.

6.2.2 Back-End

Back-End	Selected Technology
Language	Python
Framework	FastAPI
Runtime Server	Uvicorn
Development Tool	Visual Studio Code
Description	Python will be used for server-side processing and integration with the application's AI and data-processing components. FastAPI will provide REST API endpoints for receiving

Back-End	Selected Technology
	submissions, coordinating analysis, applying usage limits and returning structured results. Uvicorn will run the FastAPI application during development and deployment.
Why We Chose Them	Python is well suited to AI, natural-language processing and image-analysis libraries. FastAPI provides request validation, clear routing and automatically generated API documentation, making it practical for a five-person FYP team. Using Python for both the back-end and AI components also avoids maintaining a separate Node.js service.

6.2.3 Database

Database	Selected Technology
Database	Cloud Firestore
Database Type	NoSQL document database
Development Tool	Firebase Console
Description	Cloud Firestore will store user profiles, account roles, free and premium submission allowances, analysis results, result history, evidence records, feedback and reported incorrect results. The current Android client accesses protected account and result data through FastAPI; the backend Firebase Admin SDK performs Firestore operations. The Firebase Console will be used to inspect records, configure access rules and monitor database usage.
Why We Chose It	Cloud Firestore integrates with Firebase Authentication and can be accessed from the React Native application through the Firebase JavaScript SDK. It requires less database administration than a self-hosted database, and its document structure is suitable for user accounts, submissions and result records. Firebase Realtime Database is not required for the selected scope.

6.2.4 API Communication

API Communication	Selected Technology
Architecture	RESTful API
Data Format	JSON
React Native Library	fetch

API Communication	Selected Technology
Testing Tool	Postman
Description	The React Native application will communicate with the FastAPI back-end through HTTPS REST endpoints. fetch will send text, links, images and authentication headers to the back-end and convert JSON responses into JavaScript objects. Postman will be used to test the API independently from the mobile interface.
Why We Chose Them	The current app uses fetch for JSON requests and Firebase bearer headers, with request timeouts and friendly error handling. Server-only credentials and analysis logic remain outside the Android client.

6.2.5 Authentication

Authentication	Selected Technology
Authentication Service	Firebase Authentication
React Native Integration	Firebase JavaScript SDK
Initial Sign-In Method	Email and password
Description	Firebase Authentication will manage account registration, login, logout, password reset and user sessions through the Firebase JavaScript SDK. The React Native application will obtain a Firebase identity token, and the FastAPI back-end will verify that token before allowing protected operations. Application roles and access tiers will be checked by the back-end using the associated account record.
Why We Chose It	Firebase Authentication works with the selected React Native and Cloud Firestore stack and removes the need to build password storage and session handling from the beginning. Email-and-password authentication is sufficient for the initial prototype and keeps registration familiar and straightforward for the target users.

6.3 Current analysis components

Component	Implemented method	Important limitation
Input preparation	Validate at most 5,000 characters; normalize Unicode/whitespace without discarding meaning; verify English; warn about instruction-like content.	A warning is not a complete prompt-injection defence.
Claim classification	Concurrent Ollama Cloud calls to gpt-oss:120b, gemma4:31b and nemotron-3-super. At least two valid replies are needed; valid votes select a majority category.	A split returns unverifiable; agreement is not a truth probability.
Claim extraction	gemma4:31b extracts a principal factual source span; validate JSON and occurrence in the submitted text.	A span match cannot prove all decisive context was retained.
Evidence retrieval	Google published-check discovery → Tavily page extraction → scoped Tavily search when needed; validate, rank and deduplicate citations.	Lexical relevance can miss paraphrases or admit related but inconclusive sources.
Evidence assessment	Supporting/contradicting/neutral stance using lexical overlap, negation, amounts and dates; heuristic quality and risk rules.	Known semantic stance errors; no calibrated probability or trained assessment model.
Orchestration	Validate handoffs, skip unnecessary stages, distinguish failures from insufficient evidence, return structured results.	One principal claim is processed; multiple-claim coverage needs evaluation.

6.3.1 Model voting, validation and prompt handling

Invalid responses and provider failures do not vote. Fewer than two valid classifiers produces a controlled failure. A winning category uses agreement divided by the three configured models, so 0.67 is agreement rather than a 67% likelihood of truth. Submitted content is placed in a separate untrusted data message; system instructions define the task. Locally enforced schemas and span checks reject malformed or fabricated extraction output.

The configured claim stage uses a 90-second overall deadline and bounded provider calls. Retrieval uses a 65-second stage deadline; the current mobile text request waits up to 180 seconds. These bounds limit waiting but do not establish a response-time service guarantee. There are no automatic provider retries in these stages.

Evidence / source: Current `claim_analysis/categories.py` and `service.py`; `evidence_retrieval service/providers`; `mobile/src/services/api.js`. Ollama Cloud reference in Section 13.

6.4 APIs, interfaces and dependencies

Endpoint / boundary	Purpose
GET / and GET /health	Basic process reachability; does not prove provider readiness.
PUT /account/me	Create/refresh the authenticated profile; verification status is enforced by subsequent protected operations.
GET /account/me	Return the verified active user's own profile and allowance.
POST /analysis/text	Verified, active account; English text payload; optional Idempotency-Key; controlled errors and owned completed/failed persistence.
GET /analysis/results	Owned result list, default page limit 20 and maximum 50; optional result-ID cursor.
GET /analysis/results/{result_id}	Owned result detail; missing/other-user records return 404.
Pipeline models	PreparedText → ClaimAnalysis → RetrievalResult → AssessmentResult → TextAnalysisResult, plus FailedAnalysisRecord.
Dependency	Purpose / credential boundary
Firebase Authentication / Firestore	Client identity and backend-owned storage. Admin JSON stays outside the repository; no password is stored in a user-profile document.
Ollama Cloud	Claim-stage inference using a server-only API key; no local model download required.
Google Fact Check Tools API	Search existing ClaimReview records; no claim-verification conclusion is implied by a search match alone.
Tavily Search / Extract	Fetch source passages and search the scoped catalogue; provider keys remain backend-only.
External source pages	Evidence/provenance; availability, publication date and meaning must be evaluated for each claim.

The app displays controlled errors for invalid input, authentication/access failure, depleted allowance, overlapping/reused requests and unavailable services. Empty successful searches can produce Not Enough Information; technical failures are not converted into a confident assessment.

Evidence / source: backend/app/routers, pipeline/shared/models.py; provider APIs in Section 13.

6.5 Hosting, deployment and configuration

Current delivery is a local Windows/Android-emulator demonstration with a cloud Firebase project and cloud AI/search providers. There is no verified public API deployment, installed release APK, Play Store listing or operational portal. Source-control use does not establish a CI/CD pipeline.

Project-root .env supplies OLLAMA_API_KEY, GOOGLE_FACT_CHECK_API_KEY, TAVILY_API_KEY and GOOGLE_APPLICATION_CREDENTIALS. The credential path points outside Git. The mobile environment contains Firebase Web configuration and EXPO_PUBLIC_API_URL; it must not contain provider keys or the Admin JSON. Real credentials are deliberately omitted from this report.

```
cd C:\Dev\uiabo\backend
.\.venv\Scripts\python.exe -m uvicorn app.main:app --host 127.0.0.1 --port 8000
```

```
cd C:\Dev\uiabo\mobile
npm.cmd ci
npx.cmd expo start --lan
```

The existing virtual environment and private configuration must be prepared first. The standard Android emulator uses http://10.0.2.2:8000 to reach this backend. This is local development HTTP; the submitted deployment design requires HTTPS. The PUM Section 2 provides the longer setup procedure.

The current workspace uses Expo 57, React Native 0.86.3, React 19.2.3, Node v24.14.0 and Python 3.14.3. Expo's SDK 57 reference lists Node 22.13.x and Android 7+ as framework minimums; these are not a UIABO device-compatibility certification. A clean-machine and physical-device matrix remains pending (Expo, SDK 57 reference).

6.6 Release work still required

- Choose and document deployment targets, HTTPS endpoints, environment separation and release artifact/version.
- Provide reproducible Firebase rules/IAM configuration and a tested recovery/backup procedure.
- Implement and evaluate required static-image/deepfake functions, then payments and operational workflows as agreed.
- Validate assessment accuracy, latency, capacity, accessibility and failure behavior before stronger release claims.

Evidence / source: Current READMEs, configuration templates, package files and PUM; official Expo SDK 57 reference accessed 7 September 2026.

7. Risk Analysis

The original PRD risk ratings below are retained as the planning baseline. They are qualitative estimates, not current measured likelihoods or completed compliance assessments. The update after the table records observed risks and remaining actions.

Risk Category	Identified Risk	Risk Elements		
		Impact	Probability	Risk Rating
Communication	Group members are required to be proactive in terms of communicating with stakeholders	H	L	M
	Inadequate user feedback to improve verdict explanations and interface	M	M	M
Resources	No project budget; reliance on free-tier APIs subject to rate limits	M	H	M
	Lack of a labelled Singapore-context scam and misinformation dataset	H	H	H
	Lack of access to professional fact-checkers for ground-truth labels	M	M	M
	Limited team experience in machine learning and mobile development	M	L	M
Technical	Language model produces confident but incorrect explanations	H	M	H
	OCR errors on uploaded screenshots corrupt the text analysed	H	M	M
	Free-tier hosting cannot serve the required model size or GPU inference	M	L	M
	Inference latency too high for acceptable mobile response time	M	H	M
User	Users over-trusting an incorrect verdict and acting on it	H	M	H
	Low user engagement due to a complex or unfamiliar interface	M	M	M
	Privacy concerns discourage users from uploading screenshots	M	L	M
Maintenance	Blacklists and trusted news sources becoming stale	H	M	H
	Scam and misinformation tactics evolving after deployment	M	H	M
Security and Legal	Prompt injection in submitted content manipulating the model's verdict	H	M	H
	Non-compliance with PDPA in storing and processing user uploads	H	M	H
	Third-party personal data contained in uploaded chat screenshots	M	L	H

7.1 Current risk treatment

Risk / evidence	Treatment in place	Remaining action / responsible area
Assessment misreads debunking or unrelated negation.	Uncertainty, citations and prototype limitations are displayed.	Assessment owner + integration: evaluate semantic stance with quoted evidence and held-out labels.
Cloud/search provider failure or quota.	Bounded calls, fallback retrieval, sanitized errors; failed runs do not charge.	Integration: record provider usage and representative outage/recovery behavior.
Prompt injection or malformed AI output.	Untrusted-data prompts, local schema validation, source-span checks.	Claim/assessment owners: broader adversarial evaluation; no immunity claim.
Another user reads a saved result or double charge occurs.	Server ownership, transactions, idempotency, per-user lease; focused tests passed.	Backend: broader review of rules/IAM, recovery and deployment controls.
Evidence is outdated, related or duplicated.	Scoped sources, URL/content deduplication and dates retained when known.	Retrieval/assessment: temporal context and independent-source evaluation.
Required features exceed available schedule.	September prototype narrowed to the basic text workflow; image tools labelled unavailable.	Project leader/team: confirm owners and dates for required deepfake/image delivery.
User over-trust or difficult navigation.	Readable TDM-based screen layout and evidence/uncertainty messaging.	UI/testing: older-user usability study and false-reassurance analysis.
Privacy/compliance documentation incomplete.	Provider secrets separated; owned data access checked; no payment-card storage.	Team: final privacy/terms, retention/deletion, processor and policy review. No legal-compliance certification claimed.

Evidence / source: PRD 4.3; observed assessment and app-integration reports; current implementation. Action allocation is a proposed follow-up by responsibility area.

8. System Test Summary

8.1 Overview

The former PTD reported 154 tests and disconnected claim/retrieval stages. The latest recorded integration snapshot reports 262 passing backend tests, a successful Android export and live authenticated Firestore checks. Claim analysis and retrieval are now connected. These results verify behavior and integration; they do not establish misinformation-detection accuracy.

8.2 Scope and methodology

- Unit/contract tests cover input preparation, language checks, AI response validation, retrieval filtering/fallback, lexical assessment and orchestrator stopping rules.
- FastAPI tests verify authenticated access, owner-only history/detail, quota, retries, conflicting keys and controlled errors.
- Transaction/lease tests exercise duplicate completion, concurrent requests, stale leases, rollback, calendar resets and denial of client-selected Premium.
- Offline tests use mocked providers and in-memory/fake stores. Separate opt-in live smoke tests consume provider quota and use isolated integration accounts.
- Android export verifies bundling; emulator captures and API observations establish selected user flows. They are not physical-device or elderly-user acceptance tests.

8.3 Dated result summary

Date / record	Observed result	Limit
6 Sep — Matthew claim report	207 backend tests at that point; 7/8 synthetic claim-stage examples matched; satire disagreed.	Not a held-out benchmark or factual-veracity accuracy.
7 Sep — retrieval report	247 backend tests at that point; 45 retrieval tests. Three live text-pipeline requests completed with real providers.	Operational success includes an observed incorrect stance on a myth passage.
7 Sep — app integration	262 backend tests passed; one existing Starlette/httpx deprecation warning. Android production export succeeded.	Dated recorded run; no new full test execution was needed for this documentation update.
7 Sep — live Firestore report	9 recorded checks: replay, owned history/detail, other-user denial, quota, stored evidence and direct-read denial.	Focused checks with isolated accounts, not a complete security audit.
7 Sep — Android captures	Premium Home, private history and saved result opened in emulator.	No paid subscription, physical-device matrix or older-user study proven.

8.4 Live integration observations

Input / experiment	Observed output	Interpretation
Great Wall visible from the Moon	Needs Caution, score 67; Snopes and two NASA sources; recorded 6.36 s.	A NASA myth-description passage was wrongly treated as supporting. Operational pass; assessment error remains.
Singapore independence on 9 August 1965	Low Concern, score 25; NUS/NLB sources; recorded 14.90 s in the pipeline check.	One factual example, not a general accuracy or latency measure.
Chicken rice preference	Not Enough Information, null score; retrieval skipped; recorded 12.89 s.	Non-checkable stopping path observed.
Authenticated Firestore smoke flow	Saved independent-Singapore claim with evidence; repeat returned same result; next free submission rejected with 429.	Checks ownership and charging; 11.72 s report duration covers the smoke sequence, not a single inference latency.

8.5 Evaluation still required

Area	Evidence needed
Assessment accuracy	Team-labelled, held-out claims/passages; per-class precision/recall/F1, confusion matrix and false Low Concern rate.
Retrieval quality	Relevant decisive evidence found, passage context retained, date/entity alignment and source independence.
Performance/capacity	Repeated latency measurements including p50/p95, concurrent-load testing against 100 users / 100 requests per minute targets.
Usability/accessibility	Task completion, comprehension of uncertainty, errors and navigation with older users or justified representatives.
Security/privacy/recovery	Broader rules/IAM review, adversarial prompts, deployment transport, backup/recovery and retention/deletion validation.
Image and release acceptance	Selected detector evaluation, OCR/context tests, physical devices and signed release package.

8.6 Proposed assessment-improvement experiment

Compare the current lexical baseline with an evidence-grounded LLM or evaluated entailment model. Require per-source stance, an exact evidence quote and a short justification; validate the quote against the passage. Check entities, dates, amounts and eligibility, and retain a neutral/insufficient-evidence outcome. Deduplicate dependent sources before aggregation. This is a proposed experiment, not an implemented fix.

Start with 60–100 balanced claims including myths, unrelated negation, dated announcements and missing evidence. Two teammates should label and reconcile disagreements. Separate development examples from held-out evaluation and report coverage as well as errors, so returning Not Enough Information for everything is not mistaken for good accuracy. No final numeric acceptance target has been approved.

Evidence / source: evaluation/reports/matthew_claim_evaluation.md; chu_retrieval_evaluation.md; text_pipeline_live_2026-09-07.json; app-integration/README.md and firestore-live.json. Existing synthetic samples are not independent ground truth.

9. Project Communication

9.1 Overview

The five-person team uses GitHub for source control and shared implementation, and maintains PRD, URS, TDM, meeting notes and Sprint 1 handoff samples in the project workspace. Team discussions are used to confirm scope, assign ownership, review interfaces and identify blockers. Shared JSON contracts allow pipeline components to be developed and tested without waiting for another member's unfinished implementation.

- GitHub repository for code review, change history and integration.
- Meeting notes and presentation material for decisions and progress reporting.
- PRD, URS and TDM for requirements and design alignment.
- Sprint 1 sample JSON files for cross-member interface coordination.

9.2 Project Website

No deployed UIABO project website or public operational portal has been supplied. The current deliverables are the local Android prototype, source repository and submission documents. A website URL, hosting record and screenshots remain pending if a website is required.

9.3 Communication management plan

Proposed additions for team review. Existing evidence supports GitHub, shared documents and meeting notes; the cadence and escalation rules below are not claimed as previously approved practices.

Audience / purpose	Channel / artifact	Proposed trigger
Team implementation coordination	Repository changes, shared contracts, task list and blocker notes.	Each handoff or blocker; include owner, next action and evidence.
Supervisor progress/scope review	Dated progress summary and demonstrable build.	Agreed review meetings and material scope changes.
Assessor delivery	Submission PDFs, user manual and reproducible demonstration instructions.	Required submission milestones.
User feedback	Planned usability notes and feedback/report workflows.	Each approved study or implemented feedback submission.

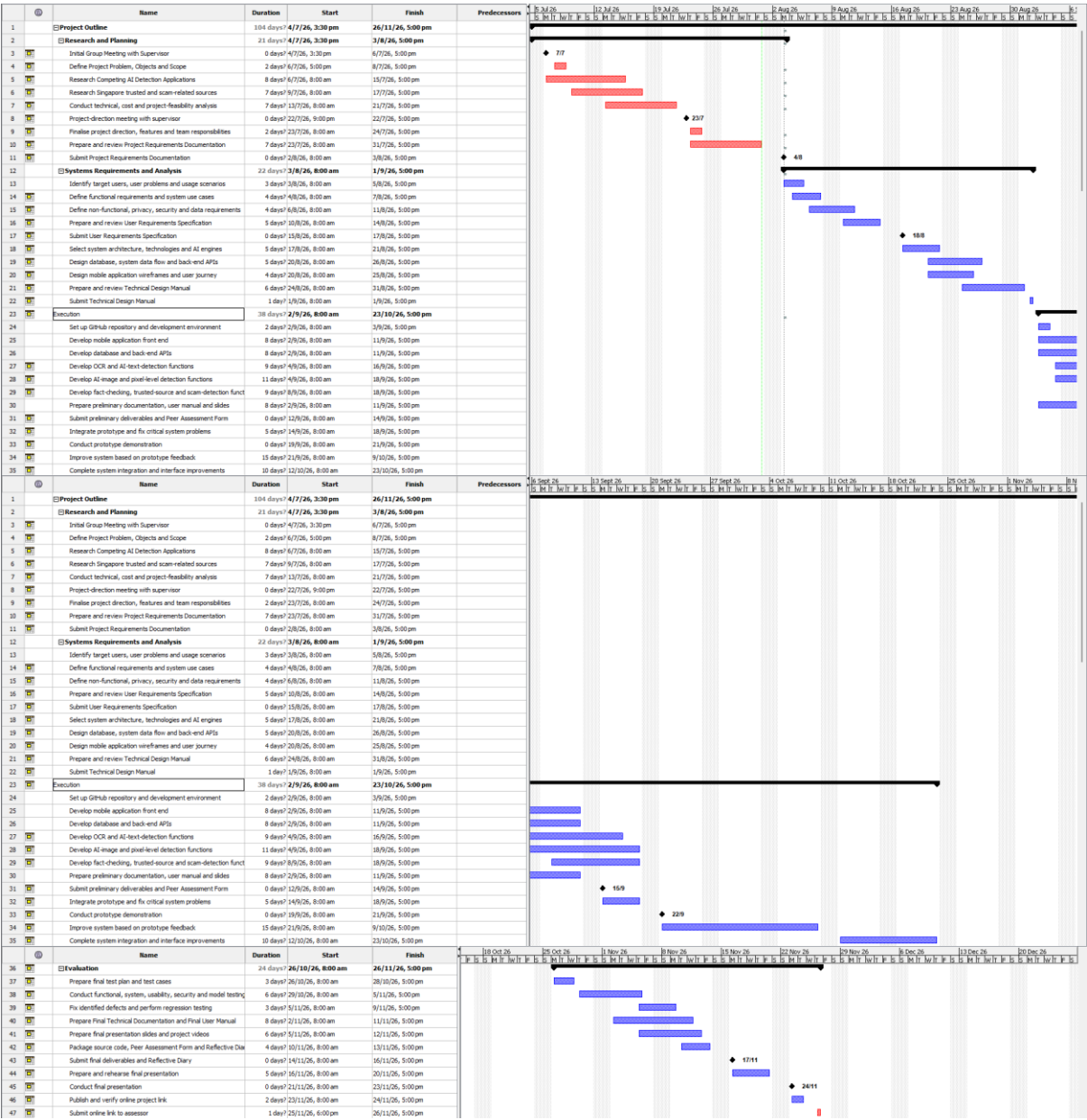
9.4 Conflict escalation and document revision

- Record the disputed requirement or technical decision, alternatives and impact in a shared note.
- The affected component owners and project leader review it against the URS/TDM and prototype goal.
- Escalate unresolved scope or academic-requirement questions to the supervisor; record the decision without inventing approval.
- For interface changes, update schema, examples, tests and dependent components together.
- For document revisions, retain the original, record the date/reason/source, refresh contents and review the exported PDF.

Evidence / source: Existing Section 9, MeetingNotes and Sprint1/UIABO_SPRINT_1_TEAM_TASKS.md. The PUM and this PTD update provide dated documentation artifacts.

10. Project Schedule

The Gantt images below are the original PRD planning baseline. They do not reflect a newly approved rebaseline or establish actual completion. The dated milestones and work breakdown after the figures show the evidence available for this update.



10.1 Milestones and current progress

Milestone	Date / state	Evidence or action
Initial backend/Firebase foundation	12–13 Aug 2026, recorded	Meeting brief records basic API tests and Firestore connectivity.
Claim analysis integration	6 Sep 2026, recorded	Ollama live report and validation changes.
Retrieval integration	7 Sep 2026, recorded	Google/Tavily live requests and pipeline results.
Android and owned persistence	7 Sep 2026, recorded	Emulator, Firestore history/detail and quota checks.
Preliminary documentation	7 Sep 2026, draft	PUM section files/PDF and this updated PTD.
Prototype submission	12 Sep 2026, user-provided deadline	Basic Sprint 1 text demonstration; scope clarity, known limits and reproducible setup needed.
Final submission	November 2026, exact date pending	Complete required image/deepfake scope and evaluation; confirm official deadline and rebaseline.

10.2 Work breakdown structure

Work package	Deliverable / responsibility area	State
W1 Requirements/design	PRD, URS, TDM; team and documentation.	Existing baseline; scope differences recorded.
W2 Input preparation	English validation and preparation; Yi Da stage.	Integrated.
W3 Claim analysis	Classification/extraction; Matthew stage.	Integrated; additional evaluation needed.
W4 Evidence retrieval	Provider discovery/search/citations; Chu stage.	Live completion exists in the current workspace; confirm contribution attribution.
W5 Assessment	Evidence stance/risk/explanation; Poon stage.	Integrated baseline; semantic accuracy improvement outstanding.
W6 App/API/persistence	Android flow, auth, results and quota; Donovan-led integration.	Basic text journey working.
W7 Required image scope	OCR, image/caption/context, deepfake.	Not implemented; confirm named owners and dates.
W8 Remaining product functions	Payments, deletion, reports, public sharing and portals.	Planned; prioritization and delivery dates needed.
W9 Evaluation/release	Dataset, usability, load/security, packaging and documentation.	Partial tests/drafts; acceptance evidence incomplete.

10.3 Proposed next work order

Before 12 September, prioritize reproducible text demonstration, clear unavailable-feature labels, meaningful accuracy examples and consistent documentation. For the remaining project, prioritize required image/deepfake implementation and assessment evaluation, then agreed operational/subscription functions and release validation. These priorities do not assign invented sprint dates or certify supervisor acceptance.

Evidence / source: PRD 4.5 Gantt; dated meeting/integration reports; user-provided September and November milestones.

11. Roles and Responsibilities

Name	Roles	Responsibilities	Contact Information
HO SZE WEI, DONOVAN	Project Leader, Back-End Developer, Database Engineer, Documentation	Monitor the overall progress of the project and delegate tasks to each team member. Design, develop, and maintain the back-end functions of the system. Design and manage the project database. Ensure that project documentation is properly prepared and updated.	swdho001@mymail.sim.edu.sg
MATTHEW ALEXANDER PEERIS	Documentation, Database Engineer, Software Analyst, Tester	Manage and maintain the project documentation. Design, develop, and manage the project database. Analyse the system requirements and ensure that the system meets its intended business goals. Prepare test cases and conduct testing to identify errors and verify that the system functions correctly.	peeris001@mymail.sim.edu.sg
CHU WAI CHUNG	Lead Documentation, Database Engineer, Software Tester, Software Analyst	Lead and manage the preparation of project documentation, including meeting minutes and progress records. Design, develop, and maintain the project database. Analyse the system requirements and ensure that the system meets its intended objectives. Prepare test cases and conduct software testing to verify the quality and functionality of the system.	wcchu001@mymail.sim.edu.sg
POON CHUN PING	Database Engineer, Documentation, Software Analyst	Design, develop, and maintain the project database. Assist in preparing and updating the project documentation. Analyse the system requirements and ensure that the proposed system meets its intended functions and business goals.	cppoon001@mymail.sim.edu.sg
WONG YI DA	Front-End Developer, Back-End Developer, Software Tester, Security Tester	Design and develop the front-end and back-end components of the system. Prepare and conduct software tests to ensure that the system functions correctly. Perform security testing to identify vulnerabilities and implement suitable security measures to protect the system and user data.	ydwong003@mymail.sim.edu.sg

11.1 Sprint 1 component ownership and contribution record

Member	Agreed component handoff	Current evidence / attribution limit
Wong Yi Da	PreparedText: input preparation	Input validation and English gate are connected.

Member	Agreed component handoff	Current evidence / attribution limit
Matthew Alexander Peeris	ClaimAnalysis: extraction/classification	User confirmed his work was ready; cloud integration/validation results are recorded.
Chu Wai Chung	RetrievalResult: evidence retrieval	Original filtering/test hook retained; unfinished live provider work was completed in the current integration workspace. Do not equate ownership with sole authorship.
Poon Chun Ping	AssessmentResult: evidence assessment	Lexical baseline integrated; observed stance errors remain.
Ho Sze Wei, Donovan	Orchestration, API, app and persistence integration	Coordinates completed pipeline handoffs, Android results, Firestore ownership and allowances, and documentation updates.

The original role table records planned responsibilities. Before final submission, the team should reconcile actual contributions with repository history and reassigned tasks; no unverified percentage contribution or individual sign-off is included.

Evidence / source: Sprint 1 team-task document; existing PRD role table; dated integration reports and project discussion.

12. Meeting Minutes

Historical meeting records are retained as recorded; they are not rewritten to reflect later progress. The earlier note treating deepfake as scope-sensitive is superseded for current planning by the user's confirmation that deepfake is required and audio excluded. This later clarification is not claimed as a recorded supervisor approval at the August meeting.

12.1 Implementation Planning Meeting – 13 August 2026

Objective

Review the current UIABO implementation, agree on the first evidence-backed misinformation pipeline and divide Sprint 1 work.

Recorded Progress

- Initial FastAPI application, health endpoint and text-analysis endpoint created.
- Request and response validation and initial automated API tests completed.
- Firebase project and Cloud Firestore connection established, with a successful write-and-read smoke test.
- The original fixed mock result was identified as unsuitable for a real misinformation assessment.

Decisions and Actions

- Use a staged pipeline: input preparation, claim analysis, evidence retrieval, evidence assessment and final integration.
- Develop against shared Pydantic and JSON interfaces so members can work independently.
- Return Not Enough Information rather than inventing a risk score when a claim is not checkable or evidence is insufficient.
- Treat deepfake or AI-generated-image detection as scope-sensitive until it is confirmed with the supervisor.
- Create a small evaluation dataset and test each component before complete integration.

12.2 Available Supporting Records

Record	Purpose
MeetingNotes/uiabo_meeting_notes.md	Detailed implementation review, pipeline proposal and decisions required.
MeetingNotes/team_meeting_brief_2026-08-13.md	Condensed meeting brief and suggested next-sprint goal.
MeetingNotes/Meeting1Slides.md	Slides used to present the Sprint 1 plan.
MeetingNotes/Meeting1_Speaker_Notes.md	Slide-by-slide presentation guidance.
MeetingNotes/Sprint1/UIABO_SPRINT_1_TEAM_TASKS.md	Team ownership, interface examples and completion checklist.

12.3 Later progress records and missing minute fields

The September provider and app-integration reports are technical progress evidence, not formal meeting minutes. No meeting time, attendance, minute taker or approval has been invented from those reports. Complete these fields from actual team records before final submission.

13. References

The existing PRD market-research references are retained with their original recorded access dates; that historical comparison was not re-audited during this documentation update. Technical references below were consulted for the current integration description.

- Corney, D. (2025, February 11). *How AI can help fact checkers*. Full Fact AI. <https://fullfact.ai/blog/how-ai-can-help-fact-checkers/>
- Digital for Life. (2025, July 8). *Seniors Go Digital: Digital skills for seniors*. <https://www.digitalforlife.gov.sg/About/Our-Projects/Seniors-Go-Digital>
- Factiveverse. (n.d.-a). *Factiveverse API*. Retrieved July 31, 2026, from <https://www.factiveverse.ai/solutions/api>
- Factiveverse. (n.d.-b). *Factiveverse Gather*. Retrieved July 31, 2026, from <https://www.factiveverse.ai/solutions/gather>
- Factiveverse. (n.d.-c). *Factiveverse Live*. Retrieved July 31, 2026, from <https://www.factiveverse.ai/solutions/live>
- Full Fact AI. (n.d.). *How we do it*. Retrieved July 31, 2026, from <https://fullfact.ai/product/>
- Google. (2024, August 8). *Fact Check Tools API*. Google for Developers. <https://developers.google.com/fact-check/tools/api/reference/rest>
- Johnston, C. (2025, October 1). *How we're using generative AI to transcribe challenging audio*. Full Fact AI. <https://fullfact.ai/blog/using-generative-ai-to-transcribe-challenging-audio/>
- National Population and Talent Division. (2026, July 8). *Longevity*. <https://www.population.gov.sg/our-population/population-trends/longevity/>
- NewsGuard. (n.d.-a). *False Claim Fingerprints*. Retrieved July 31, 2026, from <https://www.newsguardtech.com/solutions/false-claim-fingerprints/>
- NewsGuard. (n.d.-b). *False Claim Fingerprints process*. Retrieved July 31, 2026, from <https://www.newsguardtech.com/ratings/false-claim-fingerprints-process/>
- NewsGuard. (n.d.-c). *News Reliability Ratings*. Retrieved July 31, 2026, from <https://www.newsguardtech.com/solutions/news-reliability-ratings/>
- Soon, C., Bala Krishnan, N., & Goh, S. (2022, March 17). *Why are some people more immune to falsehoods and what can we learn from them?* Institute of Policy Studies. <https://lkyspp.nus.edu.sg/ips/publications/details/commentary-why-are-some-people-more-immune-to-falsehoods-and-what-can-we-learn-from-them>
- WeVerify. (n.d.-a). *Context aggregation and analysis*. Retrieved July 31, 2026, from <https://weverify.eu/tools/caa/>
- WeVerify. (n.d.-b). *Image Verification Assistant*. Retrieved July 31, 2026, from <https://weverify.eu/tools/image-verification-assistant/>
- WeVerify. (n.d.-c). *Tools*. Retrieved July 31, 2026, from <https://weverify.eu/tools/>
- WeVerify. (n.d.-d). *Verification plugin*. Retrieved July 31, 2026, from <https://weverify.eu/verification-plugin/>
- World Wide Web Consortium. (2025, November 20). *Older users and web accessibility: Meeting the needs of ageing web users*. <https://www.w3.org/WAI/older-users/>

13.1 Implementation references

- Expo. SDK 57 reference. Accessed 7 September 2026. <https://docs.expo.dev/versions/v57.0.0/>
- Firebase. Verify ID tokens. Accessed 7 September 2026. <https://firebase.google.com/docs/auth/admin/verify-id-tokens>

Firebase. Transactions and batched writes. Accessed 7 September 2026.

<https://firebase.google.com/docs/firestore/manage-data/transactions>

Ollama. Cloud. Accessed 7 September 2026. <https://docs.ollama.com/cloud>

Google. Fact Check Tools API: claims.search. Accessed 7 September 2026.

<https://developers.google.com/fact-check/tools/api/reference/rest/v1alpha1/claims/search>

Tavily. Search API. Accessed 7 September 2026. <https://docs.tavily.com/documentation/api-reference/endpoint/search>

Tavily. Extract API. Accessed 7 September 2026. <https://docs.tavily.com/documentation/api-reference/endpoint/extract>

13.2 Internal evidence and submission references

PRD/FYP-26-S3-30_PRD.docx; URS/FYP-26-S3-30_URS.pdf; TDM/FYP-26-S3-30_TDM.pdf.

MeetingNotes/team_meeting_brief_2026-08-13.md and

Sprint1/UIABO_SPRINT_1_TEAM_TASKS.md.

uiabo/evaluation/reports/matthew_claim_evaluation.md and matthew_claim_live_2026-09-06.json.

uiabo/evaluation/reports/chu_retrieval_evaluation.md and text_pipeline_live_2026-09-07.json.

uiabo/evaluation/reports/app-integration/README.md, firestore-live.json and Android captures.

uiabo/evaluation/trusted_sources/README.md and pipeline component READMEs.

Submission1/PUM/FYP-26-S3-30_PrelimUserManual_DRAFT.docx and

PUM_MISSING_INFORMATION.md.

14. Data Collection and Evidence Management

This chapter adds the data-collection coverage present in the comparison sample. UIABO currently retrieves evidence at request time and uses existing cloud models. It has not trained a misinformation model on a completed proprietary dataset, and the planned data-ingestion portal is not implemented.

14.1 Data sources

Source	Data / role	Limit
User submission	English text and original wording.	May contain personal information, false claims or instructions; treat as untrusted.
Google Fact Check Tools	Previously reviewed claim metadata and review URLs.	A reviewed claim match is discovery, not sufficient evidence by itself.
Tavily extraction/search	Page passages, titles, URLs and available dates from scoped web sources.	Pages may be unavailable, truncated or out of context.
Runtime source catalogue	gov.sg, nasa.gov, snopes.com, fullfact.org, factcheck.org, reuters.com, apnews.com, channelnewsasia.com, nus.edu.sg.	Initial English allowlist; inclusion does not guarantee truth, independence or peer review.
Team evaluation samples	Fixed handoff fixtures and synthetic claim/assessment examples.	Development tests; insufficient as independent evaluation or a training corpus.

14.2 Types of data and provenance

Data group	Fields / treatment
Identity and access	Firestore UID, name/email, verification, account status and role; passwords remain in Firestore Authentication.
Claim	Original/normalized input, extracted source span, category, checkability, classification reason and agreement.
Evidence	Evidence ID, title, source URL, publisher/domain, optional author/date, passage, source type, relevance and retrieval timestamp.
Assessment/result	Stance, quality, concern label, heuristic score or null, uncertainty, explanation, next action, warnings and pipeline version.
Usage and recovery	Owner, result/request identifiers, fingerprint, timestamps, lease and applicable allowance period.

14.3 Collection methods

For checkable content, search up to five English reviewed claims; accept catalogue URLs and extract up to three review pages. Preserve a labelled claim/verdict block when it agrees with provider metadata. If fewer than two distinct passages remain, run scoped advanced search

requesting up to eight results. Validate and rank up to six final evidence items. Provider-generated answers are disabled in that search request.

Google failure or unusable page extraction can trigger the search fallback. A successful empty search is distinguished from provider failure. Partial failure with usable evidence retains warnings. These are current bounded request-time operations, not a scheduled crawling or ingestion service.

14.4 Preprocessing and quality controls

- Preserve original input; normalize Unicode composition and whitespace while preserving meaning-bearing wording and punctuation.
- Validate English, input length and usable content. Flag instruction-like content instead of deleting arbitrary claim text.
- Validate source URL scheme/domain and citation fields; retain meaningful URL query identifiers while removing tracking fragments/parameters.
- Remove duplicate URLs/content. This reduces duplication but does not establish independent corroboration.
- Keep unknown publication dates null; do not substitute event dates or claim dates.
- Retain evidence snapshots with the result so later source changes do not silently rewrite the original explanation.

Current relevance thresholds use lexical coverage and provider relevance. These scores are not truth probabilities. Related passages may omit a decisive condition, and older sources may remain useful for historical claims; recency must be interpreted in context.

14.5 Privacy, access and data lifecycle

Analysis text is sent to the configured cloud claim service, and the extracted claim is used in external evidence searches. The system therefore has external processing dependencies. The final user-facing privacy notice, retention schedule, deletion workflow and review of provider terms/processing arrangements are outstanding. Public accessibility of a source does not establish unrestricted reuse rights.

Current safeguards include backend-only provider secrets, verified/active-account checks, owned result access and no payment-card data collection. Do not present these focused measures as a complete privacy or legal-compliance assessment. Demonstrations should use dedicated accounts and non-sensitive example text.

14.6 Evaluation dataset plan

Build a versioned claim-and-evidence dataset with the original claim, date/context, expected verdict, decisive cited passages and annotation rationale. Record provenance and usage conditions. Label/reconcile independently, separate development from held-out evaluation, and retain failure cases. The proposed 60–100-case starting set in Section 8.6 is not a completed dataset or final sample-size justification.

Evidence / source: Current retrieval/input implementations, shared models, source catalogue and saved integration reports; provider documentation in Section 13.

15. Requirement Definition and Traceability

15.1 Stakeholder identification and evidence gathering

Requirements evidence consists of the PRD's documented product research, the URS user stories/use cases, TDM diagrams and screens, available meeting notes, and subsequent user scope clarifications. The primary audience is older Singapore Android users; operational roles and academic reviewers are identified in Section 2.10. No completed user interview, survey or focus-group study is claimed without its records.

15.2 Requirement analysis and priority

The URS distinguishes base requirements PF-B01–B16 and additional requirements PF-A01–A04. The team's Sprint 1 goal is the basic text pipeline. The user clarified a September Android-emulator prototype and confirmed deepfake as required while excluding audio. These statements narrow the immediate increment; they do not silently remove required final-project features.

The analysis approach maps each user action to a use case, component boundary, data record and verification artifact. Feasibility is judged against evidence quality, external dependencies, UI effort and schedule. Changes affecting scope or shared field names should be recorded and reviewed with affected owners; unresolved academic requirements need supervisor clarification.

15.3 Functional hierarchy and access

Module	Functions / roles
Individual account management	Free/premium registration/access, verification, profile, allowance; deletion and billing planned.
Content analysis	Text now; public-link and premium image/OCR/context/deepfake planned.
Results and evidence	Owned history/detail, citations, uncertainty and native share now; deletion/public links/reports planned.
Administration	Planned authorised account queries, creation, suspension and feedback review.
Data engineering	Planned incorrect-result investigation and ingestion monitoring.

15.4 Requirement-to-evidence matrix

Requirement	Purpose	Implementation / evidence	Status / gap
PF-B01; UC-01/02/04–06	Individual account access/profile	AuthScreen, VerifyEmailScreen, ProfileScreen; account/auth tests.	Available subset; operational equivalents remain planned.
PF-B02; UC-03/16/18	Registration and account deletion	Free registration/verification exists.	Deletion and paid registration not implemented.
PF-B03; UC-07	English text analysis	All text stages, POST /analysis/text, emulator flow.	Operationally demonstrated; accuracy not established.
PF-B04; UC-08	Direct public-link analysis	TDM link screen; copied-text alternative in app.	Direct fetching/screening pending.

Requirement	Purpose	Implementation / evidence	Status / gap
PF-B05; UC-09–11/13	History, details, allowance and reports	Owned GET routes, quota tests, live Firestore.	History/detail/allowance available; reporting pending.
PF-B06/07; UC-17/19/20	Paid upgrade, renewal, cancellation	TDM designs and Premium server role.	Payment subscription flow pending.
PF-B08/09; UC-21/23	Image+caption and OCR	TDM designs and shared intended results.	Pipelines/format limits/evaluation pending.
PF-B10/11; UC-22/24	Context and deepfake/AI image	Required static-image scope in URS/TDM.	Detector and context pipeline pending.
PF-B12–14; UC-30–34	Operational account management	URS cases and TDM portal designs.	Role-restricted portal pending.
PF-B15/16; UC-41–43	Reports and ingestion monitoring	URS cases and TDM portal designs.	Dashboard, investigation and ingestion workflows pending.
PF-A01/02/04; UC-12/14/35	Delete history and review feedback	URS additional requirements.	Not implemented.
PF-A03; UC-15	Public shareable result link	Native summary/citation sharing exists.	Public result-page links pending.
URS 4.1–4.4	Capacity, UI speed, storage and result-time targets	Selected live timings and bundle success.	No load/physical-device/storage-warning certification.
URS 4.5 / 2.5	Access, privacy and security	Focused auth/ownership/idempotency tests.	Inactivity expiry, broader security and compliance review pending.

15.5 Acceptance and change control

A feature is operationally complete only when its intended user flow, authorization, data persistence and failure behavior are demonstrated. Quality claims require their own evaluation. The 262-test record cannot substitute for labelled fact-check accuracy or older-user usability. Acceptance thresholds and final sign-off remain to be agreed and documented.

Evidence / source: URS Sections 2.2, 3 and 4; existing Section 5 facsimiles contain detailed use cases; Section 8 supplies dated verification records.

16. Project Charter, Scope and Delivery Controls

This charter summary is a proposed consolidation of existing requirements and schedule information. It is not a signed authorization or newly approved scope baseline.

Charter field	Draft record
Project	uiabo — AI-Based Misinformation Detection for Short-Form Content; FYP-26-S3-30.
Purpose	Help older Singapore Android users review suspicious short-form content using evidence and clear uncertainty.
Team / oversight	Five members listed on the cover; supervisor Mr Liaw Chun Huei; assessor Mr Terrence Chew.
Immediate milestone	12 September 2026 prototype, as stated by the user; basic Sprint 1 text demonstration.
Final milestone	November 2026; exact official deadline not supplied.
Resources	Existing development machines, Firebase project and provider accounts. No agreed funded budget or committed provider spend documented.
Constraints	English Android scope, external service access/quotas, small evaluation base and limited remaining development time.
Decision / approval	Team review and supervisor confirmation of unresolved acceptance/scope items required; no signatures inferred.

16.1 Scope statement

Category	Scope
Implemented increment	Individual authentication, English text pipeline, cited result, Android Home/text/history/detail and server-enforced allowances.
Required/planned product	Public links with screening; image+caption, OCR, context and deepfake; agreed accounts/subscriptions and operational functions.
Additional features	History deletion, reviews, public result links and operational feedback as identified by the URS.
Excluded	Audio/speech, video and non-English analysis; private/paywalled inaccessible content. No iOS release or public API licensing is part of the current working increment.

16.2 Deliverables and dependencies

- Product: Android application, authenticated API, Firebase configuration, evidence pipeline, source code and reproducible setup.
- Documentation: PRD, URS, TDM, this PTD, PUM, source/evaluation records and actual meeting minutes.
- Verification: automated results, labelled evaluation, usability/performance/security records and final demonstration evidence.

- Dependencies: provider availability, appropriate credentials, source access, detector selection and approved final deadlines.

16.3 Acceptance and unresolved approval

The local text demonstration has evidence of completion, but the full planned product is incomplete. Final acceptance must explicitly address accuracy limits, required image/deepfake delivery, unfinished user/operational flows and non-functional targets. Confirm whether a public project website is an assessed deliverable. Approval dates, signatures and exact final submission dates remain pending rather than fabricated.

Evidence / source: PRD scope/business model/schedule; URS; team-task document; user-provided prototype scope and dates.

17. Glossary

Term	Meaning in uiabo
API	The interface used by the Android client to request backend operations.
Authentication / authorization	Establishing identity / deciding what that identity is allowed to access.
Firebase UID / ID token	Account identifier / signed client identity token verified by the backend.
Firestore	Document database used for account, allowance and owned result records.
Claim	The factual assertion selected from submitted content for evidence checking.
ClaimReview	Published fact-check metadata used to discover existing reviews.
Evidence stance	Whether a passage supports, contradicts or is neutral toward the specific claim.
Provenance	Where evidence came from and the source/time/context needed to inspect it.
Not Enough Information	No suitable checkable claim or decisive evidence; not equivalent to false.
Risk score	Current heuristic concern indicator, not a calibrated probability of falsehood.
Uncertainty	A statement of limitations, weak evidence or disagreement in an assessment.
OCR	Extraction of visible text from an image, with user correction planned.
Deepfake / AI-image analysis	Planned static-image generation/manipulation likelihood assessment, separate from factual truth.
Idempotency	Repeating the same successful request returns the same result without another charge.
Transaction / lease	Atomic group of storage changes / temporary per-user reservation preventing overlapping work.
Ground truth / held-out set	Reviewed reference labels / evaluation examples excluded from development tuning.
WBS	Work breakdown structure: deliverables divided into manageable work packages.
Traceability	Mapping a requirement to design, implementation and verification evidence.

18. Conclusion and Remaining Deliverables

UIABO has progressed from a mock API and disconnected components to a working authenticated text-check journey in an Android emulator. Claim analysis, evidence retrieval, baseline assessment, cited results, owned Firestore history and server-enforced free/premium allowances are connected. Dated tests support those operational claims.

The most important remaining quality gap is evidence interpretation: the current lexical stance component can misread myth quotations and unrelated negation. Required static-image/deepfake work, direct links, payments and operational portals remain unfinished. Documentation now describes these as planned, and it does not equate integration with validated factual accuracy.

18.1 Remaining information and acceptance evidence

Item	Required next evidence / decision
Assessment evaluation	Labelled dataset, independent review, held-out metrics and agreed acceptance criteria.
Image/deepfake delivery	Named owners, detector/services, input limits, implementation and evaluation.
Remaining user/operational functions	Link screening, subscriptions, deletion, reports, public result links and portals according to agreed priority.
Deployment/privacy	Release package, public addresses if required, configuration/rules/IAM, privacy/terms, retention/deletion and recovery procedures.
Non-functional validation	Load, response-time distribution, physical-device testing, inactivity rules, accessibility and older-user comprehension.
Management records	Official November deadline, rebaselined Gantt/WBS dates, approved learning objectives/charter, complete meeting minutes and contribution attribution.

18.2 Appendix and evidence locator

Detailed use cases, diagrams and proposed database/UI designs are retained in Section 5. Dated test records and their limits are summarized in Section 8 and located in Section 13.2.

PTD_UPDATE_SUMMARY.md maps the comparison sample to the revised chapters.

PTD_MISSING_INFORMATION.md is the living checklist. The original PDF and prior checklist are preserved in backups/.